```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

```
    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    #  开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
#  ==== app/__init__.py ====
#  -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    #  模板里展示时间统一走 localdt：库里存 UTC，展示成本地时间
    from app.utils.timefmt import to_local
    app.jinja_env.filters["localdt"] = to_local
    #  注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    #  全局登录保护：除登录页与静态资源外，均需登录
```

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from flask import request, redirect, url_for,                    101
send_from_directory                                              102
from flask_login import current_user, login_user                 103
PUBLIC_ENDPOINTS = {"auth.login", "auth.register",               104
"static", "favicon"}                                             105
#  浏览器默认会请求根路径下的 /favicon.ico。                      106
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会  107
#  回退到这个路径，补一条路由免得日志里一直刷 404。               108
@app.route("/favicon.ico")                                       109
def favicon():                                                   110
    return send_from_directory(app.static_folder,                111
"favicon.ico",                                                   112
                                       mimetype="image/vnd.micro 113
soft.icon")                                                      114
@app.before_request                                              115
def require_login():                                             116
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录    117
    if app.config.get("SCREENSHOT_MODE"):                        118
        if (request.endpoint not in ("auth.login",               119
"auth.register", "static")                                       120
                and not current_user.is_authenticated):          121
            admin = User.query.filter_by(role="admin").f         122
irst()                                                           123
            if admin:                                            124
                login_user(admin)                                125
        return None                                              126
    if request.endpoint in PUBLIC_ENDPOINTS or                   127
request.endpoint is None:                                        128
        return None                                              129
    if not current_user.is_authenticated:                        130
        return redirect(url_for("auth.login"))                   131
    return None                                                  132
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。  133
#  不使用模块编号，避免出现与实际流程不一致的代号。              134
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）  135
#  与可选的待办角标（该步骤还没做完的图斑数）。                  136
FLOW_LINKS = [                                                   137
    ("项目与矿山档案", "/project/", "folder"),                   138
    ("政策与标准知识库", "/standard/", "book"),                  139
    ("限制性指标判定", "/constraint/", "shield"),                140
    ("微地质单元", "/unit/", "layers"),                          141
    ("指标体系与权重", "/ahp/", "sliders"),                      142
    ("指标数据采集", "/indicator/", "database"),                 143
    ("适宜性定量评价", "/evaluate/", "target"),                  144
    ("报告与结果输出", "/report/", "file"),                      145
]                                                                146
#  「用户与操作日志」已并入「个人设置」，不再是独立菜单，        147
#  因此系统管理组只留一个入口。                                  148
ADMIN_LINKS = [                                                  149
    ("个人设置", "/profile", "user"),                            150
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]
@app.context_processor
def inject_nav():
    path = request.path or "/"
    def hit(prefix):
        """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
        return path.startswith(prefix.rstrip("/"))
    badges = {}
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        badges = dash.nav_badges()
    def build(links):
        return [{"label": label, "href": href, "icon": icon,
                 "active": hit(href), "badge": badges.get(href, 0)}
                for label, href, icon in links]
    groups = [
        {"title": "评价流程", "links": build(FLOW_LINKS)},
        {"title": "系统管理", "links": build(ADMIN_LINKS)},
    ]
    # 面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
    if path == "/":
        page_title = "工作台"
    elif path.startswith("/profile"):
        page_title = "个人设置"
    elif path.startswith(("/users", "/logs")):
        page_title = "账号与日志"
    elif path.startswith("/search"):
        page_title = "搜索结果"
    else:
        page_title = "工作台"
        for label, href, _icon in FLOW_LINKS:
            if hit(href):
                page_title = label
                break
    # 层级引导用：项目是归档单位，图斑才是评价对象。
    # 下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
    from app.models import Project, MinePatch
    projects = Project.query.order_by(Project.id).all()
    return {
        "nav_groups": groups,
        "home_active": path == "/",
        "page_title": page_title,
        "today": date.today().strftime("%Y-%m-%d"),
        "search_q": request.args.get("q", ""),
        "role_labels": ROLE_LABELS,
        "project_count": len(projects),
        "patch_count": MinePatch.query.count(),
        "projects_without_patch": [p for p in projects
```

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                                                    if p.patches.count()
    == 0],
        }
    # 操作审计：统一在响应阶段记录成功的写操作（操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 模板改动立即生效，不必重启服务。
    # Flask 在非 debug 下默认把模板缓存到进程里，改完模板刷新浏览器还是旧页面，
    # 容易被误当成「改了没生效」。本系统只在本机运行，这点开销可以忽略。
    TEMPLATES_AUTO_RELOAD = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
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    32 MB                                                                  251
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",                      252
    "webp", "bmp", "tif", "tiff"}                                          253
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}                              254
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积                  255
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",                     256
    "webp"}                                                                257
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024          # 4 MB                    258
# ==== app/models.py ====                                                  259
# -*- coding: utf-8 -*-                                                    260
"""数据模型：系统全部数据表。                                                  261
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。                          262
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。                       263
"""                                                                        264
import json                                                                265
from datetime import datetime                                              266
from flask_login import UserMixin                                          267
from flask_sqlalchemy import SQLAlchemy                                    268
db = SQLAlchemy()                                                          269
class User(UserMixin, db.Model):                                           270
    """用户（账户与权限）。"""                                                 271
    __tablename__ = "users"                                                272
    id = db.Column(db.Integer, primary_key=True)                           273
    username = db.Column(db.String(64), unique=True,                       274
    nullable=False, index=True)                                            275
    password_hash = db.Column(db.String(256),                              276
    nullable=False)                                                        277
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator          278
    role = db.Column(db.String(20), nullable=False,                        279
    default="operator")                                                    280
    active = db.Column(db.Boolean, nullable=False,                         281
    default=True)                                                          282
    created_at = db.Column(db.DateTime,                                    283
    default=datetime.utcnow)                                               284
    # ---- 个人资料（用户可在「个人设置」里自行维护）----                        285
    display_name = db.Column(db.String(64), nullable=True)                 286
    # 昵称                                                                 287
    email = db.Column(db.String(128), nullable=True)                       288
    phone = db.Column(db.String(32), nullable=True)                        289
    gender = db.Column(db.String(8), nullable=True)                        290
    # 男 / 女 / 保密                                                       291
    avatar = db.Column(db.String(256), nullable=True)                      292
    # 头像，相对 static 的路径                                             293
    @property                                                              294
    def is_admin(self):                                                    295
        return self.role == "admin"                                        296
    @property                                                              297
    def shown_name(self):                                                  298
        """界面上显示的名字：优先用昵称，没填就用账号名。"""                      299
        return self.display_name or self.username                          300
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    @property                                                          301
    def initial(self):                                                 302
        """没有头像时用首字占位。"""                                    303
        return (self.shown_name or "?")[0].upper()                     304
    def to_dict(self):                                                 305
        return {"id": self.id, "username": self.username,              306
                "display_name": self.display_name,                     307
    "gender": self.gender,                                             308
                "role": self.role, "active": self.active}              309
class OperationLog(db.Model):                                          310
    """操作日志。"""                                                   311
    __tablename__ = "operation_logs"                                   312
    id = db.Column(db.Integer, primary_key=True)                       313
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"           314
    ), nullable=True)                                                  315
    action = db.Column(db.String(128), nullable=False)                 316
    # 操作类型                                                         317
    target_type = db.Column(db.String(64), nullable=True)              318
    # 对象类型                                                         319
    target_id = db.Column(db.Integer, nullable=True)                   320
    # 对象主键                                                         321
    detail = db.Column(db.Text, nullable=True)                         322
    # 操作详情                                                         323
    created_at = db.Column(db.DateTime,                                324
    default=datetime.utcnow, index=True)                               325
class Project(db.Model):                                               326
    """评价项目（项目与矿山档案）。"""                                 327
    __tablename__ = "projects"                                         328
    id = db.Column(db.Integer, primary_key=True)                       329
    name = db.Column(db.String(128), nullable=False)                   330
    code = db.Column(db.String(64), nullable=True)                     331
    # 项目编号                                                         332
    description = db.Column(db.Text, nullable=True)                    333
    status = db.Column(db.String(20), default="进行中")                334
    # 进行中 / 已完成 / 已归档                                         335
    created_by = db.Column(db.Integer,                                 336
    db.ForeignKey("users.id"), nullable=True)                          337
    created_at = db.Column(db.DateTime,                                338
    default=datetime.utcnow)                                           339
    patches = db.relationship("MinePatch",                             340
    backref="project", lazy="dynamic",                                 341
                                  cascade="all, delete-orphan")        342
class MinePatch(db.Model):                                             343
    """矿山图斑。"""                                                   344
    __tablename__ = "mine_patches"                                     345
    id = db.Column(db.Integer, primary_key=True)                       346
    project_id = db.Column(db.Integer,                                 347
    db.ForeignKey("projects.id"), nullable=False)                      348
    name = db.Column(db.String(128), nullable=False)                   349
    admin_region = db.Column(db.String(128), nullable=True)            350
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    # 所在行政区                                                    351
    center_x = db.Column(db.Float, nullable=True)                   352
    # 中心经度 / 横坐标                                             353
    center_y = db.Column(db.Float, nullable=True)                   354
    # 中心纬度 / 纵坐标                                             355
    area_ha = db.Column(db.Float, nullable=True)                    356
    # 图斑面积（公顷）                                              357
    mineral_type = db.Column(db.String(64), nullable=True)          358
    # 开采矿种                                                      359
    mining_method = db.Column(db.String(64), nullable=True)         360
    # 开采方式                                                      361
    stop_years = db.Column(db.Float, nullable=True)                 362
    # 停产年限                                                      363
    responsible_entity_missing = db.Column(db.Boolean,              364
    default=True)  # 责任主体是否缺失                               365
    # 遥感底图 / 无人机正射影像挂接                                 366
    image_path = db.Column(db.String(256), nullable=True)           367
    # 相对 static 的路径                                            368
    image_note = db.Column(db.String(256), nullable=True)           369
    # 影像说明（如来源、时相）                                      370
    image_uploaded_at = db.Column(db.DateTime,                      371
    nullable=True)                                                  372
    created_at = db.Column(db.DateTime,                             373
    default=datetime.utcnow)                                        374
    units = db.relationship("MicroUnit", backref="patch",           375
    lazy="dynamic",                                                 376
                                cascade="all, delete-orphan")       377
class ReferenceEcosystem(db.Model):                                 378
    """参照生态系统：标准化基准值，与图斑绑定。"""                  379
    __tablename__ = "reference_ecosystems"                          380
    id = db.Column(db.Integer, primary_key=True)                    381
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        382
    ches.id"), nullable=False)                                      383
    coverage_benchmark = db.Column(db.Float, default=75.0)          384
    # 植被覆盖度基准 %                                              385
    shannon_benchmark = db.Column(db.Float, default=0.7)            386
    # Shannon-Wiener 基准                                           387
    dominant_species = db.Column(db.String(256),                    388
    nullable=True)  # 建群种                                       389
    soil_benchmark = db.Column(db.String(256),                      390
    nullable=True)     # 土壤基准                                   391
class Standard(db.Model):                                           392
    """地方标准 / 政策文件（政策与标准知识库）。"""                 393
    __tablename__ = "standards"                                     394
    id = db.Column(db.Integer, primary_key=True)                    395
    province = db.Column(db.String(32), nullable=False)             396
    # 省份                                                          397
    name = db.Column(db.String(256), nullable=False)                398
    # 名称                                                          399
    doc_number = db.Column(db.String(128), nullable=True)           400
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    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
    __tablename__ = "standard_rules"
    id = db.Column(db.Integer, primary_key=True)
    standard_id = db.Column(db.Integer,
    db.ForeignKey("standards.id"), nullable=False)
    item = db.Column(db.String(128), nullable=False)
    # 判定项
    operator = db.Column(db.String(8), nullable=False)
    # 比较符：> >= < <= ==
    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")  # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录。"""
    __tablename__ = "constraint_checks"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    policy_pass = db.Column(db.Boolean, nullable=True)
    # 政策约束
    location_pass = db.Column(db.Boolean, nullable=True)
    # 区位约束
    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    farmland_pass = db.Column(db.Boolean, nullable=True)
    # 耕地属性
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime,
    default=datetime.utcnow)
```

```
class MicroUnit(db.Model):
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
```

```
    level = db.Column(db.Integer, nullable=False,                    501
    default=1)          #  1=一级指标  2=二级指标                    502
    is_default = db.Column(db.Boolean, default=False)                503
    #  是否默认（论文表 3）                                          504
    created_at = db.Column(db.DateTime,                              505
    default=datetime.utcnow)                                         506
class WeightItem(db.Model):                                          507
    """权重明细：方案内的各项权重。"""                              508
    __tablename__ = "weight_items"                                   509
    id = db.Column(db.Integer, primary_key=True)                     510
    scheme_id = db.Column(db.Integer,                                511
    db.ForeignKey("weight_schemes.id"), nullable=False)              512
    indicator_code = db.Column(db.String(32),                        513
    nullable=False)         #  指标编码                             514
    weight = db.Column(db.Float, nullable=False)                     515
    #  权重值                                                        516
class ExpertMatrix(db.Model):                                        517
    """专家判断矩阵：多专家打分的原始矩阵。"""                      518
    __tablename__ = "expert_matrices"                                519
    id = db.Column(db.Integer, primary_key=True)                     520
    scheme_id = db.Column(db.Integer,                                521
    db.ForeignKey("weight_schemes.id"), nullable=True)               522
    #  直接标注适用对象，便于无需先建权重方案即可录入专家矩阵      523
    unit_type = db.Column(db.String(32), nullable=True)              524
    #  微地质单元类型                                                525
    l1_name = db.Column(db.String(64), nullable=True)                526
    #  所属一级指标（二级矩阵用）                                    527
    expert_name = db.Column(db.String(64), nullable=True)            528
    #  专家姓名                                                      529
    level = db.Column(db.Integer, nullable=False,                    530
    default=1)          #  矩阵对应指标层级                          531
    matrix_json = db.Column(db.Text, nullable=False)                 532
    #  判断矩阵（JSON 二维数组）                                     533
    cr = db.Column(db.Float, nullable=True)                          534
    #  该专家矩阵的一致性比例                                        535
    created_at = db.Column(db.DateTime,                              536
    default=datetime.utcnow)                                         537
class IndicatorData(db.Model):                                       538
    """指标实测数据：按微地质单元组织。"""                          539
    __tablename__ = "indicator_data"                                 540
    id = db.Column(db.Integer, primary_key=True)                     541
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni         542
    ts.id"), nullable=False, unique=True)                            543
    soil_ratio = db.Column(db.Float, nullable=True)                  544
    #  土石比                                                        545
    thickness = db.Column(db.Float, nullable=True)                   546
    #  土层厚度 cm                                                   547
    ph = db.Column(db.Float, nullable=True)                          548
    organic_matter = db.Column(db.Float, nullable=True)              549
    #  有机质 g/kg                                                   550
```

```
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null
    heavy_metals = db.Column(db.Text, nullable=True)
    coverage = db.Column(db.Float, nullable=True)
    # 植被覆盖度 %
    shannon = db.Column(db.Float, nullable=True)
    # Shannon-Wiener 多样性指数
    slope = db.Column(db.Float, nullable=True)
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡 / 阴阳坡 / 阴坡
    def heavy_metals_dict(self):
        return json.loads(self.heavy_metals) if
    self.heavy_metals else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
```

```
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
#  邮箱格式：够用的轻量校验。
#  WTForms 自带的 Email() 需要额外安装 email_validator，
#  为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  #  动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。
    校验消息用中文写明：登录框带 novalidate，浏览器不会代填提示，
    字段为空时若不自己给消息，用户点了「登录」会以为没反应。
    """
    username = StringField("账户名称",
                           validators=[DataRequired(message=
    "请输入账户名称"),
```

```
                                                    Length(max=64,
    message="账户名称不能超过 64 个字符")])
    password = PasswordField("密码",
                                validators=[DataRequired(messag
    e="请输入密码")])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称
    "),
                                                    Length(min=2, max=64,
                                                           message="账号名称
    2–64 个字符")])
    password = PasswordField("密码",
                                validators=[DataRequired("请设置密码
    "),
                                                      Length(min=6,
    max=64,

    message="密码至少 6 位")])
    confirm = PasswordField("确认密码",
                            validators=[DataRequired(),
                                                      EqualTo("password",

    message="两次输入的密码不一致")])
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
```

```
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_reg = SubmitField("注册并登录")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
    old_password = PasswordField("当前密码",
    validators=[DataRequired()])
    new_password = PasswordField("新密码",
                                 validators=[DataRequired(),
    Length(min=6, max=64,

    message="新密码至少 6 位")])
    confirm = PasswordField("确认新密码",
                            validators=[DataRequired(),

    EqualTo("new_password",

    message="两次输入的新密码不一致")])
    submit_pwd = SubmitField("修改密码")
#  ==== app/utils/constants.py ====
#  -*- coding: utf-8 -*-
"""评价指标常量表。
全部数值取自论文，作为系统计算的唯一基准：
- 论文 ４．２．２ 节：指标标准化理想极值与取值依据
- 论文 ４．２．３ 节 / 表 ４：适宜性等级划分
- 论文表 ３：按微地质单元分别确定的一级、二级指标权重
- 论文 ４．２．１ 节：平均随机一致性指标 ＲＩ
"""
#  ------------------------------------------------------------------
    ------------------
#  一、微地质单元类型
#  ------------------------------------------------------------------
    ------------------
#  论文 ４．１ 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
#  三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
#  每个单元类型启用的指标类别（论文表 ３ 隐含：岩质边坡无土壤指标，平台无地形指标）
#  供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                    # 不设地形指标
}
#  ------------------------------------------------------------------
    ------------------
#  二、指标标准化理想极值（论文 ４．２．２ 节）
#  ------------------------------------------------------------------
    ------------------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （cm）
```

```
IDEAL_PH_RANGE = (5.0, 9.0)      # pH 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0             # 有机质理想值
    (g/kg)，全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0            # 植被覆盖度参照生态系统 (%)
IDEAL_SHANNON = 0.7              # Shannon-Wiener 多样性指数参照生态系统
# 重金属极值 (mg/kg)，《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 4.2.2 节："阳坡、阴阳坡、阴坡分别赋值 0.2、0.6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# pH 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
PH_TOL_HIGH = 10.0
# ---------------------------------------------------------------
    ----------------
# 三、适宜性等级划分（论文表 4）
# ---------------------------------------------------------------
    ----------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
```

```
        #  s c o r e  恰好等于  0 . 7  时落入 " 适宜 "                                  801
        r e t u r n  G R A D E _ L E V E L S [ - 1 ]                                  802
#  - - -
```

```
#  ----------------------------------------------------------------------  851
    ------------------                                                      852
#  论文摘要与  4．1  节：四项限制性指标；流程段另明确＂是否为耕地＂为约束条件  853
CONSTRAINT_ITEMS  =  （＂政策＂，  ＂区位约束＂，  ＂土壤污染＂，  ＂地质灾害＂，  ＂耕地属性＂）  854
#  ----------------------------------------------------------------------  855
    ------------------                                                      856
#  七、账号职务                                                             857
#  ----------------------------------------------------------------------  858
    ------------------                                                      859
#  只分两种：管理员可管理账号、维护基础数据并查看操作日志，操作员用业务功能。  860
ROLE_LABELS  =  {＂admin＂:  ＂管理员＂，  ＂operator＂:  ＂操作员＂}        861
ROLE_CODES  =  （＂admin＂，  ＂operator＂）                                 862
DEFAULT_ROLE  =  ＂operator＂                #  新注册账号的默认职务         863
ROLE_ORDER  =  （＂admin＂，  ＂operator＂）    #  列表排序用：管理员在前     864
#  性别（注册与个人资料中选填）                                             865
GENDERS  =  （＂男＂，  ＂女＂，  ＂保密＂）                                 866
#  ====  app/utils/schema.py  ====                                         867
#  -*-  coding:  utf-8  -*-                                                868
＂＂＂轻量结构迁移。                                                        869
SQLite  下  `db.create_all()`  只建缺失的表，不会给已有表补列。             870
本模块在启动时检查目标列是否存在，缺失则  ALTER  TABLE  补上，               871
使旧数据库无需重建即可用上新字段。                                          872
＂＂＂                                                                      873
from  sqlalchemy  import  text                                              874
#  （表名，  列名，  列定义）                                               875
REQUIRED_COLUMNS  =  [                                                      876
    #  遥感影像挂接                                                         877
    （＂mine_patches＂，  ＂image_path＂，  ＂VARCHAR(256)＂），             878
    （＂mine_patches＂，  ＂image_note＂，  ＂VARCHAR(256)＂），             879
    （＂mine_patches＂，  ＂image_uploaded_at＂，  ＂DATETIME＂），          880
    #  指标维护                                                             881
    （＂indicators＂，  ＂enabled_units＂，  ＂VARCHAR(128)＂），            882
    （＂indicators＂，  ＂description＂，  ＂TEXT＂），                      883
    #  多专家矩阵                                                           884
    （＂expert_matrices＂，  ＂unit_type＂，  ＂VARCHAR(32)＂），           885
    （＂expert_matrices＂，  ＂l1_name＂，  ＂VARCHAR(64)＂），             886
    （＂expert_matrices＂，  ＂cr＂，  ＂FLOAT＂），                         887
    #  用户个人资料（账户页）                                               888
    （＂users＂，  ＂display_name＂，  ＂VARCHAR(64)＂），                   889
    （＂users＂，  ＂email＂，  ＂VARCHAR(128)＂），                         890
    （＂users＂，  ＂phone＂，  ＂VARCHAR(32)＂），                          891
    （＂users＂，  ＂gender＂，  ＂VARCHAR(8)＂），                          892
    （＂users＂，  ＂avatar＂，  ＂VARCHAR(256)＂），                        893
]                                                                           894
#  职务体系由「管理员  /  评价员  /  查看者」三种收敛为「管理员  /  操作员」两种，  895
#  旧数据里的后两种统一归为操作员。                                         896
ROLE_MIGRATION  =  {＂evaluator＂:  ＂operator＂，  ＂viewer＂:             897
    ＂operator＂}                                                           898
def  _add_columns(db):                                                      899
    added  =  []                                                            900
```

```
    with db.engine.connect() as conn:
        for table, column, ddl in REQUIRED_COLUMNS:
            rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
            if not rows:
                continue                                #
    表还不存在，create_all 会建
            existing = {r[1] for r in rows}
            if column in existing:
                continue
            conn.execute(text(
                "ALTER TABLE %s ADD COLUMN %s %s" % (table,
    column, ddl)))
            conn.commit()
            added.append("%s.%s" % (table, column))
    return added
def _migrate_roles(db):
    """把旧角色名统一为新的两种，返回改动的行数。"""
    changed = 0
    with db.engine.connect() as conn:
        rows = conn.execute(text("PRAGMA
    table_info(users)")).fetchall()
        if not rows:
            return 0
        for old, new in ROLE_MIGRATION.items():
            r = conn.execute(text("UPDATE users SET role =
    :new "
                                  "WHERE role = :old"),
                                 {"new": new, "old": old})
            changed += r.rowcount or 0
        conn.commit()
    return changed
def ensure_schema(db):
    """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""
    try:
        changed = _add_columns(db)
        n = _migrate_roles(db)
        if n:
            changed.append("用户角色归一为管理员 / 操作员 %d 条" % n)
        return changed
    except Exception as e:                              #
    迁移失败不应阻断启动
        print("[schema] 结构检查跳过：%s" % e)
        return []
# ==== app/utils/uploads.py ====
# -*- coding: utf-8 -*-
"""上传文件处理（遥感影像、批量导入）。"""
import os
import re
import uuid
```

```
from datetime import datetime
from werkzeug.utils import secure_filename
def _safe_ext(filename):
    return os.path.splitext(filename)[1].lower().lstrip(".")
def ext_allowed(filename, allowed):
    return _safe_ext(filename) in allowed
def save_upload(file_storage, subdir, allowed_ext,
    upload_root):
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。
    """
    if file_storage is None or not file_storage.filename:
        return None, "未选择文件"
    ext = _safe_ext(file_storage.filename)
    if ext not in allowed_ext:
        return None, "不支持的文件类型：.%s" % ext
    target_dir = os.path.join(upload_root, subdir)
    os.makedirs(target_dir, exist_ok=True)
    # 仅保留安全字符，并加时间戳与随机串防重名
    base = secure_filename(os.path.splitext(file_storage.fil
    ename)[0]) or "file"
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or
    "file"
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M
    %S"),
                         uuid.uuid4().hex[:6], ext)
    abs_path = os.path.join(target_dir, name)
    file_storage.save(abs_path)
    rel = os.path.join("uploads", subdir,
    name).replace("\\", "/")
    return rel, None
def remove_upload(rel_path, upload_root):
    """删除已保存的上传文件（相对 static 的路径）。"""
    if not rel_path:
        return
    abs_path = os.path.join(upload_root,
    rel_path.replace("uploads/", "", 1))
    try:
        if os.path.isfile(abs_path):
            os.remove(abs_path)
    except OSError:
        pass
# ==== app/utils/charts.py ====
# -*- coding: utf-8 -*-
"""轻量图表。
用纯 SVG 输出，不依赖任何绘图库或前端脚本：
- 服务端渲染，页面与打印（PDF）中都能稳定显示
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）
- 单色方案，符合界面「一个视图一个强调色」的约定
"""
```

```
from html import escape
ACCENT = "#1f3a5f"
ACCENT_SOFT = "#7f93ad"
INK = "#23262b"
MUTED = "#6b7280"
LINE = "#dfe3e8"
#  灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",
    "#bcc9da", "#dbe3ec"]
def _fmt(v, digits=4):
    try:
        return ("%%.%df" % digits) % float(v)
    except (TypeError, ValueError):
        return str(v)
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,
    value_w=76):
    """水平条形图。items: [(标签, 数值, 显示文本)]"""
    if not items:
        return '<p class="muted">暂无数据</p>'
    n = len(items)
    height = n * (bar_h + gap) + 10
    max_v = max((abs(float(v)) for _, v, _ in items),
    default=1) or 1
    body_w = width - label_w - value_w
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"
    height="%d" '
             'role="img" aria-label="条形图" '
             'style="max-width:%dpx;font-family:inherit;">'
    % (width, height, height, width)]
    for i, (label, value, disp) in enumerate(items):
        y = i * (bar_h + gap) + 5
        frac = (abs(float(value)) / max_v) if max_v else 0
        w = max(2.0, body_w * frac)
        fill = GRAYS[i % len(GRAYS)]
        parts.append(
            '<text x="%d" y="%d" text-anchor="end"
    font-size="12" fill="%s">%s</text>'
            % (label_w - 10, y + bar_h * 0.72, INK,
    escape(str(label))))
        parts.append(
            '<rect x="%d" y="%d" width="%.1f" height="%d"
    rx="3" fill="%s"/>'
            % (label_w, y, w, bar_h, fill))
        parts.append(
            '<text x="%d" y="%d" font-size="12" fill="%s" '
            'style="font-variant-numeric:tabular-nums;">%s</
    text>'
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,
    escape(str(disp))))
    parts.append('</svg>')
```

```
    return "".join(parts)                                                1051
def donut(items, size=210, thickness=34):                                1052
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]                1053
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。                1054
    """                                                                  1055
    vals = [(l, float(v), d) for l, v, d in items if                     1056
    float(v) > 0]                                                        1057
    if not vals:                                                         1058
        return '<p class="muted">暂无数据</p>'                           1059
    total = sum(v for _, v, _ in vals) or 1                              1060
    r = (size - thickness) / 2                                           1061
    cx = cy = size / 2                                                   1062
    import math                                                          1063
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                        1064
    height="%d" role="img" '                                             1065
             'aria-label="权重分布环形图">' % (size, size, size,         1066
    size)]                                                               1067
    angle = -math.pi / 2                                                 1068
    for i, (label, v, disp) in enumerate(vals):                          1069
        sweep = 2 * math.pi * (v / total)                                1070
        a0, a1 = angle, angle + sweep                                    1071
        x0, y0 = cx + r * math.cos(a0), cy + r *                         1072
    math.sin(a0)                                                         1073
        x1, y1 = cx + r * math.cos(a1), cy + r *                         1074
    math.sin(a1)                                                         1075
        large = 1 if sweep > math.pi else 0                              1076
        parts.append(                                                    1077
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f                1078
    %.2f" '                                                              1079
            'fill="none" stroke="%s" stroke-width="%d"/>'                1080
           % (x0, y0, r, r, large, x1, y1, GRAYS[i %                     1081
    len(GRAYS)], thickness))                                             1082
        angle = a1                                                       1083
    parts.append('<text x="%d" y="%d" text-anchor="middle"               1084
    font-size="12" '                                                     1085
                 'fill="%s">合计</text>' % (cx, cy - 2,                  1086
    MUTED))                                                              1087
    parts.append('<text x="%d" y="%d" text-anchor="middle"               1088
    font-size="15" '                                                     1089
                 'fill="%s" style="font-variant-numeric:tabu             1090
    lar-nums;">%s</text>'                                                1091
                 % (cx, cy + 18, INK, _fmt(total, 4)))                   1092
    parts.append('</svg>')                                               1093
    legend = ['<table style="width:auto;margin-top:10px;"><t             1094
    head><tr>'                                                           1095
              '<th scope="col">项目</th><th                              1096
    scope="col">数值</th>'                                               1097
              '<th scope="col">占比</th></tr></thead><tbody>']           1098
    for i, (label, v, disp) in enumerate(vals):                          1099
        legend.append(                                                   1100
```

```
                '<tr><td style="text-align:left;">'                    1101
                '<span style="display:inline-block;width:10px;he       1102
    ight:10px;'                                                        1103
                'border-radius:2px;background:%s;margin-right:6p       1104
    x;"></span>%s</td>'                                                1105
                '<td class="mono">%s</td><td                           1106
    class="mono">%.1f%%</td></tr>'                                     1107
                % (GRAYS[i % len(GRAYS)], escape(str(label)),          1108
    escape(str(disp)),                                                 1109
                    100.0 * v / total))                                1110
    legend.append('</tbody></table>')                                  1111
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p           1112
    x;align-items:center;">'                                           1113
                + "".join(parts) + "".join(legend) + '</div>')         1114
# ==== app/utils/browser.py ====                                       1115
# -*- coding: utf-8 -*-                                                1116
"""HTML 转 PDF（报告导出）。                                           1117
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、        1118
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，     1119
避免中文路径在 file:// URL 下出问题。                                  1120
"""                                                                    1121
import os                                                              1122
import shutil                                                          1123
import subprocess                                                      1124
import tempfile                                                        1125
import uuid                                                            1126
CANDIDATES = [                                                         1127
    r"C:\Program Files\Google\Chrome\Application\chrome.exe"           1128
    ,                                                                  1129
    r"C:\Program Files (x86)\Google\Chrome\Application\chrom           1130
    e.exe",                                                            1131
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msed           1132
    ge.exe",                                                           1133
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe           1134
    ",                                                                 1135
]                                                                      1136
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器                     1137
ENV_KEY = "DSH_BROWSER_PATH"                                           1138
def browser_path():                                                    1139
    """定位可用的浏览器可执行文件，找不到返回 None。"""               1140
    override = os.environ.get(ENV_KEY)                                 1141
    if override and os.path.isfile(override):                          1142
        return override                                                1143
    for p in CANDIDATES:                                               1144
        if os.path.isfile(p):                                          1145
            return p                                                   1146
    for name in ("chrome", "msedge", "chromium"):                      1147
        found = shutil.which(name)                                     1148
        if found:                                                      1149
            return found                                               1150
```

```
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
        ]
        p = subprocess.run(cmd, capture_output=True,
    text=True,
                           timeout=timeout,
    encoding="utf-8", errors="replace")
        if not os.path.isfile(pdf_path) or
    os.path.getsize(pdf_path) < 800:
            return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
    "").strip()[:200]
        with open(pdf_path, "rb") as f:
            return f.read(), None
    except subprocess.TimeoutExpired:
        return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
        return None, "生成 PDF 失败：%s" % e
    finally:
        shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
```

```
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
```

```
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
                        "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按论文 4.2.2 节式(8) 实现：
    x' = (x - x_min) / (x_max - x_min)
```

```
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
```

```
    total = 0.0                                                          1351
    for metal, extreme in HEAVY_METAL_LIMIT.items():                     1352
        value = heavy_metals.get(metal)                                  1353
        total += std_cost(value, extreme) if value is not                1354
    None else 1.0                                                        1355
    return total / len(HEAVY_METAL_LIMIT)                                1356
def standardize_unit(unit_type, data, weights=None):                     1357
    """对单个微地质单元的全部指标做标准化。                                  1358
    参数                                                                 1359
    ----                                                                 1360
    unit_type : str，岩质边坡 / 碎石边坡 / 平台                            1361
    data : dict，实测数据，字段见接口契约：                                 1362
        soil_ratio, thickness, ph, organic_matter,                       1363
    heavy_metals(dict),                                                  1364
        coverage, shannon, slope, aspect                                 1365
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG   1366
    HTS 结构），                                                         1367
                未提供时用默认值。                                        1368
    返回                                                                 1369
    ----                                                                 1370
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }   1371
    标准化值可能为 None，表示该指标不适用或方法待定。                       1372
    """                                                                  1373
    from app.utils.constants import DEFAULT_WEIGHTS                      1374
    if weights is None:                                                  1375
        weights = DEFAULT_WEIGHTS[unit_type]                             1376
    l1_enabled = weights["l1"].keys()                                    1377
    l2w = weights["l2"]                                                  1378
    result = {}                                                          1379
    # ---- 土壤指标 ----                                                 1380
    if "土壤指标" in l1_enabled:                                          1381
        result["土壤指标"] = {                                            1382
            "土石比": (data.get("soil_ratio"),                           1383
                       std_benefit(data.get("soil_ratio"),               1384
    IDEAL_SOIL_RATIO),                                                   1385
                       l2w["土壤指标"]["土石比"]),                        1386
            "土层厚度": (data.get("thickness"),                          1387
                         std_benefit(data.get("thickness"),              1388
    IDEAL_THICKNESS),                                                    1389
                         l2w["土壤指标"]["土层厚度"]),                    1390
            "pH": (data.get("ph"),                                       1391
                   std_mid_ph(data.get("ph"),                            1392
    *IDEAL_PH_RANGE),                                                    1393
                   l2w["土壤指标"]["pH"]),                               1394
            "有机质": (data.get("organic_matter"),                       1395
                       std_benefit(data.get("organic_matter")            1396
    , IDEAL_ORGANIC),                                                    1397
                       l2w["土壤指标"]["有机质"]),                       1398
            "污染指标": (data.get("heavy_metals") or {},                 1399
                         pollution_score(data.get("heavy_meta            1400
```

```
        ls") or {}),                                                    1401
                                    l2w["土壤指标"]["污染指标"]),          1402
            }                                                           1403
        # ---- 植被指标 ----                                           1404
        if "植被指标" in l1_enabled:                                    1405
            result["植被指标"] = {                                      1406
                "植被覆盖度": (data.get("coverage"),                    1407
                                    std_benefit(data.get("coverage"),   1408
    IDEAL_COVERAGE),                                                    1409
                                    l2w["植被指标"]["植被覆盖度"]),      1410
                "生物多样性": (data.get("shannon"),                     1411
                                    std_benefit(data.get("shannon"),    1412
    IDEAL_SHANNON),                                                     1413
                                    l2w["植被指标"]["生物多样性"]),      1414
            }                                                           1415
        # ---- 地形指标 ----                                           1416
        if "地形指标" in l1_enabled:                                    1417
            result["地形指标"] = {                                      1418
                "坡度": (data.get("slope"),                             1419
                         std_slope(data.get("slope")),                  1420
                         l2w["地形指标"]["坡度"]),                      1421
                "坡向": (data.get("aspect"),                            1422
                         aspect_score(data.get("aspect")),              1423
                         l2w["地形指标"]["坡向"]),                      1424
            }                                                           1425
        return result                                                   1426
# ==== app/services/weight_service.py ====                            1427
# -*- coding: utf-8 -*-                                               1428
"""权重方案的存取与生效权重解析。                                       1429
系统有两套权重来源：                                                   1430
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），    1431
   是**权威取值来源**，案例复算与报告结论以此为准；                    1432
2. 通过层次分析法在本系统中实际算出的权重，保存在                      1433
   weight_schemes / weight_items 两张表中，可标记为「生效」。          1434
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：               1435
以论文表 3 为底，用生效方案中已保存的条目覆盖。                        1436
没有任何生效方案时，即完全使用论文表 3 的权重。                        1437
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，          1438
而不是算完即弃。                                                       1439
"""                                                                    1440
import copy                                                            1441
from app.models import db, WeightScheme, WeightItem                    1442
from app.utils.constants import DEFAULT_WEIGHTS                        1443
from app.services.indicator_service import TREE                        1444
# 方案名称：每个微地质单元类型固定一个可生效的方案                      1445
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"                          1446
def _code_maps():                                                      1447
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""      1448
    l1, l2 = {}, {}                                                    1449
    for row in TREE:                                                   1450
```

```
        code, name, level, parent = row[0], row[1], row[2],
    row[3]
        if level == 1:
            l1[code] = name
        elif level == 2:
            l2[code] = (l1.get(parent, parent), name)
    return l1, l2
def effective_weights(unit_type):
    """返回该单元类型当前生效的权重。
    返回 (weights, source_label, scheme)
        weights      嵌套权重结构，可直接交给标准化与评价引擎
        source_label 权重来源说明，用于界面与报告标注
        scheme       生效的权重方案对象，无自定义方案时为 None
    """
    base = DEFAULT_WEIGHTS.get(unit_type)
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,

    name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.
    id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
    merged["l2"][l1_name]:
                merged["l2"][l1_name][l2_name] = w
                applied += 1
    if applied == 0:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme
def save_weights(unit_type, code_weight_pairs, note=None):
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）
    分次录入而无需一次填完。
    """
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,
```

```
                                                                    1501
    name=ACTIVE_NAME).first()                                       1502
    if scheme is None:                                              1503
        scheme = WeightScheme(unit_type=unit_type,                  1504
    name=ACTIVE_NAME,                                               1505
                                    level=1, is_default=True)       1506
        db.session.add(scheme)                                      1507
        db.session.flush()                                          1508
    if note:                                                        1509
        scheme.level = scheme.level or 1                            1510
    if not scheme.is_default:                                       1511
        scheme.is_default = True                                    1512
    written = 0                                                     1513
    for code, w in code_weight_pairs:                               1514
        if w is None or w <= 0:                                     1515
            continue                                                1516
        item = WeightItem.query.filter_by(scheme_id=scheme.i        1517
    d,                                                              1518
                                                                    1519
    indicator_code=code).first()                                    1520
        if item is None:                                            1521
            item = WeightItem(scheme_id=scheme.id,                  1522
    indicator_code=code)                                            1523
            db.session.add(item)                                    1524
        item.weight = float(w)                                      1525
        written += 1                                                1526
    db.session.commit()                                             1527
    return scheme, written                                          1528
def reset_weights(unit_type):                                       1529
    """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""                1530
    schemes = WeightScheme.query.filter_by(unit_type=unit_ty        1531
    pe,                                                             1532
                                                                    1533
    name=ACTIVE_NAME).all()                                         1534
    n = 0                                                           1535
    for s in schemes:                                               1536
        WeightItem.query.filter_by(scheme_id=s.id).delete(          1537
            synchronize_session=False)                              1538
        db.session.delete(s)                                        1539
        n += 1                                                      1540
    db.session.commit()                                             1541
    return n                                                        1542
def scheme_items(unit_type):                                        1543
    """返回该单元类型已保存的自定义权重条目 {指标编码: 权重}。"""                1544
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ        1545
    e,                                                              1546
                                                                    1547
    name=ACTIVE_NAME).first()                                       1548
    if scheme is None:                                              1549
        return {}, None                                             1550
```

```
    items = {i.indicator_code: i.weight                                    1551
             for i in WeightItem.query.filter_by(scheme_id=s               1552
    cheme.id).all()}                                                       1553
    return items, scheme                                                   1554
def summary():                                                             1555
    """各单元类型的权重来源概览，供权重首页展示。"""                                  1556
    rows = []                                                              1557
    for unit_type in DEFAULT_WEIGHTS:                                      1558
        weights, label, scheme = effective_weights(unit_type               1559
    )                                                                      1560
        rows.append({                                                      1561
            "unit_type": unit_type,                                        1562
            "label": label,                                                1563
            "custom": scheme is not None,                                  1564
        })                                                                 1565
    return rows                                                            1566
# ==== app/services/evaluation_service.py ====                             1567
# -*- coding: utf-8 -*-                                                    1568
"""适宜性评价引擎。                                                             1569
按论文式（９） 实现：                                                            1570
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）                          1571
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比                              1572
- 等级判定按论文表 4。                                                          1573
"""                                                                        1574
from app.utils.constants import DEFAULT_WEIGHTS, grade_of                  1575
from app.services import standardization as std                            1576
def score_unit(unit_type, data, weights=None):                             1577
    """计算单个微地质单元的评价因子 F'。                                            1578
    返回                                                                   1579
    ----                                                                   1580
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:          1581
    [...] }                                                                1582
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，                          1583
    则按其余项权重归一化后计算，保证单元得分始终有定义。                               1584
    """                                                                    1585
    if weights is None:                                                    1586
        weights = DEFAULT_WEIGHTS[unit_type]                               1587
    std_result = std.standardize_unit(unit_type, data,                     1588
    weights)                                                               1589
    l1_weights = weights["l1"]                                             1590
    l1_scores = {}                                                         1591
    details = []                                                           1592
    for l1_name, items in std_result.items():                              1593
        wsum = 0.0                                                         1594
        acc = 0.0                                                          1595
        for l2_name, (raw, std_value, w2) in items.items():                1596
            if std_value is None:                                          1597
                # 不适用项：不计入                                             1598
                details.append({                                           1599
                    "l1": l1_name, "l2": l2_name, "raw":                   1600
```

```
        raw,                                                                    1601
                            "std_value": None, "weight": w2,                    1602
        "contribution": None,                                                   1603
                        })                                                      1604
                        continue                                                1605
                    contribution = std_value * w2                               1606
                    acc += contribution                                         1607
                    wsum += w2                                                  1608
                    details.append({                                            1609
                        "l1": l1_name, "l2": l2_name, "raw": raw,               1610
                        "std_value": std_value, "weight": w2,                   1611
                        "contribution": contribution,                           1612
                    })                                                          1613
            l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0                1614
        #  一级指标权重按实际参与计算的指标归一化。                                1615
        #  论文表 3 的权重本身已归一（和为 1），此处为恒等变换；                    1616
        #  若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。              1617
        lw_sum = sum(l1_weights[k] for k in l1_scores if k in                    1618
        l1_weights)                                                             1619
        if lw_sum > 0:                                                          1620
            unit_score = sum(l1_weights[k] * l1_scores[k]                       1621
                             for k in l1_scores if k in                         1622
        l1_weights) / lw_sum                                                    1623
        else:                                                                   1624
            unit_score = 0.0                                                    1625
        return {"l1_scores": l1_scores, "unit_score":                           1626
        unit_score, "details": details}                                         1627
def score_mine(units, weights_map=None):                                        1628
        """计算整个矿山的评价因子 F 与适宜性等级。                               1629
        参数                                                                    1630
        ----                                                                    1631
        units : list[dict]，每个元素：                                          1632
            {unit_type, ratio, data}                                            1633
            ratio 为该单元面积占比（0~1），data 为实测数据 dict。               1634
        weights_map : dict，{单元类型： 权重结构}。未提供的单元类型              1635
            使用论文表 3 的默认权重。权重可来自 AHP 计算并经                    1636
            weight_service 保存生效，从而真正参与评价。                         1637
        返回                                                                    1638
        ----                                                                    1639
        dict:{ final_score, grade, grade_info, unit_results }                   1640
        """                                                                     1641
        weights_map = weights_map or {}                                         1642
        total = 0.0                                                             1643
        unit_results = []                                                       1644
        for u in units:                                                         1645
            r = score_unit(u["unit_type"], u["data"],                           1646
                           weights_map.get(u["unit_type"]))                     1647
            weighted = u["ratio"] * r["unit_score"]                             1648
            total += weighted                                                   1649
            unit_results.append({                                               1650
```

```
                "unit_type": u["unit_type"],
                "ratio": u["ratio"],
                "unit_score": r["unit_score"],
                "weighted": weighted,
                "l1_scores": r["l1_scores"],
                "details": r["details"],
            })
    grade_info = grade_of(total)
    return {
        "final_score": total,
        "grade": grade_info["grade"],
        "grade_info": grade_info,
        "unit_results": unit_results,
    }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
from app.utils.timefmt import to_local, now_utc
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
        run.font.name = "Times New Roman"
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")
    return h
def _table(doc, headers, rows, widths=None):
```

```
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
        _set_run(hdr[i].paragraphs[0].add_run(), htext,
    10.5, bold=True)
    for row in rows:
        cells = t.add_row().cells
        for i, val in enumerate(row):
            _set_run(cells[i].paragraphs[0].add_run(),
    str(val), 10.5)
    return t
def build_report(patch, mine_result):
    """生成评价报告 Word 文档，返回 BytesIO。
    参数
    ----
    patch : MinePatch，图斑对象
    mine_result : evaluation_service.score_mine 的返回 dict
    """
    doc = Document()
    # 页边距
    for section in doc.sections:
        section.top_margin = Cm(2.5)
        section.bottom_margin = Cm(2.5)
        section.left_margin = Cm(2.5)
        section.right_margin = Cm(2.5)
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0",
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)
    _para(doc, "生成时间：%s" % to_local(now_utc(), "%Y-%m-%d
    %H:%M:%S"),
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,
    space_after=12)
    # 图斑信息
    _heading(doc, "一、图斑基本信息", 1)
    info = [
        ("图斑名称", patch.name),
        ("所在行政区", patch.admin_region or "—"),
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is
    not None else "—"),
        ("开采矿种", patch.mineral_type or "—"),
        ("开采方式", patch.mining_method or "—"),
    ]
    _table(doc, ["项目", "内容"], info)
    # 评价结论
    _heading(doc, "二、评价结论", 1)
    gi = mine_result["grade_info"]
    concl = [
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),
        ("适宜性等级", mine_result["grade"]),
```

```
            ("判定依据", gi["meaning"]),
            ("推荐措施", gi["measure"]),
        ]
        _table(doc, ["项目", "结果"], concl)
        # 各单元得分
        _heading(doc, "三、各微地质单元得分", 1)
        rows = []
        for u in mine_result["unit_results"]:
            rows.append((u["unit_type"], "%.2f%%" % (u["ratio"] * 100),
                         "%.4f" % u["unit_score"], "%.4f" % u["weighted"]))
        _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"], rows)
        # 完整得分明细
        _heading(doc, "四、得分明细", 1)
        for u in mine_result["unit_results"]:
            _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"], u["ratio"] * 100),
                  size=11, bold=True, space_after=4)
            detail_rows = []
            for d in u["details"]:
                raw = "—" if d["raw"] is None else d["raw"]
                std = "—" if d["std_value"] is None else "%.4f" % d["std_value"]
                w = "%.4f" % d["weight"]
                c = "—" if d["contribution"] is None else "%.4f" % d["contribution"]
                detail_rows.append((d["l1"], d["l2"], raw, std, w, c))
            _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重", "贡献"],
                   detail_rows)
            # 一级指标得分
            l1_row = []
            for k, v in u["l1_scores"].items():
                l1_row.append("%s=%.4f" % (k, v))
            _para(doc, "一级指标得分：" + "，".join(l1_row), size=10.5, space_after=10)
        buf = io.BytesIO()
        doc.save(buf)
        buf.seek(0)
        return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
```

```
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
    # ---- 二级指标：土壤 ----
    ("soil_ratio", "土石比", 2, "soil", "—",
    str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),
    ("thickness", "土层厚度", 2, "soil", "cm",
    str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 20 cm", False),
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
    "mid", "碎石边坡,平台",
     "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %
    IDEAL_PH_RANGE, False),
    ("organic_matter", "有机质", 2, "soil", "g/kg",
    str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
     "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),
    ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
    "cost", "碎石边坡,平台",
     "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
    # ---- 二级指标：植被 ----
    ("coverage", "植被覆盖度", 2, "veg", "%",
    str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
     "论文 4.2.2 节：参照生态系统取值 75%", False),
    ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
    "benefit", ALL_UNITS,
     "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
    # ---- 二级指标：地形 ----
    ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
    SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
     "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°
    为陡坡重要分界", False),
    ("aspect", "坡向", 2, "terrain", "—",
```

```
        "阳坡０．２/阴阳坡０．６/阴坡１．０",  "ｄｉｓｃｒｅｔｅ",  "岩质边坡,碎石边坡",
          "论文　４．２．２　节：阴坡及半阴坡水分条件更优，故赋值高于阳坡",  Ｆａｌｓｅ),
]
#  污染指标下的三级指标（８　种重金属）
ｆｏｒ　_ｍ,　_ｌｉｍｉｔ　ｉｎ　ＨＥＡＶＹ_ＭＥＴＡＬ_ＬＩＭＩＴ.ｉｔｅｍｓ():
      ＴＲＥＥ.ａｐｐｅｎｄ((
            "ｈｍ_"　+　_ｍ,　_ｍ,　３,　"ｐｏｌｌｕｔｉｏｎ",　"ｍｇ/ｋｇ",
      ｓｔｒ(_ｌｉｍｉｔ),　"ｃｏｓｔ",
            "碎石边坡,平台",
            "《绿化种植土壤》ＣＪ/Ｔ　３４０—２０１６　土壤重金属含量　ＩＶ　级最高值　%ｇ　ｍｇ/ｋｇ"　%
      _ｌｉｍｉｔ,
            Ｆａｌｓｅ,
      ))
ｄｅｆ　ｓｅｅｄ_ｉｎｄｉｃａｔｏｒｓ():
      """写入/更新指标定义（幂等）。返回写入条数。"""
      ｃｏｄｅ_ｍａｐ　=　{
            "ｂｅｎｅｆｉｔ":　"越大越优",　"ｃｏｓｔ":　"越小越优",
            "ｍｉｄ":　"适中型",　"ｄｉｓｃｒｅｔｅ":　"离散赋值",
      }
      ｎ　=　０
      ｆｏｒ　(ｃｏｄｅ,　ｎａｍｅ,　ｌｅｖｅｌ,　ｐａｒｅｎｔ,　ｕｎｉｔ,　ｉｄｅａｌ,　ｉｔｙｐｅ,
      ｕｎｉｔｓ,　ｄｅｓｃ,　ｉｓ_ｃｏｎ)　ｉｎ　ＴＲＥＥ:
            ｉｎｄ　=　Ｉｎｄｉｃａｔｏｒ.ｑｕｅｒｙ.ｆｉｌｔｅｒ_ｂｙ(ｃｏｄｅ=ｃｏｄｅ).ｆｉｒｓｔ()
            ｉｆ　ｉｎｄ　ｉｓ　Ｎｏｎｅ:
                  ｉｎｄ　=　Ｉｎｄｉｃａｔｏｒ(ｃｏｄｅ=ｃｏｄｅ)
                  ｄｂ.ｓｅｓｓｉｏｎ.ａｄｄ(ｉｎｄ)
            ｉｎｄ.ｎａｍｅ　=　ｎａｍｅ
            ｉｎｄ.ｌｅｖｅｌ　=　ｌｅｖｅｌ
            ｉｎｄ.ｐａｒｅｎｔ_ｃｏｄｅ　=　ｐａｒｅｎｔ
            ｉｎｄ.ｕｎｉｔ　=　ｕｎｉｔ
            ｉｎｄ.ｉｄｅａｌ_ｖａｌｕｅ　=　ｉｄｅａｌ
            ｉｎｄ.ｉｄｅａｌ_ｔｙｐｅ　=　ｉｔｙｐｅ
            ｉｎｄ.ｅｎａｂｌｅｄ_ｕｎｉｔｓ　=　ｕｎｉｔｓ
            ｉｎｄ.ｄｅｓｃｒｉｐｔｉｏｎ　=　ｄｅｓｃ
            ｉｎｄ.ｉｓ_ｃｏｎｓｔｒａｉｎｔ　=　ｉｓ_ｃｏｎ
            ｉｎｄ.ｉｓ_ｅｖａｌｕａｔｉｖｅ　=　(ｌｅｖｅｌ　ｉｎ　(２,　３))
            ｎ　+=　１
      ｄｂ.ｓｅｓｓｉｏｎ.ｃｏｍｍｉｔ()
      ｒｅｔｕｒｎ　ｎ
ｄｅｆ　ｉｄｅａｌ_ｔｙｐｅ_ｌａｂｅｌ(ｃｏｄｅ):
      ｒｅｔｕｒｎ　{"ｂｅｎｅｆｉｔ":　"越大越优",　"ｃｏｓｔ":　"越小越优",
                  "ｍｉｄ":　"适中型",　"ｄｉｓｃｒｅｔｅ":　"离散赋值"}.ｇｅｔ(ｃｏｄｅ,
      ｃｏｄｅ　ｏｒ　"—")
#　====　ａｐｐ/ｓｅｒｖｉｃｅｓ/ｉｍｐｏｒｔ_ｓｅｒｖｉｃｅ.ｐｙ　====
#　-*-　ｃｏｄｉｎｇ:　ｕｔｆ-８　-*-
"""指标数据批量导入。
支持　ＣＳＶ　与　Ｅｘｃｅｌ（.ｘｌｓｘ/.ｘｌｓ）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
      单元类型，　土石比，　土层厚度，　ｐＨ，　有机质，
```

```
    Hg, As, Ni, Cu, Cd, Pb, Cr, Zn,
    植被覆盖度, Shannon, 坡度, 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
# 表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
```

```
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
None)
                        for i in range(len(header)) if
header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict, {单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict], 每项 {unit_type, fields{...},
    heavy_metals{}}
      errors : list[str], 逐行问题说明
    """
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    # 表头校验
    unknown = [h for h in (rows[0].keys() if rows else [])
    if h not in FIELD_ALIASES]
```

```
    if unknown:                                                    2001
        errors.append("存在无法识别的列，将被忽略：%s" %           2002
"、".join(map(str, unknown)))                                      2003
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头 2004
        rec = {}                                                   2005
        for k, v in raw.items():                                   2006
            key = FIELD_ALIASES.get(str(k).strip())                2007
            if key:                                                2008
                rec[key] = v                                       2009
        unit_type = str(rec.get("unit_type") or "").strip()        2010
        if not unit_type:                                          2011
            errors.append("第 %d 行：缺少「单元类型」" % idx)      2012
            continue                                               2013
        if unit_type not in unit_type_map:                         2014
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"  2015
                          % (idx, unit_type,                       2016
"、".join(unit_type_map) or "无"))                                 2017
            continue                                               2018
        if unit_type not in UNIT_TYPES:                            2019
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"  2020
                          % (idx, unit_type))                      2021
            continue                                               2022
        fields, heavy = {}, {}                                     2023
        ok = True                                                  2024
        for key, val in rec.items():                               2025
            if key in ("unit_type",):                              2026
                continue                                           2027
            if key.startswith("hm_"):                              2028
                n, err = _num(val)                                 2029
                if err:                                            2030
                    errors.append("第 %d 行：%s %s" % (idx,        2031
key[3:], err))                                                     2032
                    ok = False                                     2033
                elif n is not None:                                2034
                    heavy[key[3:]] = n                             2035
                continue                                           2036
            if key == "aspect":                                    2037
                s = str(val).strip() if val is not None            2038
else ""                                                            2039
                if s == "":                                        2040
                    fields["aspect"] = None                        2041
                elif s in ASPECT_TYPES:                            2042
                    fields["aspect"] = s                           2043
                else:                                              2044
                    errors.append("第 %d 行：坡向「%s」无效（应为 %s）"  2045
                                  % (idx, s,                       2046
"、".join(ASPECT_TYPES)))                                          2047
                    ok = False                                     2048
                continue                                           2049
            n, err = _num(val)                                     2050
```

```
                if err:
                    errors.append("第 %d 行:%s %s" % (idx, key,
    err))
                    ok = False
                else:
                    fields[key] = n
            if ok:
                parsed.append({"unit_type": unit_type,
    "fields": fields,
                               "heavy_metals": heavy})
    return parsed, errors
# ==== app/services/workflow.py ====
# -*- coding: utf-8 -*-
"""评价流程前置条件（定量评价与报告输出的流程门禁）。
论文 4.1 节确定的评价流程存在严格的先后关系：
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                        EvaluationResult)
#  指标数据非空字段（用于判断某单元是否已录入数据）
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
               "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
        return True
    return bool(ind.heavy_metals_dict())
def patch_progress(patch_id):
    """检查某图斑的评价流程完成情况。
    返回 dict：
        units        {done, count, ratio_sum, detail}
        constraint   {done, passed, detail}
        data         {done, ready, missing, detail}
        blocking     未满足的前置步骤列表（按流程顺序）
        ready        是否可进入定量评价
    """
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
```

```
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    # ---- 步骤一：微地质单元 ----
    units_step = {
        "done": bool(units),
        "count": len(units),
        "ratio_sum": ratio_sum,
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"
                   % (len(units), ratio_sum * 100))
                  if units else "尚未划分微地质单元",
    }
    # ---- 步骤二：限制性条件判定 ----
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id) \
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
        constraint_step = {"done": True, "passed": False,
                           "detail": "限制性条件判定未通过（%s）" %
    (cc.reason or "")}
    # ---- 步骤三：指标数据 ----
    missing = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        if not _has_data(ind):
            missing.append(u.unit_type)
    if not units:
        data_step = {"done": False, "ready": False,
    "missing": [],
                     "detail": "尚无可录入数据的微地质单元"}
    elif missing:
        data_step = {"done": False, "ready": False,
    "missing": missing,
                     "detail": "以下微地质单元尚未录入指标数据：%s"
                               % "、".join(missing)}
    else:
        data_step = {"done": True, "ready": True,
    "missing": [],
                     "detail": "全部 %d 个微地质单元的指标数据已录入" %
    len(units)}
    ratio_ok = ratio_sum > 0.999 if units else False
    if units and not ratio_ok:
        units_step["detail"] += "（占比合计不等于 １００%，请检查面积录入）"
    # ---- 汇总未满足的前置条件（按论文 ４．１ 节的流程顺序：限制性判定优先） ----
    # reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
```

```
    # 缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
    blocking = []
    if not constraint_step["done"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未执行",
            "reason": "五项限制性条件尚未逐项判定。",
            "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去执行判定",
        })
    elif not constraint_step["passed"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未通过",
            "reason": cc.reason or "",
            "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去复核判定结果",
        })
    if not units_step["done"]:
        blocking.append({
            "step": "units", "label": "未划分微地质单元",
            "reason": "该图斑下没有任何微地质单元。",
            "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去划分",
        })
    elif not ratio_ok:
        blocking.append({
            "step": "units", "label": "面积占比合计不为 100%",
            "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),
            "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去核对面积",
        })
    if units_step["done"] and not data_step["ready"]:
        blocking.append({
            "step": "data", "label": "指标数据未录入完整",
            "reason": data_step["detail"],
            "fix_url": "/indicator/", "fix_label": "去录入",
        })
    return {
        "units": units_step,
        "constraint": constraint_step,
        "data": data_step,
        "blocking": blocking,
        "ready": not blocking,
    }
def latest_result(patch_id):
    """取该图斑最近一次的评价记录。"""
    return EvaluationResult.query.filter_by(patch_id=patch_i
    d) \
        .order_by(EvaluationResult.id.desc()).first()
def report_progress(patch_id):
    """报告输出的前置条件：评价流程齐备，且已有评价记录。
```

```
    报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
    导出时读取已保存的评价结果，而不是在导出环节重新计算。
    否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
    同一份评价出现两个数字，结论无法追溯。
    返回 dict：{ready, blocking, result, flow}
        result 为最近一次评价记录（EvaluationResult），无记录时为空。
    """
    flow = patch_progress(patch_id)
    blocking = list(flow["blocking"])
    record = latest_result(patch_id)
    if not blocking and record is None:
        blocking.append({
            "step": "evaluate", "label": "尚未执行适宜性评价",
            "reason": "该图斑还没有评价记录。",
            "fix_url": "/evaluate/%d" % patch_id,
            "fix_label": "去执行评价",
        })
    return {
        "ready": not blocking,
        "blocking": blocking,
        "result": record,
        "flow": flow,
    }
# ==== app/services/dashboard.py ====
# -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）
#    key          步骤标识
#    no           步骤编号（01~08）
#    title        名称
#    icon         图标名（见 templates/_macros.html 的 ICONS）
#    desc         一句话说明
#    href         入口地址
STEPS = [
    dict(key="project", no="01", title="项目与矿山档案",
    icon="folder",
         href="/project/",
         desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),
    dict(key="standard", no="02", title="政策与标准知识库",
    icon="book",
         href="/standard/",
```

```
            desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),
        dict(key="constraint", no="03", title="限制性指标判定",
        icon="shield",
            href="/constraint/",
            desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),
        dict(key="unit", no="04", title="微地质单元", icon="layers",
            href="/unit/",
            desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),
        dict(key="ahp", no="05", title="指标体系与权重",
        icon="sliders",
            href="/ahp/",
            desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),
        dict(key="indicator", no="06", title="指标数据采集",
        icon="database",
            href="/indicator/",
            desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),
        dict(key="evaluate", no="07", title="适宜性定量评价",
        icon="target",
            href="/evaluate/",
            desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),
        dict(key="report", no="08", title="报告与结果输出",
        icon="file",
            href="/report/",
            desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),
]
def _has_data(ind):
    return wf._has_data(ind)
# 系统的九个功能模块（登录页介绍用，含不在评价流程内的账号与权限管理）
MODULES = [
    ("01", "账号与权限管理", "自助注册与登录注销、管理员与操作员两种职务、写操作自动留痕"),
    ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),
    ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),
    ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),
    ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),
    ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),
    ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),
    ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),
    ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),
]
def snapshot():
    """一次算出工作台需要的全部数据（同一请求内只算一次）。
    返回 dict：
        stats       {projects, patches, unit_types, standards}
        steps       八个步骤，附 available / pending / note
        badges      {key: 待办数量}，仅含有待办的项
        progress    {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
```

```
            cached = _compute()                                          2301
            g._dash_snapshot = cached                                    2302
        return cached                                                    2303
    def _compute():                                                      2304
        projects = Project.query.count()                                 2305
        patches = MinePatch.query.all()                                  2306
        patch_ids = [p.id for p in patches]                              2307
        units = MicroUnit.query.filter(                                  2308
            MicroUnit.patch_id.in_(patch_ids)).all() if                  2309
        patch_ids else []                                                2310
        unit_by_patch = {}                                               2311
        for u in units:                                                  2312
            unit_by_patch.setdefault(u.patch_id, []).append(u)           2313
        unit_ids = [u.id for u in units]                                 2314
        data_unit_ids = set()                                            2315
        if unit_ids:                                                     2316
            for ind in IndicatorData.query.filter(                       2317
                    IndicatorData.unit_id.in_(unit_ids)).all():          2318
                if _has_data(ind):                                       2319
                    data_unit_ids.add(ind.unit_id)                       2320
        passed_constraint = set()                                        2321
        if patch_ids:                                                    2322
            for c in ConstraintCheck.query.filter(                       2323
                    ConstraintCheck.patch_id.in_(patch_ids)).all2324
        ():                                                              2325
                if c.passed:                                             2326
                    passed_constraint.add(c.patch_id)                    2327
        evaluated = set()                                                2328
        if patch_ids:                                                    2329
            for r in EvaluationResult.query.filter(                      2330
                    EvaluationResult.patch_id.in_(patch_ids)).al2331
        l():                                                             2332
                evaluated.add(r.patch_id)                                2333
        # ---- 各步骤的待办数量 ----                                      2334
        pending = {                                                      2335
            "constraint": sum(1 for p in patches if p.id not in 2336
        passed_constraint),                                              2337
            "unit": sum(1 for p in patches if not                        2338
        unit_by_patch.get(p.id)),                                        2339
            "indicator": 0,                                              2340
            "evaluate": 0,                                               2341
        }                                                                2342
        ready_patches = 0                                                2343
        for p in patches:                                                2344
            us = unit_by_patch.get(p.id, [])                             2345
            if us and any(u.id not in data_unit_ids for u in             2346
        us):                                                             2347
                pending["indicator"] += 1                                2348
            pr = wf.patch_progress(p.id)                                 2349
            if pr["ready"]:                                              2350
```

```
                ready_patches += 1                                  2351
                if p.id not in evaluated:                           2352
                    pending["evaluate"] += 1                        2353
        # ---- 步骤可用性：前置数据不具备时置灰 ----                  2354
        avail = {                                                   2355
            "project": True,                                        2356
            "standard": True,                                       2357
            "constraint": bool(patches),                            2358
            "unit": bool(patches),                                  2359
            "ahp": True,                                            2360
            "indicator": bool(units),                               2361
            "evaluate": ready_patches > 0,                          2362
            "report": bool(evaluated),                              2363
        }                                                           2364
        notes = {                                                   2365
            "constraint": "需先建立矿山图斑",                       2366
            "unit": "需先建立矿山图斑",                             2367
            "indicator": "需先划分微地质单元",                      2368
            "evaluate": "需前置步骤齐备的图斑",                     2369
            "report": "需先执行适宜性评价",                         2370
        }                                                           2371
        steps = []                                                  2372
        for s in STEPS:                                             2373
            steps.append(dict(                                      2374
                s,                                                  2375
                available=avail[s["key"]],                          2376
                pending=pending.get(s["key"], 0),                   2377
                note=None if avail[s["key"]] else                   2378
        notes.get(s["key"], ""),                                    2379
            ))                                                      2380
        unit_types = len({u.unit_type for u in units})              2381
        stats = {                                                   2382
            "projects": projects,                                   2383
            "patches": len(patches),                                2384
            "unit_types": unit_types,                               2385
            "standards": Standard.query.count(),                    2386
        }                                                           2387
        return {                                                    2388
            "stats": stats,                                         2389
            "steps": steps,                                         2390
            "badges": {k: v for k, v in pending.items() if v},      2391
            "progress": {                                           2392
                "patches": len(patches),                            2393
                "units": len(units),                                2394
                "data_units": len(data_unit_ids),                   2395
                "ready": ready_patches,                             2396
                "evaluated": len(evaluated),                        2397
            },                                                      2398
        }                                                           2399
def nav_badges():                                                   2400
```

```
    """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""
    snap = snapshot()
    return {
        "/constraint/": snap["badges"].get("constraint", 0),
        "/unit/": snap["badges"].get("unit", 0),
        "/indicator/": snap["badges"].get("indicator", 0),
        "/evaluate/": snap["badges"].get("evaluate", 0),
    }
# ==== app/services/audit.py ====
# -*- coding: utf-8 -*-
"""操作审计。
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
"""
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
# 端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.register": "注册账号",
    "auth.logout": "退出登录",
    "auth.add_user": "新增账号",
    "auth.edit_user": "编辑账号",
    "auth.delete_user": "删除账号",
    "auth.reset_password": "重置用户密码",
    "auth.profile": "修改个人资料",
    "auth.upload_avatar": "更新头像",
    "auth.delete_avatar": "移除头像",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
```

```
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
#  无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    #  个人设置页的两个表单共用一个端点，按提交的动作区分
    if endpoint == "auth.profile":
        return "修改密码" if request.form.get("action") ==
    "password" \
            else "修改个人资料"
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += "；结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
            target_type=(endpoint or "").split(".")[0] or
    None,
            target_id=extract_target_id(),
            detail=detail,
        ))
        db.session.commit()
    except Exception:
        #  审计失败不应影响业务响应
        db.session.rollback()
#  ==== app/views/auth.py ====
#  -*- coding: utf-8 -*-
"""账号与权限 —— 登录、注册、个人资料、用户维护、操作日志。
```

```
页面组织：
   /login          登录页（含「点我注册」入口）
   /register       自助注册（默认「操作员」）
   /profile        个人设置，分类切换：info / security / users / logs
                   后两个分类仅管理员可见，由本模块统一管控。
"""
from flask import (Blueprint, render_template, redirect,
    url_for, request,
                   flash, abort, current_app)
from flask_login import login_user, logout_user,
    login_required, current_user
from werkzeug.security import check_password_hash,
    generate_password_hash
from app.forms import (LoginForm, EmptyForm, ProfileForm,
    PasswordForm,
                       RegisterForm, email_ok)
from app.models import db, User, OperationLog
from app.utils import uploads as up
from app.utils.constants import ROLE_LABELS, DEFAULT_ROLE,
    GENDERS
bp = Blueprint("auth", __name__)
#  「个人设置」的四个分类；后两个是管理员专属
PROFILE_TABS = ("info", "security", "users", "logs")
ADMIN_TABS = ("users", "logs")
def require_admin():
    """非管理员直接 403，避免误操作基础数据。"""
    if current_user.role != "admin":
        abort(403)
def _intro_data():
    """登录页下方的系统介绍所用的静态内容。"""
    from app.services import dashboard as dash
    return {"modules": dash.MODULES}
@bp.route("/")
def index():
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        return render_template("auth/home.html",
    dash=dash.snapshot())
    return redirect(url_for("auth.login"))
@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.d
    ata).first()
        if user and check_password_hash(user.password_hash,
    form.password.data):
            if not user.active:
```

```
                    flash("该账号已停用，请联系管理员", "toast-error")
                else:
                    login_user(user, remember=form.remember.data)
                    # 用 toast 类别：渲染为页面中上方的浮层，淡入停留后淡出。
                    # 登录成功只是告知性反馈，不需要用户停下来阅读。
                    flash("登录成功", "toast")
                    return redirect(url_for("auth.index"))
            else:
                flash("账户名称或密码错误", "toast-error")
        # 字段为空之类的校验失败，错误留在表单里逐项显示，不走浮层：
        # 那类消息要对着输入框看，一闪而过反而让人以为「点了没反应」。
        return render_template("auth/login.html", form=form,
        **_intro_data())
@bp.route("/register", methods=["GET", "POST"])
def register():
    """自助注册。新账号一律为「操作员」，权限只能由管理员事后调整。"""
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = RegisterForm()
    if form.validate_on_submit():
        username = form.username.data.strip()
        if User.query.filter_by(username=username).first():
            form.username.errors.append("该账户名称已被使用")
        else:
            user = User(username=username,
                        role=DEFAULT_ROLE,
                        password_hash=generate_password_hash
        (form.password.data),
                        display_name=(form.display_name.data
        or "").strip() or None,
                        gender=(form.gender.data or
        "").strip() or None,
                        email=(form.email.data or
        "").strip() or None,
                        phone=(form.phone.data or
        "").strip() or None)
            rel, err = _save_avatar(request.files.get("avata
        r"))
            if err:
                flash(err, "error")
            else:
                user.avatar = rel
                db.session.add(user)
                db.session.commit()
                login_user(user)
                flash("注册成功，欢迎使用", "toast")
                return redirect(url_for("auth.index"))
    return render_template("auth/register.html", form=form)
@bp.route("/logout")
```

```
@login_required                                                    2601
def logout():                                                      2602
    logout_user()                                                  2603
    flash("已退出登录", "ok")                                      2604
    return redirect(url_for("auth.login"))                         2605
# ---------------------------- 个人设置 ----------------------------2606
    ----                                                           2607
# 任何登录用户在这里维护自己；管理员额外能管理账号与查看日志。    2608
# 「用户与操作日志」不再是独立菜单，而是本页的管理员分类。        2609
@bp.route("/profile", methods=["GET", "POST"])                     2610
@login_required                                                    2611
def profile():                                                     2612
    is_admin = current_user.role == "admin"                        2613
    tab = request.values.get("tab") or "info"                      2614
    if tab not in PROFILE_TABS or (tab in ADMIN_TABS and           2615
    not is_admin):                                                 2616
        tab = "info"                                               2617
    # 表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名 2618
    info_form = ProfileForm(prefix="info", obj=current_user)       2619
    pwd_form = PasswordForm(prefix="pwd")                          2620
    if request.method == "POST":                                   2621
        action = request.form.get("action")                        2622
        # 校验失败要重新渲染本页，此时必须回到提交表单所在的那个分类，  2623
        # 否则错误提示会出现在另一个分类里，用户根本看不到。        2624
        if action == "password":                                   2625
            tab = "security"                                       2626
        elif action == "info":                                     2627
            tab = "info"                                           2628
        if action == "info" and info_form.validate_on_submit       2629
    ():                                                            2630
            current_user.display_name =                            2631
    (info_form.display_name.data or "").strip() or None            2632
            current_user.gender = (info_form.gender.data or        2633
    "").strip() or None                                            2634
            current_user.email = (info_form.email.data or          2635
    "").strip() or None                                            2636
            current_user.phone = (info_form.phone.data or          2637
    "").strip() or None                                            2638
            db.session.commit()                                    2639
            flash("个人资料已保存", "ok")                          2640
            return redirect(url_for("auth.profile",                2641
    tab="info"))                                                   2642
        if action == "password" and pwd_form.validate_on_sub       2643
    mit():                                                         2644
            if not check_password_hash(current_user.password       2645
    _hash,                                                         2646
                                                                   2647
    pwd_form.old_password.data):                                   2648
                pwd_form.old_password.errors.append("当前密码不正确 2649
    ")                                                             2650
```

```
                else:                                                  2651
                    current_user.password_hash =                       2652
        generate_password_hash(                                        2653
                        pwd_form.new_password.data)                    2654
                    db.session.commit()                                2655
                    flash("密码已修改", "ok")                          2656
                    return redirect(url_for("auth.profile",            2657
        tab="security"))                                               2658
        ctx = dict(info_form=info_form, pwd_form=pwd_form,             2659
        avatar_form=EmptyForm(),                                       2660
                   tab=tab, role_labels=ROLE_LABELS,                   2661
        genders=GENDERS)                                               2662
        if tab == "users":                                             2663
            ctx.update(_users_context())                               2664
        elif tab == "logs":                                            2665
            ctx.update(_logs_context())                                2666
        return render_template("auth/profile.html", **ctx)             2667
def _users_context():                                                  2668
    """管理员分类「账号」所需的数据。"""                               2669
    return {"users": User.query.order_by(User.id).all(),               2670
            "form": EmptyForm()}                                       2671
def _logs_context():                                                   2672
    """管理员分类「操作日志」所需的数据。"""                           2673
    page = request.args.get("page", 1, type=int)                       2674
    per_page = 50                                                      2675
    q = OperationLog.query.order_by(OperationLog.id.desc())            2676
    total = q.count()                                                  2677
    records = q.offset((page - 1) * per_page).limit(per_page           2678
    ).all()                                                            2679
    users_map = {u.id: u.username for u in User.query.all()}           2680
    pages = max(1, (total + per_page - 1) // per_page)                 2681
    return {"records": records, "page": page, "pages":                 2682
    pages,                                                             2683
            "total": total, "users_map": users_map}                    2684
# ---------------------------- 头像 ----------------------------       2685
--                                                                     2686
def _avatar_length_ok(file_storage):                                   2687
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""  2688
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 *            2689
    1024 * 1024)                                                       2690
    stream = file_storage.stream                                       2691
    pos = stream.tell()                                                2692
    stream.seek(0, 2)                                                  2693
    size = stream.tell()                                               2694
    stream.seek(pos)                                                   2695
    return size <= limit, size                                         2696
def _save_avatar(file):                                                2697
    """保存头像文件，返回 （相对路径， 错误信息）。没有文件时返回 （None， None）。"""  2698
    if file is None or not file.filename:                              2699
        return None, None                                              2700
```

```
    ok, size = _avatar_length_ok(file)                                  2701
    if not ok:                                                          2702
        return None, ("头像不能超过 %.0f MB（当前 %.1f MB）"              2703
                      % (current_app.config["MAX_AVATAR_LENG            2704
    TH"] / 1024 / 1024,                                                 2705
                         size / 1024 / 1024))                           2706
    return up.save_upload(file, "avatars",                              2707
                          current_app.config["ALLOWED_AVATAR            2708
    _EXT"],                                                             2709
                          current_app.config["UPLOAD_FOLDER"            2710
    ])                                                                  2711
@bp.route("/profile/avatar", methods=["POST"])                          2712
@login_required                                                         2713
def upload_avatar():                                                    2714
    form = EmptyForm()                                                  2715
    back = url_for("auth.profile", tab=request.form.get("tab            2716
    ") or "info")                                                       2717
    if not form.validate_on_submit():                                   2718
        flash("表单校验失败，请重试", "error")                           2719
        return redirect(back)                                           2720
    rel, err = _save_avatar(request.files.get("avatar"))                2721
    if err:                                                             2722
        flash(err, "error")                                             2723
        return redirect(back)                                           2724
    if rel is None:                                                     2725
        flash("请选择头像图片", "error")                                 2726
        return redirect(back)                                           2727
    # 换新头像时删掉旧文件，避免残留                                    2728
    up.remove_upload(current_user.avatar,                               2729
    current_app.config["UPLOAD_FOLDER"])                                2730
    current_user.avatar = rel                                           2731
    db.session.commit()                                                 2732
    flash("头像已更新", "ok")                                           2733
    return redirect(back)                                               2734
@bp.route("/profile/avatar/delete", methods=["POST"])                   2735
@login_required                                                         2736
def delete_avatar():                                                    2737
    form = EmptyForm()                                                  2738
    back = url_for("auth.profile", tab=request.form.get("tab            2739
    ") or "info")                                                       2740
    if form.validate_on_submit():                                       2741
        up.remove_upload(current_user.avatar,                           2742
    current_app.config["UPLOAD_FOLDER"])                                2743
        current_user.avatar = None                                      2744
        db.session.commit()                                             2745
        flash("头像已移除", "ok")                                       2746
    return redirect(back)                                               2747
# ---------------------------- 账号维护（仅管理员）                     2748
    ----------------------------                                        2749
@bp.route("/users")                                                     2750
```

```
@login_required
def users():
    """旧地址保留，统一并入「个人设置 · 账号」。"""
    return redirect(url_for("auth.profile", tab="users"))
@bp.route("/users/add", methods=["GET", "POST"])
@login_required
def add_user():
    require_admin()
    if request.method == "POST":
        username = request.form.get("username", "").strip()
        password = request.form.get("password", "")
        role = request.form.get("role", DEFAULT_ROLE)
        if role not in ROLE_LABELS:
            role = DEFAULT_ROLE
        if not username or not password:
            flash("账户名称和密码不能为空", "error")
            return redirect(url_for("auth.add_user"))
        if User.query.filter_by(username=username).first():
            flash("账户名称已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(
            username=username, role=role,
            password_hash=generate_password_hash(password),
            display_name=request.form.get("display_name",
    "").strip() or None,
            gender=request.form.get("gender", "").strip()
    or None,
            email=request.form.get("email", "").strip() or
    None,
            phone=request.form.get("phone", "").strip() or
    None))
        db.session.commit()
        flash("账号已添加", "ok")
        return redirect(url_for("auth.profile",
    tab="users"))
    return render_template("auth/add_user.html",
    role_labels=ROLE_LABELS,
                           genders=GENDERS,
    default_role=DEFAULT_ROLE,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if request.method == "POST":
        role = request.form.get("role", user.role)
        if role not in ROLE_LABELS:
            role = user.role
```

```
        active = request.form.get("active") == "1"
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统
        if user.role == "admin" and (role != "admin" or not
    active):
            others = User.query.filter(User.role == "admin",
                                       User.id != user.id,

    User.active.is_(True)).count()
            if others == 0:
                flash("系统至少需保留一名启用状态的管理员", "error")
                return redirect(url_for("auth.edit_user",
    user_id=user.id))
        # 个人资料字段（用户自己也能在「个人设置」里改，管理员在此代改）
        email = request.form.get("email", "").strip()
        if not email_ok(email):
            flash("邮箱格式不正确", "error")
            return redirect(url_for("auth.edit_user",
    user_id=user.id))
        user.role = role
        user.active = active
        user.display_name = request.form.get("display_name",
    "").strip() or None
        user.gender = request.form.get("gender",
    "").strip() or None
        user.email = email or None
        user.phone = request.form.get("phone", "").strip()
    or None
        new_password = request.form.get("password",
    "").strip()
        if new_password:
            user.password_hash = generate_password_hash(new_
    password)
        db.session.commit()
        flash("账号信息已保存", "ok")
        return redirect(url_for("auth.profile",
    tab="users"))
    return render_template("auth/edit_user.html", user=user,
                           role_labels=ROLE_LABELS,
    genders=GENDERS,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/delete", methods=["POST"])
@login_required
def delete_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if user.id == current_user.id:
        flash("不能删除当前登录的账号", "error")
        return redirect(url_for("auth.profile",
    tab="users"))
    if user.role == "admin":
```

```
        others = User.query.filter(User.role == "admin",
                                   User.id != user.id,
                                   User.active.is_(True)).co
    unt()
        if others == 0:
            flash("系统至少需保留一名启用状态的管理员", "error")
            return redirect(url_for("auth.profile",
    tab="users"))
    db.session.delete(user)
    db.session.commit()
    flash("账号「%s」已删除" % user.username, "ok")
    return redirect(url_for("auth.profile", tab="users"))
# ------------------------------ 操作日志（仅管理员）
    ------------------------------
@bp.route("/logs")
@login_required
def logs():
    """旧地址保留，统一并入「个人设置 · 操作日志」。"""
    return redirect(url_for("auth.profile", tab="logs"))
# ==== app/views/project.py ====
# -*- coding: utf-8 -*-
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""
from datetime import datetime
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, current_app)
from flask_login import login_required, current_user
from app.forms import EmptyForm
from app.models import db, Project, MinePatch,
    ReferenceEcosystem
bp = Blueprint("project", __name__, url_prefix="/project")
@bp.route("/")
@login_required
def index():
    projects = Project.query.order_by(Project.id.desc()).all
    ()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or
    None,
            description=request.form.get("description",
```

```
        "").strip() or None,                                          2901
                created_by=current_user.id,                           2902
            )                                                         2903
            if not p.name:                                            2904
                flash("项目名称不能为空", "error")                    2905
            else:                                                     2906
                db.session.add(p)                                     2907
                db.session.commit()                                   2908
                flash("项目已创建", "ok")                             2909
                return redirect(url_for("project.index"))             2910
        return render_template("project/edit.html", form=form,        2911
        project=None)                                                 2912
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])                 2913
@login_required                                                       2914
def edit(pid):                                                        2915
        p = db.get_or_404(Project, pid)                               2916
        form = EmptyForm()                                            2917
        if request.method == "POST" and form.validate_on_submit(      2918
        ):                                                            2919
            p.name = request.form.get("name", "").strip()             2920
            p.code = request.form.get("code", "").strip() or          2921
        None                                                          2922
            p.description = request.form.get("description",           2923
        "").strip() or None                                           2924
            p.status = request.form.get("status", "进行中")           2925
            db.session.commit()                                       2926
            flash("项目已保存", "ok")                                 2927
            return redirect(url_for("project.index"))                 2928
        return render_template("project/edit.html", form=form,        2929
        project=p)                                                    2930
@bp.route("/<int:pid>/delete", methods=["POST"])                      2931
@login_required                                                       2932
def delete(pid):                                                      2933
        p = db.get_or_404(Project, pid)                               2934
        db.session.delete(p)                                          2935
        db.session.commit()                                           2936
        flash("项目已删除", "ok")                                     2937
        return redirect(url_for("project.index"))                     2938
# ---------------------------- 矿山图斑 ----------------------------2939
        ----                                                          2940
@bp.route("/<int:pid>/patches")                                       2941
@login_required                                                       2942
def patches(pid):                                                     2943
        p = db.get_or_404(Project, pid)                               2944
        return render_template("project/patches.html",                2945
        project=p,                                                    2946
                                    form=EmptyForm())                 2947
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])           2948
@login_required                                                       2949
def patch_delete(patch_id):                                           2950
```

```
    """删除矿山图斑，及其微地质单元与指标数据。"""
    from app.models import MicroUnit, IndicatorData,
    EvaluationResult, \
        ScoreDetail, ConstraintCheck, ReferenceEcosystem
    patch = db.get_or_404(MinePatch, patch_id)
    pid = patch.project_id
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patc
    h_id=patch.id).all()]
    if unit_ids:
        IndicatorData.query.filter(IndicatorData.unit_id.in_
    (unit_ids)).delete(
            synchronize_session=False)
    for r in EvaluationResult.query.filter_by(patch_id=patch
    .id).all():
        ScoreDetail.query.filter_by(result_id=r.id).delete(
            synchronize_session=False)
    EvaluationResult.query.filter_by(patch_id=patch.id).dele
    te(
        synchronize_session=False)
    ConstraintCheck.query.filter_by(patch_id=patch.id).delet
    e(
        synchronize_session=False)
    MicroUnit.query.filter_by(patch_id=patch.id).delete(sync
    hronize_session=False)
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).de
    lete(
        synchronize_session=False)
    db.session.delete(patch)
    db.session.commit()
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")
    return redirect(url_for("project.patches", pid=pid))
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])
@login_required
def patch_add(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=fv("area_ha", float),
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=fv("stop_years", float),
            responsible_entity_missing=bool(request.form.get
```

```
    ("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
        else:
            db.session.add(patch)
            db.session.flush()
            # 参照生态系统默认取论文值
            db.session.add(ReferenceEcosystem(
                patch_id=patch.id, coverage_benchmark=75.0,
    shannon_benchmark=0.7))
            db.session.commit()
            flash("图斑已创建", "ok")
            return redirect(url_for("project.patches",
    pid=p.id))
    return render_template("project/patch_edit.html",
    form=form, project=p, patch=None)
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",
    "POST"])
@login_required
def patch_edit(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch.name = request.form.get("name", "").strip()
        patch.admin_region = fv("admin_region")
        patch.area_ha = fv("area_ha", float)
        patch.mineral_type = fv("mineral_type")
        patch.mining_method = fv("mining_method")
        patch.stop_years = fv("stop_years", float)
        patch.responsible_entity_missing =
    bool(request.form.get("responsible_entity_missing"))
        # 参照生态系统
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa
    tch.id).first()
        if ref is None:
            ref = ReferenceEcosystem(patch_id=patch.id)
            db.session.add(ref)
        ref.coverage_benchmark = fv("coverage_benchmark",
    float) or 75.0
        ref.shannon_benchmark = fv("shannon_benchmark",
    float) or 0.7
        ref.dominant_species = fv("dominant_species")
        patch.image_note = fv("image_note")
        db.session.commit()
        flash("图斑已保存", "ok")
```

```
            return redirect(url_for("project.patches",                3051
        pid=patch.project_id))                                        3052
        return render_template("project/patch_edit.html",             3053
        form=form,                                                    3054
                                    project=patch.project,            3055
        patch=patch,                                                  3056
                                    reference=ReferenceEcosystem.quer3057
        y.filter_by(                                                  3058
                                        patch_id=patch.id).first())   3059
# ------------------------------ 遥感影像挂接                        3060
        ------------------------------                                3061
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])            3062
@login_required                                                       3063
def upload_image(patch_id):                                           3064
        """上传遥感底图 / 无人机正射影像并挂接到图斑。"""                3065
        from app.utils import uploads as up                           3066
        patch = db.get_or_404(MinePatch, patch_id)                    3067
        form = EmptyForm()                                            3068
        if not form.validate_on_submit():                             3069
            flash("表单校验失败，请重试", "error")                      3070
            return redirect(url_for("project.patch_edit",             3071
        patch_id=patch.id))                                           3072
        rel, err = up.save_upload(request.files.get("image"),         3073
        "patches",                                                    3074
                                      current_app.config["ALLOWED_IM3075
        AGE_EXT"],                                                    3076
                                      current_app.config["UPLOAD_FOL3077
        DER"])                                                        3078
        if err:                                                       3079
            flash(err, "error")                                       3080
            return redirect(url_for("project.patch_edit",             3081
        patch_id=patch.id))                                           3082
        # 替换旧影像时清理原文件，避免残留                              3083
        if patch.image_path:                                          3084
            up.remove_upload(patch.image_path,                        3085
        current_app.config["UPLOAD_FOLDER"])                          3086
        patch.image_path = rel                                        3087
        patch.image_uploaded_at = datetime.utcnow()                   3088
        note = request.form.get("image_note", "").strip()             3089
        if note:                                                      3090
            patch.image_note = note                                   3091
        db.session.commit()                                           3092
        flash("遥感影像已挂接", "ok")                                  3093
        return redirect(url_for("project.patch_edit",                 3094
        patch_id=patch.id))                                           3095
@bp.route("/patch/<int:patch_id>/image/delete",                       3096
      methods=["POST"])                                               3097
@login_required                                                       3098
def delete_image(patch_id):                                           3099
        from app.utils import uploads as up                           3100
```

```
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        up.remove_upload(patch.image_path,
    current_app.config["UPLOAD_FOLDER"])
        patch.image_path = None
        patch.image_uploaded_at = None
        db.session.commit()
        flash("遥感影像已移除", "ok")
    return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
# ==== app/views/unit.py ====
# -*- coding: utf-8 -*-
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import UNIT_TYPES
bp = Blueprint("unit", __name__, url_prefix="/unit")
def recalc_ratio(patch_id):
    """按面积重算该图斑各单元面积占比。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    if total > 0:
        for u in units:
            u.ratio = round((u.area or 0.0) / total, 6)
    return total
@bp.route("/")
@login_required
def list_patches():
    """微地质单元入口：列出各图斑及其单元划分情况。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    rows = []
    for p in patches:
        units = MicroUnit.query.filter_by(patch_id=p.id).all
    ()
        rows.append({
            "patch": p,
            "count": len(units),
            "ratio_sum": sum(u.ratio or 0.0 for u in units),
        })
    return render_template("unit/list.html", rows=rows)
@bp.route("/<int:patch_id>")
@login_required
def index(patch_id):
```

```
    patch = db.get_or_404(MinePatch, patch_id)
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
                           unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=float(area) if area else None,
                      description=request.form.get("descript
    ion", "").strip() or None)
        db.session.add(u)
        db.session.flush()
        recalc_ratio(patch.id)
        db.session.commit()
        flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
```

```
    if request.method == "POST" and form.validate_on_submit(
    ):
        new_type = request.form.get("unit_type",
    u.unit_type)
        if new_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.edit",
    unit_id=u.id))
        # 改类型时同样不允许与本图斑已有的其他单元重复
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,


    unit_type=new_type).first()
        if dup is not None and dup.id != u.id:
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,
    "error")
            return redirect(url_for("unit.edit",
    unit_id=u.id))
        u.unit_type = new_type
        area = request.form.get("area", "").strip()
        u.area = float(area) if area else None
        u.description = request.form.get("description",
    "").strip() or None
        recalc_ratio(u.patch_id)
        db.session.commit()
        flash("微地质单元已保存", "ok")
        return redirect(url_for("unit.index",
    patch_id=u.patch_id))
    return render_template("unit/edit.html", unit=u,
    patch=u.patch,
                           unit_types=UNIT_TYPES, form=form)
@bp.route("/delete/<int:unit_id>", methods=["POST"])
@login_required
def delete(unit_id):
    u = db.get_or_404(MicroUnit, unit_id)
    pid = u.patch_id
    name = u.unit_type
    # 一并删除该单元的指标数据，避免残留
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch
    ronize_session=False)
    db.session.delete(u)
    db.session.flush()
    recalc_ratio(pid)
    db.session.commit()
    flash("微地质单元「%s」已删除" % name, "ok")
    return redirect(url_for("unit.index", patch_id=pid))
# ==== app/views/standard.py ====
# -*- coding: utf-8 -*-
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
```

```
from flask_login import login_required                                  3251
from app.forms import EmptyForm                                         3252
from app.models import db, Standard, StandardRule                       3253
bp = Blueprint("standard", __name__, url_prefix="/standard")            3254
@bp.route("/")                                                          3255
@login_required                                                         3256
def index():                                                            3257
    standards = Standard.query.order_by(Standard.category,              3258
    Standard.province).all()                                            3259
    local = [s for s in standards if s.category == "地方标准"]          3260
    policy = [s for s in standards if s.category != "地方标准"]         3261
    return render_template("standard/index.html",                       3262
    local=local, policy=policy,                                         3263
                                 form=EmptyForm())                      3264
@bp.route("/add", methods=["GET", "POST"])                              3265
@login_required                                                         3266
def add():                                                              3267
    form = EmptyForm()                                                  3268
    if request.method == "POST" and form.validate_on_submit(            3269
    ):                                                                  3270
        s = Standard(                                                   3271
            province=request.form.get("province",                       3272
    "").strip(),                                                        3273
            name=request.form.get("name", "").strip(),                  3274
            doc_number=request.form.get("doc_number",                   3275
    "").strip() or None,                                                3276
            publish_date=request.form.get("publish_date",               3277
    "").strip() or None,                                                3278
            implement_date=request.form.get("implement_date"            3279
    , "").strip() or None,                                              3280
            category=request.form.get("category", "地方标准"),          3281
            summary=request.form.get("summary", "").strip()             3282
    or None,                                                            3283
        )                                                               3284
        if not s.province or not s.name:                                3285
            flash("省份与名称不能为空", "error")                        3286
        else:                                                           3287
            db.session.add(s)                                           3288
            db.session.commit()                                         3289
            flash("标准已添加", "ok")                                   3290
            return redirect(url_for("standard.index"))                  3291
    return render_template("standard/edit.html", form=form,             3292
    standard=None)                                                      3293
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])                   3294
@login_required                                                         3295
def edit(sid):                                                          3296
    s = db.get_or_404(Standard, sid)                                    3297
    form = EmptyForm()                                                  3298
    if request.method == "POST" and form.validate_on_submit(            3299
    ):                                                                  3300
```

```
            s.province = request.form.get("province",                3301
      "").strip()                                                    3302
            s.name = request.form.get("name", "").strip()            3303
            s.doc_number = request.form.get("doc_number",            3304
      "").strip() or None                                            3305
            s.publish_date = request.form.get("publish_date",        3306
      "").strip() or None                                            3307
            s.implement_date = request.form.get("implement_date"     3308
      , "").strip() or None                                          3309
            s.category = request.form.get("category", "地方标准")      3310
            s.summary = request.form.get("summary", "").strip()      3311
      or None                                                        3312
            db.session.commit()                                      3313
            flash("已保存", "ok")                                      3314
            return redirect(url_for("standard.rules", sid=s.id))     3315
      return render_template("standard/edit.html", form=form,        3316
      standard=s)                                                    3317
@bp.route("/<int:sid>/delete", methods=["POST"])                     3318
@login_required                                                      3319
def delete(sid):                                                     3320
      s = db.get_or_404(Standard, sid)                               3321
      StandardRule.query.filter_by(standard_id=sid).delete()         3322
      db.session.delete(s)                                           3323
      db.session.commit()                                            3324
      flash("已删除", "ok")                                            3325
      return redirect(url_for("standard.index"))                     3326
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])               3327
@login_required                                                      3328
def rules(sid):                                                      3329
      s = db.get_or_404(Standard, sid)                               3330
      form = EmptyForm()                                             3331
      if request.method == "POST" and form.validate_on_submit(       3332
      ):                                                             3333
            r = StandardRule(                                        3334
                  standard_id=s.id,                                  3335
                  item=request.form.get("item", "").strip(),         3336
                  operator=request.form.get("operator", ">="),       3337
                  threshold=request.form.get("threshold",            3338
      "").strip() or None,                                           3339
                  condition_type=request.form.get("condition_type"   3340
      , "必要条件"),                                                   3341
                  region=request.form.get("region", "").strip()      3342
      or None,                                                       3343
                  note=request.form.get("note", "").strip() or       3344
      None,                                                          3345
            )                                                        3346
            if not r.item:                                           3347
                  flash("判定项不能为空", "error")                      3348
            else:                                                    3349
                  db.session.add(r)                                  3350
```

```
                db.session.commit()
                flash("规则已添加", "ok")
            return redirect(url_for("standard.rules", sid=s.id))
        rs = StandardRule.query.filter_by(standard_id=s.id).all()
        return render_template("standard/rules.html",
        form=form, standard=s, rules=rs)
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
        r = db.get_or_404(StandardRule, rid)
        sid = r.standard_id
        db.session.delete(r)
        db.session.commit()
        flash("规则已删除", "ok")
        return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
@login_required
def compare():
        """省际口径比对：同一指标在不同省份的阈值横向对比。"""
        rules = StandardRule.query.join(Standard).all()
        groups = {}
        for r in rules:
            groups.setdefault(r.item, []).append(r)
        return render_template("standard/compare.html",
        groups=groups)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""限制性指标判定 —— 一票否决判定。
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, ConstraintCheck
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
FIELDS = [
    ("policy_pass", "政策约束"),
    ("location_pass", "区位约束"),
    ("pollution_pass", "土壤污染"),
    ("geo_pass", "地质灾害"),
    ("farmland_pass", "耕地属性"),
]
@bp.route("/")
@login_required
```

```
def index():                                                    3401
    patches = MinePatch.query.order_by(MinePatch.id.desc()).    3402
    all()                                                       3403
    latest = {}                                                 3404
    for p in patches:                                           3405
        latest[p.id] = ConstraintCheck.query.filter_by(patch    3406
    _id=p.id) \                                                 3407
            .order_by(ConstraintCheck.id.desc()).first()        3408
    return render_template("constraint/index.html",             3409
    patches=patches, latest=latest)                             3410
@bp.route("/<int:patch_id>", methods=["GET", "POST"])           3411
@login_required                                                 3412
def check(patch_id):                                            3413
    patch = db.get_or_404(MinePatch, patch_id)                  3414
    form = EmptyForm()                                          3415
    record = ConstraintCheck.query.filter_by(patch_id=patch_    3416
    id) \                                                       3417
        .order_by(ConstraintCheck.id.desc()).first()            3418
    if request.method == "POST" and form.validate_on_submit(    3419
    ):                                                          3420
        if record is None:                                      3421
            record = ConstraintCheck(patch_id=patch.id)         3422
            db.session.add(record)                              3423
        for field, _ in FIELDS:                                 3424
            setattr(record, field, request.form.get(field)      3425
    == "1")                                                     3426
        record.passed = all(getattr(record, f) for f, _ in      3427
    FIELDS)                                                     3428
        failed = [name for f, name in FIELDS if not             3429
    getattr(record, f)]                                         3430
        record.reason = ("全部通过" if record.passed            3431
                         else "未通过：" + "、".join(failed))   3432
        db.session.commit()                                     3433
        flash("判定结果已保存", "ok")                           3434
        return redirect(url_for("constraint.check",             3435
    patch_id=patch.id))                                         3436
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）  3437
    field_states = [(f, name, None if record is None else       3438
    getattr(record, f))                                         3439
                    for f, name in FIELDS]                      3440
    return render_template("constraint/check.html",             3441
    patch=patch, form=form,                                     3442
                           fields=FIELDS,                       3443
    field_states=field_states,                                  3444
                           record=record)                       3445
# ==== app/views/ahp.py ====                                    3446
# -*- coding: utf-8 -*-                                         3447
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""                   3448
import json                                                     3449
from flask import Blueprint, render_template, request,          3450
```

```
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
# 一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 (指标名称列表, 默认权重列表)。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
```

```
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form,
                                   weight_rows=ws.summary())
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form,
                           weight_rows=ws.summary())
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
    level = int(request.values.get("level", 1))
    l1 = normalize_l1(unit_type, level,
    request.values.get("l1"))
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
```

```
    n = len(names)
    form = EmptyForm()
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
range(n)] for i in range(n)]
    def compute(upper):
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        bad = ahp.find_most_inconsistent(A)
        return {
            "weights": [round(v, 4) for v in r["weights"]],
            "lambda_max": round(r["lambda_max"], 4),
            "ci": r["ci"],
            "cr": r["cr"],
            "consistent": r["consistent"],
            "matrix": [[round(A[i, j], 4) for j in
range(n)] for i in range(n)],
            "most_inconsistent": bad,
        }
    result = None
    if request.method == "POST" and form.validate_on_submit(
):
        action = request.form.get("action", "calc")
        try:
            upper = [float(request.form.get("m_%d_%d" % (i,
j), "1")) 
                     for i in range(n) for j in range(i +
1, n)]
            result = compute(upper)
        except (ValueError, ZeroDivisionError):
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",
"error")
            return redirect(url_for("ahp.matrix",
unit_type=unit_type,
                                    level=level, l1=l1))
        if action == "save":
            if not result["consistent"]:
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"
                      "请调整后再保存" % result["cr"], "error")
            else:
                pairs = to_code_pairs(unit_type, level, l1,
names,
                                      result["weights"])
                scheme, cnt = ws.save_weights(unit_type,
pairs)
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"
                      "此后该单元类型的评价将以此权重计算。"
                      % (unit_type, cnt, result["cr"]),
"ok")
                return redirect(url_for("ahp.matrix",
```

```
    unit_type=unit_type,                                                    3601
                                                level=level, l1=l1))        3602
    elif request.values.get("calc") == "1":                                 3603
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性                  3604
        result = compute([default[i][j] for i in range(n)                   3605
    for j in range(i + 1, n)])                                              3606
    _, source_label, custom_scheme = ws.effective_weights(un                3607
    it_type)                                                                3608
    return render_template(                                                 3609
        "ahp/matrix.html",                                                  3610
        form=form, unit_type=unit_type, level=level, l1=l1,                 3611
        names=names, vec=vec, default=default,                              3612
    result=result,                                                          3613
        source_label=source_label, custom_scheme=custom_sche                3614
    me,                                                                     3615
    )                                                                       3616
@bp.route("/weights/reset", methods=["POST"])                               3617
def weights_reset():                                                        3618
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""                    3619
    unit_type = request.form.get("unit_type")                               3620
    form = EmptyForm()                                                      3621
    if form.validate_on_submit():                                           3622
        if unit_type not in DEFAULT_WEIGHTS:                                3623
            flash("无效的微地质单元类型", "error")                             3624
        else:                                                               3625
            n = ws.reset_weights(unit_type)                                 3626
            if n:                                                           3627
                flash("已恢复「%s」为论文表 3 默认权重" % unit_type,                3628
    "ok")                                                                   3629
            else:                                                           3630
                flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,        3631
    "ok")                                                                   3632
    return redirect(url_for("ahp.index"))                                   3633
# ---------------------------- 指标维护 ----------------------------        3634
    ----                                                                    3635
@bp.route("/indicators")                                                    3636
def indicators():                                                           3637
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""                          3638
    if Indicator.query.count() == 0:                                        3639
        ind_svc.seed_indicators()                                           3640
    all_ind = Indicator.query.order_by(Indicator.level,                     3641
    Indicator.id).all()                                                     3642
    by_parent = {}                                                          3643
    for i in all_ind:                                                       3644
        by_parent.setdefault(i.parent_code, []).append(i)                   3645
    return render_template("ahp/indicators.html",                           3646
    by_parent=by_parent,                                                    3647
                                    all_ind=all_ind,                        3648
    unit_types=UNIT_TYPES,                                                  3649
                                    type_label=ind_svc.ideal_type_lab       3650
```

```
    el,                                                                3651
                                form=EmptyForm())                      3652
@bp.route("/indicators/<code>/edit", methods=["GET",                   3653
    "POST"])                                                           3654
def indicator_edit(code):                                              3655
    ind = Indicator.query.filter_by(code=code).first_or_404(           3656
    )                                                                  3657
    form = EmptyForm()                                                 3658
    if request.method == "POST" and form.validate_on_submit(           3659
    ):                                                                 3660
        name = request.form.get("name", "").strip()                    3661
        if not name:                                                   3662
            flash("指标名称不能为空", "error")                          3663
            return redirect(url_for("ahp.indicator_edit",              3664
    code=code))                                                        3665
        ind.name = name                                                3666
        ind.unit = request.form.get("unit", "").strip() or             3667
    None                                                               3668
        ind.ideal_value = request.form.get("ideal_value",              3669
    "").strip() or None                                                3670
        ind.ideal_type = request.form.get("ideal_type") or             3671
    ind.ideal_type                                                     3672
        units = request.form.getlist("enabled_units")                  3673
        ind.enabled_units = ",".join(units) if units else              3674
    None                                                               3675
        ind.is_constraint = request.form.get("is_constraint"           3676
    ) == "1"                                                           3677
        ind.description = request.form.get("description",              3678
    "").strip() or None                                                3679
        db.session.commit()                                            3680
        flash("指标「%s」已保存" % ind.name, "ok")                      3681
        return redirect(url_for("ahp.indicators"))                     3682
    return render_template("ahp/indicator_edit.html",                  3683
    ind=ind,                                                           3684
                                unit_types=UNIT_TYPES,                 3685
                                type_label=ind_svc.ideal_type_lab      3686
    el, form=form)                                                     3687
@bp.route("/indicators/reset", methods=["POST"])                       3688
def indicators_reset():                                                3689
    """按论文取值重置指标定义。"""                                      3690
    form = EmptyForm()                                                 3691
    if form.validate_on_submit():                                      3692
        n = ind_svc.seed_indicators()                                  3693
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")                3694
    return redirect(url_for("ahp.indicators"))                         3695
# ---------------------------- 多专家矩阵 ----------------------------  3696
    -----                                                              3697
def expert_scope():                                                    3698
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""            3699
    unit_type = request.values.get("unit_type") or                     3700
```

```
    UNIT_TYPES[0]                                                    3701
    if unit_type not in DEFAULT_WEIGHTS:                             3702
        unit_type = UNIT_TYPES[0]                                    3703
    level = int(request.values.get("level", 1))                      3704
    l1 = normalize_l1(unit_type, level,                              3705
    request.values.get("l1"))                                        3706
    return unit_type, level, l1                                      3707
def expert_query(unit_type, level, l1):                              3708
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,            3709
    level=level)                                                     3710
    if level == 2:                                                   3711
        q = q.filter_by(l1_name=l1)                                  3712
    else:                                                            3713
        q = q.filter(ExpertMatrix.l1_name.is_(None))                 3714
    return q.order_by(ExpertMatrix.id)                               3715
def default_scheme(unit_type, level, l1=None):                       3716
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""                    3717
    name = "%s · %s%s" % (unit_type,                                 3718
                          "一级指标" if level == 1 else "二级指标",          3719
                          "(%s)" % l1 if l1 else "")                 3720
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ3721
    e,                                                               3722
                                          level=level,               3723
    name=name).first()                                               3724
    if scheme is None:                                               3725
        scheme = WeightScheme(unit_type=unit_type,                   3726
    level=level, name=name,                                          3727
                              is_default=False)                      3728
        db.session.add(scheme)                                       3729
        db.session.flush()                                           3730
    return scheme                                                    3731
@bp.route("/experts", methods=["GET", "POST"])                       3732
def experts():                                                       3733
    """多专家判断矩阵录入与汇总。"""                                          3734
    unit_type, level, l1 = expert_scope()                            3735
    if unit_type not in DEFAULT_WEIGHTS:                             3736
        flash("无效的微地质单元类型", "error")                               3737
        return redirect(url_for("ahp.index"))                        3738
    names, vec = indicator_list(unit_type, level, l1)                3739
    n = len(names)                                                   3740
    form = EmptyForm()                                               3741
    if request.method == "POST" and form.validate_on_submit(3742
    ):                                                               3743
        action = request.form.get("action")                          3744
        if action == "add":                                          3745
            expert_name = request.form.get("expert_name",            3746
    "").strip() or "未具名专家"                                          3747
            try:                                                     3748
                upper = [float(request.form.get("m_%d_%d" %          3749
    (i, j), "1"))                                                    3750
```

```
                        for i in range(n) for j in range(i
+ 1, n)]
        except ValueError:
            flash("矩阵输入有误，请填入正数", "error")
            return redirect(url_for("ahp.experts",
unit_type=unit_type,
                                    level=level, l1=l1))
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        scheme = default_scheme(unit_type, level, l1)
        db.session.add(ExpertMatrix(
            scheme_id=scheme.id,
            unit_type=unit_type, l1_name=(l1 if level
== 2 else None),
            expert_name=expert_name, level=level,
            matrix_json=json.dumps([list(map(float,
row)) for row in A]),
            cr=round(r["cr"], 6)))
        db.session.commit()
        flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,
r["cr"]), "ok")
    elif action == "save_agg":
        # 把汇总权重落库并生效，使多专家结果真正参与评价计算
        weights = _aggregate(unit_type, level, l1)
        if weights is None:
            flash("暂无可汇总的专家矩阵", "error")
        elif not weights["consistent"]:
            flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"
                  "请先检查或调整专家矩阵" % weights["cr"],
"error")
        else:
            names_, _ = indicator_list(unit_type,
level, l1)
            pairs = to_code_pairs(unit_type, level, l1,
names_,
                                  weights["weights"])
            _, cnt = ws.save_weights(unit_type, pairs)
            flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"
                  % (unit_type, cnt, weights["cr"]),
"ok")
    return redirect(url_for("ahp.experts",
unit_type=unit_type,
                            level=level, l1=l1))
rows = expert_query(unit_type, level, l1).all()
default = [[round(vec[i] / vec[j], 4) for j in
range(n)] for i in range(n)]
return render_template("ahp/experts.html",
unit_type=unit_type, level=level,
                       l1=l1, names=names, rows=rows,
default=default,
```

```
                                    n=n, form=form)                          3801
@bp.route("/experts/<int:mid>/delete", methods=["POST"])                     3802
def expert_delete(mid):                                                      3803
    row = db.get_or_404(ExpertMatrix, mid)                                   3804
    unit_type, level, l1 = row.unit_type, row.level,                         3805
    row.l1_name                                                              3806
    form = EmptyForm()                                                       3807
    if form.validate_on_submit():                                            3808
        db.session.delete(row)                                               3809
        db.session.commit()                                                  3810
        flash("已删除专家矩阵", "ok")                                          3811
    return redirect(url_for("ahp.experts",                                   3812
    unit_type=unit_type,                                                     3813
                                      level=level, l1=l1))                   3814
def _aggregate(unit_type, level, l1):                                        3815
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""                3816
    rows = expert_query(unit_type, level, l1).all()                          3817
    mats = []                                                                3818
    for r in rows:                                                           3819
        try:                                                                 3820
            mats.append([list(map(float, line))                              3821
                         for line in json.loads(r.matrix_jso                 3822
    n)])                                                                     3823
        except Exception:                                                    3824
            continue                                                         3825
    if not mats:                                                             3826
        return None                                                          3827
    agg = ahp.aggregate_matrices(mats)                                       3828
    res = ahp.weights_and_consistency(agg)                                   3829
    return {                                                                 3830
        "count": len(mats),                                                  3831
        "weights": res["weights"],                                           3832
        "lambda_max": res["lambda_max"],                                     3833
        "ci": res["ci"], "cr": res["cr"],                                    3834
        "consistent": res["consistent"],                                     3835
        "most_inconsistent": ahp.find_most_inconsistent(agg)                 3836
    ,                                                                        3837
        "expert_crs": [r.cr for r in rows if r.cr is not                     3838
    None],                                                                   3839
    }                                                                        3840
@bp.route("/experts/aggregate")                                              3841
def experts_aggregate():                                                     3842
    """按几何平均汇总多专家矩阵，并做一致性检验。"""                                3843
    unit_type, level, l1 = expert_scope()                                    3844
    if unit_type not in DEFAULT_WEIGHTS:                                     3845
        flash("无效的微地质单元类型", "error")                                  3846
        return redirect(url_for("ahp.index"))                                3847
    names, vec = indicator_list(unit_type, level, l1)                        3848
    rows = expert_query(unit_type, level, l1).all()                          3849
    result = _aggregate(unit_type, level, l1)                                3850
```

```
    if result is not None:
        result["weights"] = [round(v, 4) for v in result["weights"]]
        result["lambda_max"] = round(result["lambda_max"], 4)
    return render_template("ahp/experts_aggregate.html",
    unit_type=unit_type,
                           level=level, l1=l1, names=names,
    rows=rows,
                           result=result, form=EmptyForm())
# ==== app/views/indicator.py ====
# -*- coding: utf-8 -*-
"""指标数据采集与管理 —— 路由与页面逻辑。
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标
（岩质边坡无土壤指标、平台无地形指标）。
"""
import io
import json
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, send_file)
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_ENABLED_INDICATORS
from app.services import import_service as imp
bp = Blueprint("indicator", __name__,
    url_prefix="/indicator")
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)
SOIL_FIELDS = [
    ("soil_ratio", "土石比", ""),
    ("thickness", "土层厚度", "cm"),
    ("ph", "pH", ""),
    ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
    ("coverage", "植被覆盖度", "%"),
    ("shannon", "Shannon-Wiener 多样性指数", ""),
]
TERRAIN_FIELDS = [
    ("slope", "坡度", "°"),
    ("aspect", "坡向", ""),
]
def unit_field_config(unit_type):
    """根据单元类型返回需录入的字段分组。"""
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())
    groups = []
    if "土壤指标" in enabled:
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True
```

```
    = 含重金属                                                    3901
    if "植被指标" in enabled:                                     3902
        groups.append(("植被指标", VEG_FIELDS, False))            3903
    if "地形指标" in enabled:                                     3904
        groups.append(("地形指标", TERRAIN_FIELDS, False))        3905
    return groups                                                 3906
@bp.route("/")                                                    3907
def index():                                                      3908
    patches = MinePatch.query.order_by(MinePatch.id.desc()).      3909
    all()                                                         3910
    return render_template("indicator/index.html",                3911
    patches=patches)                                              3912
@bp.route("/<int:unit_id>", methods=["GET", "POST"])              3913
def edit(unit_id):                                                3914
    unit = db.get_or_404(MicroUnit, unit_id)                      3915
    ind = IndicatorData.query.filter_by(unit_id=unit.id).fir      3916
    st()                                                          3917
    form = EmptyForm()                                            3918
    groups = unit_field_config(unit.unit_type)                    3919
    if request.method == "POST" and form.validate_on_submit(      3920
    ):                                                            3921
        def f(name, cast=float):                                  3922
            v = request.form.get(name, "").strip()                3923
            return cast(v) if v != "" else None                   3924
        heavy = {}                                                3925
        for m in HEAVY_METALS:                                    3926
            v = f("hm_" + m)                                      3927
            if v is not None:                                     3928
                heavy[m] = v                                      3929
        if ind is None:                                           3930
            ind = IndicatorData(unit_id=unit.id)                  3931
            db.session.add(ind)                                   3932
        ind.soil_ratio = f("soil_ratio")                          3933
        ind.thickness = f("thickness")                            3934
        ind.ph = f("ph")                                          3935
        ind.organic_matter = f("organic_matter")                  3936
        ind.heavy_metals = json.dumps(heavy,                      3937
    ensure_ascii=False) if heavy else None                        3938
        ind.coverage = f("coverage")                              3939
        ind.shannon = f("shannon")                                3940
        ind.slope = f("slope")                                    3941
        ind.aspect = request.form.get("aspect", "").strip()       3942
    or None                                                       3943
        db.session.commit()                                       3944
        flash("指标数据已保存", "ok")                             3945
        return redirect(url_for("indicator.index"))               3946
    # 组装当前值供模板回显                                        3947
    current = {}                                                  3948
    if ind:                                                       3949
        current = {                                               3950
```

```
                "soil_ratio": ind.soil_ratio, "thickness":
    ind.thickness,
                "ph": ind.ph, "organic_matter":
    ind.organic_matter,
                "coverage": ind.coverage, "shannon":
    ind.shannon,
                "slope": ind.slope, "aspect": ind.aspect,
            }
            current["heavy"] = ind.heavy_metals_dict()
    return render_template("indicator/edit.html",
                                unit=unit, form=form,
    groups=groups,
                                heavy_metals=HEAVY_METALS,
                                aspect_types=ASPECT_TYPES,
    current=current)
# ---------------------------- 批量导入 ----------------------------
    ----
@bp.route("/import", methods=["GET", "POST"])
def import_data():
    """批量导入指标数据（CSV / Excel）。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    form = EmptyForm()
    report = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        patch_id = request.form.get("patch_id", type=int)
        patch = db.session.get(MinePatch, patch_id) if
    patch_id else None
        if patch is None:
            flash("请选择矿山图斑", "error")
            return redirect(url_for("indicator.import_data")
    )
        file = request.files.get("datafile")
        if file is None or not file.filename:
            flash("请选择要导入的文件", "error")
            return redirect(url_for("indicator.import_data")
    )
        ext = file.filename.rsplit(".", 1)[-1].lower()
        if ext not in ("csv", "xlsx", "xls"):
            flash("仅支持 CSV 或 Excel（.xlsx/.xls）文件", "error")
            return redirect(url_for("indicator.import_data")
    )
        units = MicroUnit.query.filter_by(patch_id=patch.id)
    .all()
        unit_map = {}
        for u in units:
            unit_map.setdefault(u.unit_type, u)
        if not unit_map:
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")
```

```
            return redirect(url_for("indicator.import_data"))
    try:
        parsed, errors = imp.parse_upload(file.filename, file.stream, unit_map)
    except Exception as e:
        flash("文件解析失败：%s" % e, "error")
        return redirect(url_for("indicator.import_data"))
    saved = 0
    for rec in parsed:
        unit = unit_map[rec["unit_type"]]
        ind = IndicatorData.query.filter_by(unit_id=unit.id).first()
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        f = rec["fields"]
        for key in ("soil_ratio", "thickness", "ph", "organic_matter",
                    "coverage", "shannon", "slope", "aspect"):
            if key in f:
                setattr(ind, key, f[key])
        if rec["heavy_metals"]:
            ind.heavy_metals = json.dumps(rec["heavy_metals"],

                                          ensure_ascii=False)
        saved += 1
    db.session.commit()
    report = {"patch": patch.name, "saved": saved, "errors": errors,
              "total": len(parsed) + len(
                  [e for e in errors if e.startswith("第")])}
    if saved:
        flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")
    else:
        flash("没有可导入的数据行，请检查文件内容", "error")
    return render_template("indicator/import.html", patches=patches,
                           form=form, report=report, headers=imp.HEADERS)
@bp.route("/import/template")
def import_template():
    """下载导入模板（CSV，含一行示例）。"""
    csv_text = imp.build_template_csv()
    data = csv_text.encode("utf-8-sig")  # 带 BOM，便于 Excel 正确识别中文
```

```
    return send_file(io.BytesIO(data), as_attachment=True,
                     download_name="指标数据导入模板.csv",
                     mimetype="text/csv")
# ==== app/views/evaluate.py ====
# -*- coding: utf-8 -*-
"""适宜性定量评价引擎 —— 路由与页面逻辑。
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。
"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import EmptyForm
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail)
from app.services import evaluation_service as ev
from app.services import workflow as wf
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind
    else None,
            "heavy_metals": ind.heavy_metals_dict() if ind
    else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_id": u.id,
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else
    0.0,
            "data": data,
        })
    return result
```

```
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自 AHP 计算并保存的生效方案；未配置时用论文表 3 的默认值。
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 JSON。
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
            "unit_score": _round(u["unit_score"]),
            "weighted": _round(u["weighted"]),
            "weight_source": sources.get(u["unit_type"]),
            "l1_scores": {k: _round(v) for k, v in
u["l1_scores"].items()},
            "details": [{
                "l1": d["l1"], "l2": d["l2"],
                "raw": None if d["raw"] is None else
str(d["raw"]),
                "std_value": _round(d["std_value"]),
                "weight": _round(d["weight"]),
                "contribution": _round(d["contribution"]),
            } for d in u["details"]],
        })
    return out
def save_evaluation(patch_id, units):
    """执行评价并把结果与逐层明细落库，返回评价记录。
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用
    论文表 3 的默认权重；来源一并记录。结果 JSON 保留各单元得分、
    一级指标得分与逐条二级指标明细，结果页与报告据此复现
    记录下来的那次评价，不再重新推算。
    """
```

```
    weights, sources = load_weights(units)
    res = ev.score_mine(units, weights)
    saved = EvaluationResult(
        patch_id=patch_id,
        final_score=round(res["final_score"], 4),
        grade=res["grade"],
        measures=res["grade_info"]["measure"],
        unit_scores=json.dumps(pack_units(res["unit_results"
    ], sources),
                               ensure_ascii=False),
    )
    db.session.add(saved)
    db.session.flush()        # 先取得 result.id,再写明细
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}
    for u in res["unit_results"]:
        for d in u["details"]:
            db.session.add(ScoreDetail(
                result_id=saved.id,
                unit_id=unit_ids.get(u["unit_type"]),
                l1_name=d["l1"], l2_name=d["l2"],
                raw_value=None if d["raw"] is None else
    str(d["raw"]),
                std_value=d["std_value"],
    weight=d["weight"],
                contribution=d["contribution"]))
    db.session.commit()
    return saved
def block_reasons(progress):
    """把未满足的前置条件转成可读的拦截原因。"""
    return ["%s：%s" % (b["label"], b["reason"]) for b in
    progress["blocking"]]
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    progress = {}
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patc
    h_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
                           latest=latest, progress=progress)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
```

```
    if request.method == "POST" and form.validate_on_submit(
    ):
        if not progress["ready"]:
            for reason in block_reasons(progress):
                flash("无法执行评价 —— " + reason, "error")
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        saved = save_evaluation(patch.id, units)
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    # 展示将采用的权重来源，便于在执行前确认
    units_now = load_patch_input(patch_id)
    _, sources = load_weights(units_now) if units_now else
    ({}, {})
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                           progress=progress,
    weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                           unit_scores=unit_scores)
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
import json
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
from app.services import report_service as rs
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.utils.constants import grade_of
from app.utils.timefmt import now_utc
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查
    结果，由调用方拦截并提示先去执行评价。
```

```
    """                                                                4251
    patch = db.get_or_404(MinePatch, patch_id)                         4252
    prog = wf.report_progress(patch_id)                                4253
    if not prog["ready"]:                                              4254
        return patch, None, None, None, prog                           4255
    rec = prog["result"]                                               4256
    result = {                                                         4257
        "final_score": rec.final_score,                                4258
        "grade": rec.grade,                                            4259
        "grade_info": grade_of(rec.final_score),                       4260
        "unit_results": json.loads(rec.unit_scores) if                 4261
    rec.unit_scores else [],                                           4262
    }                                                                  4263
    reference = ReferenceEcosystem.query.filter_by(patch_id=           4264
    patch_id).first()                                                  4265
    return patch, reference, result, rec, prog                         4266
def _deny(progress, patch_id, endpoint="report.detail"):               4267
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""                    4268
    for b in progress["blocking"]:                                     4269
        flash("暂不能生成报告 —— %s：%s" % (b["label"],                   4270
    b["reason"]), "error")                                             4271
    if progress["blocking"]:                                           4272
        flash("请先完成上述步骤后再生成报告。", "error")                      4273
    return redirect(url_for("report.index"))                           4274
def _charts(result):                                                   4275
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""                      4276
    units = [(u["unit_type"], u["unit_score"],                         4277
              "%.4f" % u["unit_score"]) for u in                       4278
    result["unit_results"]]                                            4279
    chart_units = charts.hbars(units)                                  4280
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响                    4281
    acc, cnt = {}, {}                                                  4282
    for u in result["unit_results"]:                                   4283
        for k, v in (u.get("l1_scores") or {}).items():                4284
            acc[k] = acc.get(k, 0.0) + v                               4285
            cnt[k] = cnt.get(k, 0) + 1                                 4286
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /                   4287
    cnt[k])) for k in acc]                                             4288
    items.sort(key=lambda x: -x[1])                                    4289
    chart_weight = charts.donut(items)                                 4290
    return chart_units, chart_weight                                   4291
@bp.route("/")                                                         4292
def index():                                                           4293
    patches = MinePatch.query.order_by(MinePatch.id.desc()).           4294
    all()                                                              4295
    progress = {p.id: wf.report_progress(p.id) for p in                4296
    patches}                                                           4297
    return render_template("report/index.html",                        4298
    patches=patches,                                                   4299
                                progress=progress)                     4300
```

```
@bp.route("/<int:patch_id>")
def detail(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    return render_template("report/detail.html",
    patch=patch, result=result,
                           record=record,
    reference=reference,
                           chart_units=chart_units,
    chart_weight=chart_weight)
@bp.route("/<int:patch_id>/word")
def word(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    buf = rs.build_report(patch, result)
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name
    return send_file(buf, as_attachment=True,
    download_name=filename,
                     mimetype="application/vnd.openxmlformat
    s-officedocument."
                              "wordprocessingml.document")
@bp.route("/<int:patch_id>/pdf")
def pdf(patch_id):
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    html = render_template(
        "report/print.html", patch=patch,
    reference=reference, result=result,
        chart_units=chart_units, chart_weight=chart_weight,
        generated_at=now_utc())
    data, err = browser.html_to_pdf(html)
    if err:
        flash("PDF 导出失败：%s" % err, "error")
        return redirect(url_for("report.detail",
    patch_id=patch_id))
    return send_file(io.BytesIO(data), as_attachment=True,
                     download_name="自然恢复适宜性评价报告_%s.pdf" %
    patch.name,
                     mimetype="application/pdf")
# ==== app/views/search.py ====
# -*- coding: utf-8 -*-
```

```
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。
"""
from sqlalchemy import or_
from flask import Blueprint, render_template, request
from app.models import db, Project, MinePatch, Standard
bp = Blueprint("search", __name__)
LIMIT = 50              # 每类结果上限，避免一次渲染过多
@bp.route("/search")
def index():
    q = (request.args.get("q") or "").strip()
    projects, patches, standards = [], [], []
    if q:
        like = "%" + q + "%"
        projects = Project.query.filter(or_(
            Project.name.like(like),
            Project.code.like(like),
            Project.description.like(like),
        )).order_by(Project.id.desc()).limit(LIMIT).all()
        patches = MinePatch.query.filter(or_(
            MinePatch.name.like(like),
            MinePatch.admin_region.like(like),
            MinePatch.mineral_type.like(like),
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()
        standards = Standard.query.filter(or_(
            Standard.name.like(like),
            Standard.doc_number.like(like),
            Standard.province.like(like),
            Standard.summary.like(like),
        )).order_by(Standard.id).limit(LIMIT).all()
    total = len(projects) + len(patches) + len(standards)
    return render_template("search.html", q=q,
    projects=projects,
                           patches=patches,
    standards=standards, total=total)
# ==== app/templates/base.html ====
{% import "_macros.html" as ui %}<!DOCTYPE html>
<html lang="zh-CN">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
    initial-scale=1.0">
  <link rel="icon" href="{{ url_for('static',
    filename='favicon.ico') }}">
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock
    %}</title>
  <style>
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */
    :root {
```

```
    --accent: #2c6b5a;                    /* 主色：青绿，千里江山石绿偏青 */
    --accent-hover: #1e5548;
    --accent-soft: #e6f0ec;
    --side-1: #1a4a4a;                    /* 侧边栏：深青墨 */
    --side-2: #0f3338;
    --danger: #a0392a;                    /* 朱砂红 */
    --danger-soft: #fbeae6;
    --ok: #2c6b5a;
    --ok-soft: #e6f0ec;
    --todo: #c08a3e;                      /* 赭石色待办 */
    --ink: #2a2520;                       /* 深褐墨色 */
    --ink-soft: #6b5d4f;                  /* 褐灰 */
    --line: #e0d8c8;                      /* 绢本边线 */
    --surface: #fdfaf2;                   /* 暖白绢本 */
    --canvas: #f5efe0;                    /* 绢本米黄背景 */
    --radius: 10px;
    --shadow: 0 2px 8px rgba(42, 37, 32, .06);
    --header-h: 60px;
    --nav-w: 226px;
}
* { box-sizing: border-box; }
html, body { margin: 0; padding: 0; }
body {
    font-family: "Microsoft YaHei", "PingFang SC", "Noto Serif SC", "SimSun", serif;
    font-size: 14px;
    line-height: 1.6;
    color: var(--ink);
    background: var(--canvas);
}
/* ---------- 跳到主要内容（键盘可达性） ---------- */
.skip-link {
    position: absolute;
    left: 12px;
    top: -60px;
    z-index: 30;
    padding: 8px 14px;
    background: var(--accent);
    color: #fff;
    border-radius: var(--radius);
    text-decoration: none;
    transition: top .15s ease-out;
}
.skip-link:focus { top: 12px; }
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ----------
   固定为一屏高，页面滚动交给右侧内容区，
   这样左侧导航栏（含底部用户区）不会随长页面一起被拉到下面去。 */
.app-shell {
    display: flex;
    align-items: stretch;
```

```
    height: 100vh;                                                    4451
    height: 100dvh;                                                   4452
    overflow: hidden;                                                 4453
}                                                                     4454
/* ---------- 侧边栏 ---------- */                                    4455
.app-side {                                                           4456
    flex: 0 0 var(--nav-w);                                           4457
    width: var(--nav-w);                                              4458
    display: flex;                                                    4459
    flex-direction: column;                                           4460
    background: linear-gradient(180deg, var(--side-1) 0%,            4461
var(--side-2) 100%);                                                  4462
    color: #fff;                                                      4463
}                                                                     4464
.app-brand {                                                          4465
    display: flex;                                                    4466
    align-items: center;                                              4467
    gap: 11px;                                                        4468
    padding: 17px 16px 15px;                                          4469
    border-bottom: 1px solid rgba(255, 255, 255, .10);                4470
}                                                                     4471
.app-brand__mark {                                                    4472
    flex: none;                                                       4473
    display: grid;                                                    4474
    place-items: center;                                              4475
    width: 36px;                                                      4476
    height: 36px;                                                     4477
    border-radius: 10px;                                              4478
    background: rgba(255, 255, 255, .15);                             4479
}                                                                     4480
.app-brand__name {                                                    4481
    font-size: 14.5px;                                                4482
    font-weight: 600;                                                 4483
    line-height: 1.35;                                                4484
    letter-spacing: .2px;                                             4485
}                                                                     4486
.app-brand__sub {                                                     4487
    font-size: 11.5px;                                                4488
    line-height: 1.35;                                                4489
    color: rgba(255, 255, 255, .58);                                  4490
}                                                                     4491
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y:               4492
auto; }                                                               4493
.app-nav__group {                                                     4494
    margin: 12px 0 5px;                                               4495
    padding: 0 10px;                                                  4496
    font-size: 11.5px;                                                4497
    letter-spacing: .05em;                                            4498
    color: rgba(255, 255, 255, .48);                                  4499
}                                                                     4500
```

```
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 9px 11px;
  margin-bottom: 2px;
  border-radius: 8px;
  color: rgba(255, 255, 255, .80);
  text-decoration: none;
  transition: background-color .16s ease, color .16s ease;
}
.app-nav a:hover { background: rgba(255, 255, 255, .10); color: #fff; }
.app-nav a:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
  background: rgba(255, 255, 255, .17);
  color: #fff;
  font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
  margin-left: auto;
  min-width: 20px;
  padding: 0 6px;
  font-size: 11px;
  font-weight: 600;
  line-height: 18px;
  text-align: center;
  border-radius: 9px;
  background: var(--todo);
  color: #fff;
  font-variant-numeric: tabular-nums;
}
.app-user {
  display: flex;
  align-items: center;
  gap: 8px;
  padding: 10px 14px;
  border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__me {
  display: flex;
  flex: 1;
  min-width: 0;
  align-items: center;
```

```
  gap: 10px;
  padding: 4px 6px;
  border-radius: 8px;
  color: inherit;
  text-decoration: none;
  transition: background-color .16s ease;
}
.app-user__me:hover { background: rgba(255, 255, 255, .10); }
.app-user__me:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-user__avatar {
  flex: none;
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  border-radius: 50%;
  background: rgba(255, 255, 255, .16);
  font-size: 13px;
  font-weight: 600;
  object-fit: cover;
}
.app-user__name {
  display: block;
  font-size: 13px;
  line-height: 1.3;
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}
.app-user__state {
  display: flex;
  align-items: center;
  gap: 5px;
  font-size: 11.5px;
  line-height: 1.3;
  color: rgba(255, 255, 255, .60);
}
.app-user__dot {
  width: 6px;
  height: 6px;
  border-radius: 50%;
  background: #6ed49a;
}
.app-user__out {
  flex: none;
  padding: 6px;
```

```
  border-radius: 6px;                                                    4601
  color: rgba(255, 255, 255, .68);                                       4602
  text-decoration: none;                                                 4603
}                                                                        4604
.app-user__out:hover { color: #fff; background:                          4605
rgba(255, 255, 255, .12); }                                              4606
/* ---------- 右侧内容区 ----------                                       4607
   整个骨架固定一屏高，滚动只发生在内容区内部：                           4608
   左侧导航栏（含底部用户区）始终留在原位，右侧页面自己滚。 */            4609
.app-content {                                                           4610
  flex: 1;                                                               4611
  min-width: 0;                                                          4612
  display: flex;                                                         4613
  flex-direction: column;                                                4614
  overflow-y: auto;                                                      4615
  overflow-x: hidden;                                                    4616
  /* 细滚动条：默认半透明，悬停时加深，不抢视线 */                       4617
  scrollbar-width: thin;                                                 4618
  scrollbar-color: rgba(31, 107, 69, .28) transparent;                   4619
}                                                                        4620
.app-content::-webkit-scrollbar { width: 12px; }                         4621
.app-content::-webkit-scrollbar-track { background:                      4622
transparent; }                                                           4623
.app-content::-webkit-scrollbar-thumb {                                  4624
  background: rgba(31, 107, 69, .26);                                    4625
  border: 3px solid transparent;                                         4626
  background-clip: content-box;                                          4627
  border-radius: 999px;                                                  4628
}                                                                        4629
.app-content::-webkit-scrollbar-thumb:hover {                            4630
  background: rgba(31, 107, 69, .46);                                    4631
  background-clip: content-box;                                          4632
}                                                                        4633
/* ---------- 顶栏：面包屑 + 搜索 ----------                             4634
   粘在内容区顶部，滚动时面包屑与搜索框始终可用 */                        4635
.app-header {                                                            4636
  position: sticky;                                                      4637
  top: 0;                                                                4638
  z-index: 20;                                                           4639
  display: flex;                                                         4640
  align-items: center;                                                   4641
  gap: 14px;                                                             4642
  height: var(--header-h);                                               4643
  padding: 0 22px;                                                       4644
  background: var(--surface);                                            4645
  border-bottom: 1px solid var(--line);                                  4646
}                                                                        4647
.crumb {                                                                 4648
  display: flex;                                                         4649
  align-items: center;                                                   4650
```

```
    gap: 7px;
    font-size: 13px;
    color: var(--ink-soft);
    white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
    display: flex;
    align-items: center;
    gap: 8px;
    width: 268px;
    padding: 7px 12px;
    background: var(--canvas);
    border: 1px solid var(--line);
    border-radius: 9px;
    color: var(--ink-soft);
    transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
    flex: 1;
    min-width: 0;
    border: none;
    outline: none;
    background: transparent;
    font-family: inherit;
    font-size: 13px;
    color: var(--ink);
}
.search-box button {
    display: grid;
    place-items: center;
    padding: 0;
    border: none;
    background: transparent;
    color: inherit;
    cursor: pointer;
}
.icon-btn {
    display: grid;
    place-items: center;
    width: 34px;
    height: 34px;
```

```
    border-radius: 9px;
    border: 1px solid transparent;
    color: var(--ink-soft);
    text-decoration: none;
    transition: background-color .16s ease, color .16s ease,
                border-color .16s ease;
}
.icon-btn:hover {
    color: var(--accent);
    background: var(--accent-soft);
    border-color: rgba(31, 107, 69, .18);
}
.app-main {
    flex: 1;
    min-width: 0;
    padding: 22px 26px 26px;
}
.container { max-width: none; width: 100%; }
/* ---------- 页脚 ---------- */
.app-foot {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 14px;
    padding: 0 26px 18px;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 未登录时的裸页面（登录页） ---------- */
.login-shell {
    display: flex;
    flex-direction: column;
    min-height: 100dvh;
}
/* ---------- 卡片 ---------- */
.card {
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    padding: 22px 24px;
    margin-bottom: 18px;
}
.card > h2 {
    margin: 0 0 14px;
    font-size: 16px;
    font-weight: 600;
    color: var(--accent);
    text-wrap: balance;
```

```
}
.card > h3 {
  margin: 20px 0 10px;
  font-size: 14px;
  font-weight: 600;
  color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
  width: 100%;
  border-collapse: collapse;
  margin: 10px 0 16px;
  font-variant-numeric: tabular-nums;
}
caption {
  text-align: left;
  color: var(--ink-soft);
  font-size: 12px;
  padding-bottom: 6px;
}
th, td {
  border: 1px solid var(--line);
  padding: 8px 12px;
  text-align: center;
  vertical-align: middle;
}
th {
  background: var(--accent-soft);
  font-weight: 600;
  white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
  display: block;
  margin-bottom: 4px;
  color: var(--ink-soft);
  font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
  padding: 6px 9px;
  font-family: inherit;
  font-size: 14px;
  color: var(--ink);
  background: #fff;
  border: 1px solid #c6ccd4;
```

```
    border-radius: 4px;                                                    4801
    max-width: 100%;                                                       4802
}                                                                          4803
input[type="number"] {                                                     4804
    width: 104px;                                                          4805
    text-align: center;                                                    4806
    font-variant-numeric: tabular-nums;                                    4807
}                                                                          4808
textarea { width: 100%; min-height: 64px; resize:                          4809
vertical; }                                                                4810
input:focus-visible, select:focus-visible,                                 4811
textarea:focus-visible {                                                   4812
    outline: 2px solid var(--accent);                                      4813
    outline-offset: 1px;                                                   4814
    border-color: var(--accent);                                           4815
}                                                                          4816
.checkbox {                                                                4817
    display: inline-flex;                                                  4818
    align-items: center;                                                   4819
    gap: 7px;                                                              4820
    cursor: pointer;                                                       4821
    user-select: none;                                                     4822
}                                                                          4823
.checkbox input { width: 16px; height: 16px; margin: 0;                    4824
cursor: pointer; }                                                         4825
/* ---------- 按钮 ----------                                             4826
    统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */                4827
.btn {                                                                     4828
    display: inline-flex;                                                  4829
    align-items: center;                                                   4830
    justify-content: center;                                               4831
    gap: 6px;                                                              4832
    padding: 8px 18px;                                                     4833
    font-family: inherit;                                                  4834
    font-size: 14px;                                                       4835
    line-height: 1.4;                                                      4836
    color: #fff;                                                           4837
    background: rgba(31, 107, 69, .80);                                    4838
    border: 1px solid rgba(31, 107, 69, .30);                              4839
    border-radius: 8px;                                                    4840
    cursor: pointer;                                                       4841
    text-decoration: none;                                                 4842
    transition: background-color .16s ease, border-color                   4843
.16s ease,                                                                 4844
                color .16s ease, box-shadow .16s ease,                     4845
transform .16s ease;                                                       4846
}                                                                          4847
.btn:hover {                                                               4848
    background: var(--accent);                                             4849
    border-color: var(--accent);                                           4850
```

```
  box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow: none; }
.btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.btn--ghost {
  color: var(--accent);
  background: rgba(31, 107, 69, .06);
  border-color: rgba(31, 107, 69, .22);
}
.btn--ghost:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
}
.btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .26);
}
.btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 3px 12px rgba(179, 38, 30, .20);
}
.btn--sm { padding: 5px 13px; font-size: 13px; border-radius: 7px; }
.btn:disabled { opacity: .45; cursor: not-allowed; }
.btn:disabled:hover {
  background: rgba(31, 107, 69, .80);
  border-color: rgba(31, 107, 69, .30);
  box-shadow: none;
}
.btn--ghost:disabled:hover { background: rgba(31, 107, 69, .06); color: var(--accent); }
/* ---------- 表格操作列的行内小按钮 ---------- */
   取代原来「文字 | 文字 | 文字」的写法：
   圆角矩形 + 半透明底色，悬停时变实色。 */
.row-actions {
  display: inline-flex;
  flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */
  align-items: center;
  gap: 6px;
  white-space: nowrap;
}
.row-actions form { display: inline-flex; margin: 0; }
.link-btn {
  display: inline-flex;
```

```
    align-items: center;
    padding: 4px 11px;
    font-family: inherit;
    font-size: 13px;
    line-height: 1.5;
    color: var(--accent);
    background: rgba(31, 107, 69, .07);
    border: 1px solid rgba(31, 107, 69, .20);
    border-radius: 7px;
    cursor: pointer;
    text-decoration: none;
    white-space: nowrap;
    transition: background-color .16s ease, border-color .16s ease,
                color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.link-btn:hover {
    color: #fff;
    background: var(--accent);
    border-color: var(--accent);
    box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
}
.link-btn:active { transform: translateY(1px); box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.link-btn--danger {
    color: var(--danger);
    background: rgba(179, 38, 30, .07);
    border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
    color: #fff;
    background: var(--danger);
    border-color: var(--danger);
    box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
    padding: 10px 14px;
    margin-bottom: 14px;
    border-radius: var(--radius);
    border-left: 4px solid;
}
.flash.error {
    color: var(--danger);
```

```
    background: var(--danger-soft);
    border-left-color: var(--danger);
}
.flash.ok {
    color: var(--ok);
    background: var(--ok-soft);
    border-left-color: var(--ok);
}
/* ---------- 浮层提示（登录成功、各类操作完成、登录失败） ---------- */
   外层 toast-stack 固定在页面中上方，内层每条各自淡入 → 停留 → 淡出，
   多条提示会依次排开而不重叠。
   底色是浅绿半透明：既有一层底色把文字托住，又能透出后面的内容，
   不用高饱和色块，提示不该比页面内容更抢眼。
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
    position: fixed;
    top: 16px;
    left: 50%;
    z-index: 100;
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 6px;
    transform: translateX(-50%);
    pointer-events: none;
}
.toast-center {
   max-width: min(620px, 82vw);
    padding: 8px 22px;
    border-radius: 999px;
    /* 浅绿、半透明：不遮住背后的内容，也比纯透明更托得住文字 */
    background: rgba(223, 241, 231, .76);
    border: 1px solid rgba(31, 107, 69, .14);
    color: #2f4a3c;
    font-size: 16px;
    font-weight: 500;
    line-height: 1.65;
    letter-spacing: .03em;
    text-align: center;
    text-wrap: pretty;
    backdrop-filter: blur(6px) saturate(1.1);
    -webkit-backdrop-filter: blur(6px) saturate(1.1);
    box-shadow: 0 4px 16px rgba(23, 32, 27, .07);
    animation: toast-fade 2.6s ease-in-out forwards;
}
/* 失败类提示：同一套观感，换成浅红底，一眼能区分成败 */
.toast-center--error {
    background: rgba(252, 233, 230, .78);
    border-color: rgba(163, 43, 34, .16);
    color: #9c2b23;
```

```
}
@keyframes toast-fade {
  0%        { opacity: 0; transform: translateY(-6px); }
  14%, 74%  { opacity: 1; transform: none; }
  100%      { opacity: 0; transform: none; }
}
/* 系统设置里要求「减少动态效果」时，去掉位移，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
  .toast-center { animation: toast-fade-plain 2.6s
linear forwards; }
  @keyframes toast-fade-plain {
    0%        { opacity: 0; }
    14%, 74%  { opacity: 1; }
    100%      { opacity: 0; }
  }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 12.5px; }
.mono { font-family: Consolas, "Courier New",
monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
  display: inline-block;
  padding: 1px 9px;
  font-size: 12.5px;
  border-radius: 10px;
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background:
var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background:
var(--danger-soft); border-color: #eec4c0; }
.tag--muted { color: var(--ink-soft); background:
#f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */
.body--login {
  background:
    radial-gradient(1100px 640px at 50% 8%,
```

```
                    #edf6f1 0%, rgba(237, 246, 241, 0) 72%),
      linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);
}
.body--login .app-main { padding: 0; }
.body--login .container { max-width: none; width: 100%; }
/* 仅供读屏软件的隐藏文本 */
.visually-hidden {
  position: absolute !important;
  width: 1px;
  height: 1px;
  margin: -1px;
  padding: 0;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap;
  border: 0;
}
/* ---------- 工作台：页首横幅 ---------- */
.hero {
  position: relative;
  overflow: hidden;
  min-height: 176px;
  padding: 26px 28px;
  margin-bottom: 16px;
  border-radius: 12px;
  /* 左侧压深保证文字可读，右侧透出远山插画；
     背景按底部对齐，避免远山被居中裁掉 */
  background:
    linear-gradient(100deg, rgba(15, 40, 45, .90) 0%,
                            rgba(20, 55, 58, .55) 45%,
                            rgba(20, 55, 58, .12) 100%),
    url("{{ url_for('static', filename='hero_landscape.jpg') }}")
    center center / cover no-repeat;
  color: #fff;
}
.hero__body { max-width: 720px; }
.hero__title {
  margin: 0 0 10px;
  font-size: 21px;
  font-weight: 700;
  line-height: 1.4;
  letter-spacing: .3px;
}
.hero__desc {
  margin: 0;
  max-width: 640px;
  font-size: 12.5px;
```

```
    line-height: 1.85;                                                    5101
    color: rgba(255, 255, 255, .86);                                      5102
}                                                                         5103
.hero__actions {                                                          5104
    display: flex;                                                        5105
    flex-wrap: wrap;                                                      5106
    gap: 10px;                                                            5107
    margin-top: 18px;                                                     5108
}                                                                         5109
.hero-btn {                                                               5110
    display: inline-flex;                                                 5111
    align-items: center;                                                  5112
    gap: 6px;                                                             5113
    padding: 7px 15px;                                                    5114
    font-size: 13px;                                                      5115
    border-radius: 8px;                                                   5116
    text-decoration: none;                                                5117
    transition: background-color .16s ease, border-color                  5118
.16s ease,                                                                5119
                color .16s ease;                                          5120
}                                                                         5121
.hero-btn--solid {                                                        5122
    color: #14432f;                                                       5123
    background: rgba(255, 255, 255, .92);                                 5124
    border: 1px solid rgba(255, 255, 255, .92);                           5125
}                                                                         5126
.hero-btn--solid:hover { background: #fff; }                              5127
.hero-btn--ghost {                                                        5128
    color: #fff;                                                          5129
    background: rgba(255, 255, 255, .14);                                 5130
    border: 1px solid rgba(255, 255, 255, .45);                           5131
}                                                                         5132
.hero-btn--ghost:hover { background: rgba(255, 255,                       5133
255, .26); }                                                              5134
.hero-btn:focus-visible { outline: 2px solid #fff;                        5135
outline-offset: 2px; }                                                    5136
/* ---------- 工作台：数据概览 ---------- */                               5137
.stat-cards {                                                             5138
    display: grid;                                                        5139
    grid-template-columns: repeat(auto-fit, minmax(190px,                 5140
1fr));                                                                    5141
    gap: 12px;                                                            5142
    margin-bottom: 22px;                                                  5143
}                                                                         5144
.stat-card {                                                              5145
    display: block;                                                       5146
    padding: 15px 17px;                                                   5147
    background: var(--surface);                                           5148
    border: 1px solid var(--line);                                        5149
    border-radius: var(--radius);                                         5150
```

```
    box-shadow: var(--shadow);
    color: var(--ink);
    text-decoration: none;
    transition: border-color .16s ease, box-shadow .16s ease;
}
.stat-card:hover {
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
    display: block;
    font-size: 12.5px;
    color: var(--ink-soft);
}
.stat-card__value {
    display: block;
    margin: 3px 0 2px;
    font-size: 26px;
    font-weight: 700;
    line-height: 1.25;
    color: var(--accent);
    font-variant-numeric: tabular-nums;
}
.stat-card__unit {
    margin-left: 3px;
    font-size: 13px;
    font-weight: 500;
    color: var(--ink-soft);
}
.stat-card__note {
    display: block;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
    gap: 6px 14px;
    margin-bottom: 12px;
}
.section__title { margin: 0; font-size: 16px; }
.section__desc { margin: 0; font-size: 12.5px; color: var(--ink-soft); }
/* ---------- 工作台：流程步骤卡 ---------- */
.step-grid {
    display: grid;
```

```
  grid-template-columns: repeat(auto-fill,
minmax(238px, 1fr));
  gap: 12px;
}
.step-card {
  position: relative;
  display: block;
  padding: 16px 16px 15px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s
ease,
              transform .16s ease;
}
.step-card:hover {
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 5px 16px rgba(19, 60, 43, .10);
  transform: translateY(-1px);
}
.step-card:focus-visible { outline: 2px solid
var(--accent); outline-offset: 2px; }
.step-card__no {
  position: absolute;
  top: 13px;
  right: 15px;
  font-size: 17px;
  font-weight: 700;
  color: #d7e3db;
  font-variant-numeric: tabular-nums;
}
.step-card__icon {
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  margin-bottom: 10px;
  border-radius: 9px;
  background: var(--accent-soft);
  color: var(--accent);
}
.step-card__title {
  display: block;
  margin-bottom: 5px;
  font-size: 14px;
  font-weight: 600;
  color: var(--accent);
```

```
}
.step-card__desc {
  display: block;
  font-size: 12.5px;
  line-height: 1.65;
  color: var(--ink-soft);
  text-wrap: pretty;
}
.step-card__badge {
  display: inline-block;
  margin-top: 10px;
  padding: 1px 9px;
  font-size: 11.5px;
  border-radius: 9px;
  background: #fdf1e2;
  color: #9a5a12;
  border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
  background: #fafbfa;
  border-color: #eceff0;
  box-shadow: none;
  cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
  background: #f1f3f2;
  color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
  display: inline-block;
  margin-top: 10px;
  padding: 1px 9px;
  font-size: 11.5px;
  border-radius: 9px;
  background: #f1f3f2;
  color: #8b948f;
}
/* ---------- 结果摘要 ---------- */
.stat-row {
  display: flex;
  flex-wrap: wrap;
  gap: 28px;
  margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
var(--ink-soft); }
```

```
.stat__value { font-size: 20px; font-weight: 700;                    5301
font-variant-numeric: tabular-nums; }                                5302
/* ---------- 流程步骤条 ---------- */                                5303
.steps { display: flex; flex-wrap: wrap; gap: 8px;                   5304
margin: 4px 0 18px; }                                                5305
.step {                                                              5306
  display: flex; align-items: center; gap: 7px;                      5307
  padding: 8px 12px; font-size: 13px;                                5308
  background: #fff; border: 1px solid var(--line);                   5309
  border-radius: var(--radius);                                      5310
}                                                                    5311
.step__no {                                                          5312
  display: inline-flex; align-items: center;                         5313
justify-content: center;                                             5314
  width: 20px; height: 20px; border-radius: 50%;                     5315
  background: #eef1f5; color: var(--ink-soft);                       5316
  font-size: 12px; font-variant-numeric: tabular-nums;               5317
}                                                                    5318
.step__name { font-weight: 600; }                                    5319
.step__state { font-size: 12.5px; color:                             5320
var(--ink-soft); }                                                   5321
.step--done { border-color: #b8ddc6; background:                     5322
var(--ok-soft); }                                                    5323
.step--done .step__no { background: var(--ok); color:                5324
#fff; }                                                              5325
.step--done .step__state { color: var(--ok); }                       5326
.step--fail { border-color: #eec4c0; background:                     5327
var(--danger-soft); }                                                5328
.step--fail .step__no { background: var(--danger);                   5329
color: #fff; }                                                       5330
.step--fail .step__state { color: var(--danger);                     5331
font-weight: 600; }                                                  5332
.step--next { border-color: var(--accent); }                         5333
.step--next .step__no { background: var(--accent);                   5334
color: #fff; }                                                       5335
/* ---------- 提示块（中性底色，不用整块警示色压页面） ---------- */  5336
.notice {                                                            5337
  padding: 14px 16px; margin: 0 0 16px;                              5338
  background: var(--surface);                                        5339
  border: 1px solid var(--line);                                     5340
  border-left: 3px solid var(--todo);                                5341
  border-radius: var(--radius);                                      5342
}                                                                    5343
.notice__title { margin: 0 0 8px; font-weight: 600; }                5344
/* ---------- 前置条件表（与页面主表同款，只把待处理的行标成浅红） ---------- */ 5345
.prereq-table { margin: 0 0 18px; }                                  5346
.prereq-table > caption { padding-bottom: 7px; }                     5347
.prereq-row > td { background: var(--danger-soft); }                 5348
.prereq-row:hover > td { background: #f6dcd7; }                      5349
.prereq-flag { color: var(--danger); font-weight: 600; }             5350
```

```
.prereq-row .muted { color: #9a6b62; }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
  border: 1px solid var(--line);
  border-radius: var(--radius);
  background: var(--surface);
  padding: 14px 16px;
  margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin: 0 0 12px;
  padding-bottom: 9px;
  border-bottom: 1px solid var(--line);
}
.patch-panel__name {
  margin: 0;
  font-size: 15px;
  font-weight: 600;
}
.patch-panel__meta {
  font-size: 12.5px;
  color: var(--ink-soft);
}
.patch-panel__meta a { color: var(--accent); }
.patch-panel__tail { margin-left: auto; }
.patch-panel .notice { margin: 0; }
.patch-panel table { margin: 0; }
/* ---------- 窄屏适配 ---------- */
@media (max-width: 960px) {
  /* 上下布局时取消「一屏高 + 内容区内部滚动」，
     否则导航会占掉半屏、内容被挤在剩下的小格子里滚动 */
  .app-shell {
    flex-direction: column;
    height: auto;
    min-height: 100vh;
    min-height: 100dvh;
    overflow: visible;
  }
  .app-content { overflow: visible; }
  .app-header { position: static; }
  .app-side { width: auto; flex: none; }
  .app-brand { padding: 12px 14px; }
  .app-nav { padding: 4px 10px 10px; }
  .app-nav ul { display: flex; flex-wrap: wrap; gap:
4px; }
```

```
        .app-nav a { margin-bottom: 0; }
        .app-nav__group { margin: 8px 0 4px; }
        .app-user { padding: 10px 14px; }
        .app-header { padding: 0 14px; gap: 10px; }
        .search-box { width: 180px; }
        .app-main { padding: 16px 14px 20px; }
        .app-foot { padding: 0 14px 14px; }
      }
      @media (max-width: 640px) {
        .crumb { display: none; }
        .search-box { width: auto; flex: 1; }
      }
    </style>
</head>
<body class="{% if not current_user.is_authenticated %}body--login{% endif %}">
    <a class="skip-link" href="#main">跳到主要内容</a>
    {# 取一次即可（同一请求内重复调用返回相同内容）。
       类别为 toast / ok / toast-error 的渲染成中上方浮层，出现后自动淡出；
       其余仍是页面顶部的提示条 ——
       需要对照着改表单的错误，留在页面上比一闪而过更有用。 #}
    {% set flash_msgs = get_flashed_messages(with_categories=true) %}
    <div class="toast-stack">
      {% for category, message in flash_msgs if category in ("toast", "ok", "toast-error") %}
      <div class="toast-center{% if category == 'toast-error' %} toast-center--error{% endif %}"
           role="status" aria-live="polite">{{ message }}</div>
      {% endfor %}
    </div>
    {% if current_user.is_authenticated %}
    <div class="app-shell">
      <aside class="app-side">
        <div class="app-brand">
          <span class="app-brand__mark">{{ ui.icon("mountain", 21) }}</span>
          <span>
            <span class="app-brand__name">矿山自然恢复评价</span><br>
            <span class="app-brand__sub">适宜性评价系统 V1.0</span>
          </span>
        </div>
        <nav class="app-nav" aria-label="功能导航">
          <ul>
            <li>
              <a href="/"{% if home_active %}
    aria-current="page"{% endif %}>
                {{ ui.icon("home") }}<span>工作台</span>
              </a>
```

```
        </li>
      </ul>
      {% for g in nav_groups %}
      <p class="app-nav__group">{{ g.title }}</p>
      <ul>
        {% for item in g.links %}
        <li>
          <a href="{{ item.href }}"{% if item.active %}
aria-current="page"{% endif %}>
            {{ ui.icon(item.icon) }}<span>{{ item.label
}}</span>
            {% if item.badge %}<span
class="app-nav__badge">{{ item.badge }}</span>{% endif
%}
          </a>
        </li>
        {% endfor %}
      </ul>
      {% endfor %}
    </nav>
    <div class="app-user">
      <a class="app-user__me" href="/profile"
title="个人设置">
        {% if current_user.avatar %}
        <img class="app-user__avatar"
             src="{{ url_for('static',
filename=current_user.avatar) }}" alt="">
        {% else %}
        <span class="app-user__avatar"
aria-hidden="true">{{ current_user.initial }}</span>
        {% endif %}
        <span style="min-width:0;">
          <span class="app-user__name">{{
current_user.shown_name }}</span>
          <span class="app-user__state">
            <span class="app-user__dot"
aria-hidden="true"></span>
            {{ role_labels.get(current_user.role,
current_user.role) }}
          </span>
        </span>
      </a>
      <a class="app-user__out" href="/logout" title="退出登录"
         aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>
    </div>
  </aside>
  <div class="app-content">
    <header class="app-header">
      <nav class="crumb" aria-label="当前位置">
        <a href="/">首页</a>
```

```
            {{ ui.icon("chevron", 13) }}                                    5501
            <strong>{{ page_title }}</strong>                                5502
          </nav>                                                            5503
          <span class="app-header__spacer"></span>                          5504
          <form class="search-box" method="get"                             5505
    action="/search" role="search">                                         5506
            {{ ui.icon("search", 16) }}                                     5507
            <label for="q" class="visually-hidden">搜索项目、图斑、标准<    5508
    /label>                                                                 5509
            <input id="q" type="search" name="q"                            5510
    autocomplete="off"                                                      5511
                   placeholder="搜索项目、图斑、标准" value="{{             5512
    search_q }}">                                                           5513
            <button type="submit" aria-label="搜索">{{                      5514
    ui.icon("chevron", 15) }}</button>                                      5515
          </form>                                                           5516
          {% if current_user.is_admin %}                                    5517
          <a class="icon-btn" href="/profile?tab=logs"                      5518
    title="操作日志"                                                        5519
             aria-label="操作日志">{{ ui.icon("bell", 18) }}</a>            5520
          {% endif %}                                                       5521
        </header>                                                           5522
  {% else %}                                                                5523
  <div class="login-shell">                                                 5524
  {% endif %}                                                               5525
    <main id="main" class="app-main">                                       5526
      <div class="container">                                               5527
        {% for category, message in flash_msgs if category                  5528
    not in ("toast", "ok", "toast-error") %}                                5529
        <p class="flash {{ category }}" role="status">{{                    5530
    message }}</p>                                                          5531
        {% endfor %}                                                        5532
        {% block content %}{% endblock %}                                   5533
      </div>                                                                5534
    </main>                                                                 5535
  {% if current_user.is_authenticated %}                                    5536
      <footer class="app-foot">                                             5537
        <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０</span>          5538
        <span>数据截止 {{ today }}</span>                                   5539
      </footer>                                                             5540
    </div>                                                                  5541
  </div>                                                                    5542
  {% else %}                                                                5543
  </div>                                                                    5544
  {% endif %}                                                               5545
  {% block scripts %}{% endblock %}                                         5546
</body>                                                                     5547
</html>                                                                     5548
# ==== app/templates/_macros.html ====                                      5549
{# 共用宏：图标、流程状态面板 #}                                           5550
```

```
{#  ================================================================== 5551
      ================== 5552
    图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走） 5553
    不引图标字体或外部图片，避免额外请求与资源依赖； 5554
    stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。 5555
    ================================================================== 5556
      ================== #} 5557
{% set ICONS = { 5558
  "dot": '<circle cx="12" cy="12" r="3"/>', 5559
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2 5560
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>', 5561
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2 5562
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>', 5563
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8 5564
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4 5565
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>', 5566
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6 5567
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>', 5568
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path 5569
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>', 5570
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20 5571
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5 5572
    17h5"/>', 5573
  "database": '<ellipse cx="12" cy="5.5" rx="8" 5574
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4 5575
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4 5576
    8-3.2"/>', 5577
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12" 5578
    cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>', 5579
  "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0 5580
    2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14 5581
    2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>', 5582
  "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5 5583
    20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6 5584
    3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path 5585
    d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>', 5586
  "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5 5587
    21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4 5588
    4.4V21"/>', 5589
  "mail": '<rect x="3" y="5" width="18" height="14" 5590
    rx="2.5"/><path d="m3.8 7 8.2 5.8L20.2 7"/>', 5591
  "phone": '<path d="M6.4 3h2.8l1.5 4.4-2 1.4a12.4 12.4 0 0 5592
    0 6.5 6.5l1.4-2L21 14.8v2.8a2.4 2.4 0 0 1-2.6 2.4A16.6 5593
    16.6 0 0 1 3.5 5.6 2.4 2.4 0 0 1 5.9 3z"/>', 5594
  "lock": '<rect x="4" y="10.4" width="16" height="10.6" 5595
    rx="2.6"/><path d="M8 10.4V7.6a4 4 0 0 1 8 0v2.8"/>', 5596
  "briefcase": '<rect x="3" y="7.4" width="18" 5597
    height="13.1" rx="2.6"/><path d="M8.4 7.4V6A2 2 0 0 1 5598
    10.4 4h3.2a2 2 0 0 1 2 2v1.4"/><path d="M3 12.6h18"/>', 5599
  "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5 5600
```

```
      20.5-4-4"/>',
    "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
      6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
      1-3.6 0"/>',
    "plus": '<path d="M12 5v14M5 12h14"/>',
    "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
      4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
      21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
    "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
      2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
      7.5"/><path d="M5.5 12H15"/>',
    "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
    "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
      10v4.2"/><path d="M12 17.4h.01"/>',
    "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
      20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
     fill="none" stroke="currentColor" stroke-width="1.7"
     stroke-linecap="round" stroke-linejoin="round"
     aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
      {% elif progress.constraint.done %}未通过
      {% else %}待完成{% endif %}
    </span>
  </div>
  <div class="step {% if progress.units.done %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">2</span>
    <span class="step__name">划分微地质单元</span>
    <span class="step__state">
      {% if progress.units.done %}已完成{% else %}待完成{% endif
    %}
    </span>
```

```
  </div>
  <div class="step {% if progress.data.ready %}step--done{% else %}step--todo{% endif %}">
    <span class="step__no">3</span>
    <span class="step__name">录入指标数据</span>
    <span class="step__state">{% if progress.data.ready %}已完成{% else %}待完成{% endif %}</span>
  </div>
  <div class="step step--next">
    <span class="step__no">4</span>
    <span class="step__name">执行适宜性评价</span>
    <span class="step__state">当前步骤</span>
  </div>
</div>
{% endmacro %}
{# 前置条件未满足时的提示：做成与页面主表同款的表格，只把待处理的那几行标成浅红，
   说明写在 caption 里，缺什么、去哪补都在行内。 #}
{% macro workflow_blocking(progress) %}
{% if progress.blocking %}
<table class="prereq-table">
  <caption>本步骤的前置条件尚未满足，按论文 4.1 节的流程顺序先补上：</caption>
  <thead>
    <tr>
      <th scope="col">前置步骤</th>
      <th scope="col">说明</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for b in progress.blocking %}
    <tr class="prereq-row">
      <td>{{ b.label }}</td>
      <td class="muted">{{ b.reason }}</td>
      <td><a class="btn btn--sm" href="{{ b.fix_url }}">{{ b.fix_label }}</a></td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label }}</span>
```

```
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
   只在流程里**第一个**按图斑操作的页面（限制性指标判定）列一次，后面各页
   不再重复；建好图斑才会出现在下面的表里，但离「能进入本页流程」还差别的步骤，
   所以措辞只说到「出现在这里」为止。 #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
<table class="prereq-table">
  <caption>以下项目下尚未建立矿山图斑，本页按图斑组织数据，建好图斑后才会出现在这里：</caption>
  <thead>
    <tr>
      <th scope="col">所属项目</th>
      <th scope="col">矿山图斑</th>
      <th scope="col">状态</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for p in projects_without_patch %}
    <tr class="prereq-row">
      <td>项目「{{ p.name }}」</td>
      <td class="muted">未建立</td>
      <td><span class="prereq-flag">缺少矿山图斑</span></td>
      <td>
        <a class="btn btn--sm" href="/project/{{ p.id
    }}/patch/add">
          {{ icon("plus", 14) }}去新建图斑
        </a>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endif %}
{% endmacro %}
# ==== app/templates/_login_badge.svg ====
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg
    " viewBox="0 0 240 240" role="img"
    aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">
  <title>系统徽标</title>
  <g class="lp-ring" style="--k: 2.7">
    <circle cx="162.00" cy="120.00" r="2.60"
    fill="#c4f5ab"/>
    <circle cx="157.84" cy="138.22" r="2.97"
    fill="#b8f5ab"/>
    <circle cx="146.19" cy="152.84" r="3.06"
    fill="#acf5ab"/>
    <circle cx="129.35" cy="160.95" r="2.80"
```

```
    fill="#abf5b7" />                                                    5751
    <circle cx="110.65" cy="160.95" r="2.40"                             5752
    fill="#abf5c2" />                                                    5753
    <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce" />            5754
    <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da" />            5755
    <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6" />            5756
    <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da" />            5757
    <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce" />             5758
    <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2" />            5759
    <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7" />            5760
    <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab" />            5761
    <circle cx="157.84" cy="101.78" r="2.23"                             5762
    fill="#b8f5ab" />                                                    5763
  </g>                                                                   5764
  <g class="lp-ring" style="--k: 2.15">                                  5765
    <circle cx="177.29" cy="129.07" r="3.92"                             5766
    fill="#a4e28d" />                                                    5767
    <circle cx="171.68" cy="146.33" r="4.01"                             5768
    fill="#9be28d" />                                                    5769
    <circle cx="161.01" cy="161.01" r="3.87"                             5770
    fill="#91e28d" />                                                    5771
    <circle cx="146.33" cy="171.68" r="3.55"                             5772
    fill="#8de292" />                                                    5773
    <circle cx="129.07" cy="177.29" r="3.17"                             5774
    fill="#8de29c" />                                                    5775
    <circle cx="110.93" cy="177.29" r="2.88"                             5776
    fill="#8de2a6" />                                                    5777
    <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af" />            5778
    <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9" />            5779
    <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2" />            5780
    <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc" />            5781
    <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc" />            5782
    <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2" />             5783
    <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9" />             5784
    <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af" />             5785
    <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6" />            5786
    <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c" />            5787
    <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292" />            5788
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d" />            5789
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d" />            5790
    <circle cx="177.29" cy="110.93" r="3.63"                             5791
    fill="#a4e28d" />                                                    5792
  </g>                                                                   5793
  <g class="lp-ring" style="--k: 1.85">                                  5794
    <circle cx="194.00" cy="120.00" r="4.94"                             5795
    fill="#91cf72" />                                                    5796
    <circle cx="191.85" cy="137.71" r="4.92"                             5797
    fill="#88cf72" />                                                    5798
    <circle cx="185.52" cy="154.39" r="4.73"                             5799
    fill="#80cf72" />                                                    5800
```

```
    <circle cx="175.39" cy="169.07" r="4.42"                          5801
    fill="#78cf72" />                                                 5802
    <circle cx="162.04" cy="180.90" r="4.06"                          5803
    fill="#72cf73" />                                                 5804
    <circle cx="146.24" cy="189.19" r="3.73"                          5805
    fill="#72cf7b" />                                                 5806
    <circle cx="128.92" cy="193.46" r="3.51"                          5807
    fill="#72cf83" />                                                 5808
    <circle cx="111.08" cy="193.46" r="3.45"                          5809
    fill="#72cf8b" />                                                 5810
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93" />         5811
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c" />         5812
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4" />         5813
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac" />         5814
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4" />         5815
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc" />         5816
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4" />         5817
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac" />          5818
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4" />          5819
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c" />          5820
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93" />          5821
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b" />         5822
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83" />         5823
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b" />         5824
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73" />         5825
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72" />         5826
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72" />         5827
    <circle cx="191.85" cy="102.29" r="4.80"                          5828
    fill="#88cf72" />                                                 5829
  </g>                                                                5830
  <g class="lp-ring" style="--k: 1.62">                               5831
    <circle cx="209.57" cy="128.82" r="5.68"                          5832
    fill="#76bb59" />                                                 5833
    <circle cx="206.12" cy="146.13" r="5.40"                          5834
    fill="#6fbb59" />                                                 5835
    <circle cx="199.37" cy="162.43" r="5.06"                          5836
    fill="#68bb59" />                                                 5837
    <circle cx="189.57" cy="177.10" r="4.71"                          5838
    fill="#61bb59" />                                                 5839
    <circle cx="177.10" cy="189.57" r="4.41"                          5840
    fill="#5abb59" />                                                 5841
    <circle cx="162.43" cy="199.37" r="4.19"                          5842
    fill="#59bb5f" />                                                 5843
    <circle cx="146.13" cy="206.12" r="4.10"                          5844
    fill="#59bb66" />                                                 5845
    <circle cx="128.82" cy="209.57" r="4.15"                          5846
    fill="#59bb6d" />                                                 5847
    <circle cx="111.18" cy="209.57" r="4.32"                          5848
    fill="#59bb73" />                                                 5849
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a" />         5850
```

```
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81" />
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88" />
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f" />
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96" />
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d" />
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4" />
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4" />
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d" />
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96" />
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f" />
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88" />
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81" />
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a" />
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73" />
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d" />
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66" />
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f" />
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59" />
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59" />
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59" />
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59" />
    <circle cx="209.57" cy="111.18" r="5.85"
    fill="#76bb59" />
</g>
<g class="lp-ring" style="--k: 1.46">
    <circle cx="226.00" cy="120.00" r="5.94"
    fill="#65a843" />
    <circle cx="224.55" cy="137.45" r="5.62"
    fill="#5fa843" />
    <circle cx="220.26" cy="154.42" r="5.29"
    fill="#59a843" />
    <circle cx="213.22" cy="170.45" r="5.00"
    fill="#53a843" />
    <circle cx="203.65" cy="185.11" r="4.77"
    fill="#4da843" />
    <circle cx="191.79" cy="197.99" r="4.63"
    fill="#47a843" />
    <circle cx="177.98" cy="208.74" r="4.59"
    fill="#43a846" />
    <circle cx="162.58" cy="217.07" r="4.67"
    fill="#43a84c" />
    <circle cx="146.02" cy="222.76" r="4.85"
    fill="#43a852" />
    <circle cx="128.75" cy="225.64" r="5.10"
    fill="#43a858" />
    <circle cx="111.25" cy="225.64" r="5.42"
    fill="#43a85e" />
    <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864" />
    <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a" />
    <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870" />
```

```
      <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876" />
      <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c" />
      <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882" />
      <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888" />
      <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e" />
      <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894" />
      <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e" />
      <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888" />
      <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882" />
      <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c" />
      <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876" />
      <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870" />
      <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a" />
      <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864" />
      <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e" />
      <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858" />
      <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852" />
      <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c" />
      <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846" />
      <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843" />
      <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843" />
      <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843" />
      <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843" />
      <circle cx="224.55" cy="102.55" r="6.23"
      fill="#5fa843" />
    </g>
    <g class="lp-badge__mark">
      <circle cx="120" cy="120" r="27" fill="#14432f" />
      <g transform="translate(120 120) scale(1.5)
      translate(-12 -12)" fill="none" stroke="#ffffff"
      stroke-width="1.9" stroke-linecap="round"
      stroke-linejoin="round">
        <path d="m3 20 6.5-11L14 17l2.5-4L21 20z" />
        <circle cx="17" cy="6" r="2.2" />
      </g>
    </g>
</svg>
# ==== app/templates/error.html ====
{% extends "base.html" %}
{% block title %}{{ code }} · {{ message }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>{{ code }} · {{ message }}</h2>
  <p class="muted">
    请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
  </p>
  <p class="actions" style="margin-top:16px;">
    <a class="btn" href="/">返回首页</a>
  </p>
</div>
```

```
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 登录页（仿 Apple 账户页） ==================
     纯白底、居中、大量留白；色彩只用在徽标与按钮上。
     首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
     随滚动逐条弹入、往回滚时弹回。 */
  .lp {
    color: #1d1d1f;
    background:
      linear-gradient(180deg, rgba(245, 239, 224, .88) 0%,
                              rgba(245, 239, 224, .92)
    100%),
      url("../static/hero_qinglv.jpg")
      center top / cover no-repeat fixed;
  }
  /* ---------- 首屏 ---------- */
  .lp-hero {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
    position: relative;
    background:
      linear-gradient(180deg, rgba(15, 40, 45, .72) 0%,
                              rgba(20, 55, 58, .50) 50%,
                              rgba(15, 40, 45, .76) 100%),
      url("../static/hero_qinglv.jpg")
      center center / cover no-repeat;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
     --p 为 0~1 的滚动进度，由页面脚本写入；
     每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
  .lp-badge {
    width: 212px;
    height: 212px;
    margin-bottom: 36px;
    --p: 0;
    animation: lp-in .7s ease both;
  }
  .lp-badge__svg {
    display: block;
    width: 100%;
```

```
    height: 100%;                                                    6001
    overflow: visible;                    /* 散开时圆点不被画布裁掉 */   6002
    opacity: calc(1 - var(--p) * .94);                               6003
    filter: blur(calc(var(--p) * 2.6px));                            6004
  }                                                                  6005
  .lp-ring {                                                         6006
    transform-box: fill-box;                                         6007
    transform-origin: center;                                        6008
    transform: scale(calc(1 + (var(--k) - 1) * var(--p)));           6009
  }                                                                  6010
  .lp-badge__mark {                                                  6011
    transform-box: fill-box;                                         6012
    transform-origin: center;                                        6013
    transform: scale(calc(1 + .18 * var(--p)));                      6014
  }                                                                  6015
  .lp-title {                                                        6016
    margin: 0 0 18px;                                                6017
    font-size: clamp(28px, 4.2vw, 42px);                             6018
    font-weight: 700;                                                6019
    line-height: 1.3;                                                6020
    letter-spacing: -.4px;                                           6021
    color: #fff;                                                     6022
    text-shadow: 0 2px 12px rgba(0, 0, 0, .3);                       6023
    animation: lp-in .7s ease .06s both;                             6024
  }                                                                  6025
  .lp-cta {                                                          6026
    padding: 13px 52px;                                              6027
    font-family: inherit;                                            6028
    font-size: 17px;                                                 6029
    font-weight: 500;                                                6030
    color: #fff;                                                     6031
    background: var(--accent);                                       6032
    border: none;                                                    6033
    border-radius: 980px;                                            6034
    cursor: pointer;                                                 6035
    animation: lp-in .7s ease .18s both;                             6036
    transition: background-color .2s ease, box-shadow .2s            6037
    ease,                                                            6038
                transform .2s ease;                                  6039
  }                                                                  6040
  .lp-cta:hover {                                                    6041
    background: var(--accent-hover);                                 6042
    box-shadow: 0 10px 26px rgba(44, 107, 90, .4);                   6043
  }                                                                  6044
  .lp-cta:active { transform: translateY(1px); box-shadow:           6045
    none; }                                                          6046
  .lp-cta:focus-visible { outline: 3px solid rgba(31, 107,           6047
    69, .40); outline-offset: 3px; }                                 6048
  .lp-more {                                                         6049
    margin-top: 38px;                                                6050
```

```
    font-size: 13px;                                                    6051
    color: rgba(255, 255, 255, .7);                                     6052
    text-decoration: none;                                              6053
    animation: lp-in .7s ease .24s both;                                6054
  }                                                                     6055
  .lp-more:hover { color: #fff; }                                       6056
  @keyframes lp-in {                                                    6057
    from { opacity: 0; transform: translateY(12px); }                   6058
    to   { opacity: 1; transform: none; }                               6059
  }                                                                     6060
  /* ---------- 滚动进场 ---------- */                                  6061
     进入视口时从屏外飞入，离开视口时飞回，可反复。                    6062
     交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */ 6063
  .lp-reveal {                                                          6064
    opacity: 0;                                                         6065
    transition: opacity .7s ease,                                       6066
                transform 1.05s cubic-bezier(.18, 1.04, .3,             6067
    1),                                                                 6068
                border-color .62s ease, background-color                6069
    .62s ease;                                                          6070
    transition-delay: 0s;                                               6071
    will-change: opacity, transform;                                    6072
  }                                                                     6073
  /* 起始位置整体在屏外：横向按视口宽度取位移，纵向按视口高度取位移，  6074
     再带一点缩小，落位时按贝塞尔曲线的过冲轻微回弹。 */               6075
  .lp-reveal[data-dir="left"]  { transform:                             6076
    translate3d(-56vw, 5vh, 0) scale(.9); }                             6077
  .lp-reveal[data-dir="right"] { transform:                             6078
    translate3d(56vw, 5vh, 0) scale(.9); }                              6079
  .lp-reveal[data-dir="up"]    { transform: translate3d(0,              6080
    24vh, 0) scale(.94); }                                              6081
  .lp-reveal.is-in {                                                    6082
    opacity: 1;                                                         6083
    transform: none;                                                    6084
    transition-delay: calc(var(--i, 0) * 110ms);                        6085
  }                                                                     6086
  /* 标题颜色由淡到深，强化「浮现」的层次感 */                         6087
  .lp-reveal h2,                                                        6088
  .lp-reveal h3,                                                        6089
  .lp-reveal .lp-mod__name { color: #c2d2c8; transition:                6090
    color .7s ease; }                                                   6091
  .lp-reveal.is-in h3,                                                  6092
  .lp-reveal.is-in .lp-mod__name { color: var(--accent); }              6093
  .lp-reveal.is-in h2 { color: #1d1d1f; }                               6094
  /* ---------- 系统介绍 ---------- */                                 6095
  .lp-intro {                                                           6096
    max-width: none;                                                    6097
    margin: 0;                                                          6098
    padding: 56px 24px 72px;                                            6099
    background: transparent;                                            6100
```

```
  /* 卡片从屏外飞入，横向超出的部分裁掉，免得出现横向滚动条；
     用 clip 而不是 hidden，纵向滚动仍由页面本身负责 */
  overflow-x: clip;
}
.lp-intro > * { max-width: 1040px; margin-left: auto;
  margin-right: auto; }
.lp-intro__title {
  margin: 0 auto 26px;
  font-size: clamp(22px, 2.6vw, 30px);
  text-align: center;
  letter-spacing: -.3px;
}
/* 三张特点卡与九个模块卡都按三等分排版，并整体居中 */
.lp-points {
  display: grid;
  grid-template-columns: repeat(3, minmax(0, 1fr));
  gap: 16px;
  margin-bottom: 46px;
}
.lp-point {
  padding: 30px 26px;
  background: rgba(253, 250, 242, .82);
  backdrop-filter: blur(12px);
  -webkit-backdrop-filter: blur(12px);
  border: 1px solid rgba(224, 216, 200, .6);
  border-radius: 22px;
  box-shadow: 0 24px 60px rgba(42, 37, 32, .10), 0 8px
  20px rgba(42, 37, 32, .05);
  transition: transform .25s ease, box-shadow .25s ease;
}
.lp-point:hover {
  transform: translateY(-4px);
  box-shadow: 0 32px 72px rgba(42, 37, 32, .14), 0 12px
  28px rgba(42, 37, 32, .07);
}
.lp-point__icon {
  display: grid;
  place-items: center;
  width: 40px;
  height: 40px;
  margin-bottom: 16px;
  border-radius: 12px;
  background: rgba(44, 107, 90, .1);
  color: var(--accent);
}
.lp-point h3 { margin: 0 0 8px; font-size: 16px; }
.lp-point p {
  margin: 0;
  font-size: 13.5px;
  line-height: 1.85;
```

```
    color: #6e6e73;
    text-wrap: pretty;
}
.lp-sub-title {
    margin: 0 auto 20px;
    font-size: 17px;
    text-align: center;
}
.lp-mods {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: 12px;
    margin-bottom: 46px;
    align-items: start;
}
/* 模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明 */
.lp-mod {
    position: relative;
    padding: 17px 18px 17px 48px;
    background: rgba(253, 250, 242, .75);
    backdrop-filter: blur(8px);
    -webkit-backdrop-filter: blur(8px);
    border: 1px solid rgba(224, 216, 200, .6);
    border-radius: 14px;
    outline: none;
    transition: border-color .3s ease, box-shadow .3s ease;
}
.lp-mod:hover,
.lp-mod:focus-within,
.lp-mod:focus {
    border-color: rgba(44, 107, 90, .5);
    box-shadow: 0 8px 22px rgba(42, 37, 32, .08);
}
.lp-mod__no {
    position: absolute;
    left: 17px;
    top: 17px;
    font-size: 13px;
    font-weight: 700;
    color: #c8d6cd;
    font-variant-numeric: tabular-nums;
}
.lp-mod__name {
    display: block;
    font-size: 14px;
    font-weight: 600;
}
/* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */
.lp-mod__body {
    display: grid;
```

```
    grid-template-rows: 0fr;                                          6201
    transition: grid-template-rows .34s ease, margin-top              6202
    .34s ease;                                                        6203
  }                                                                   6204
  .lp-mod:hover .lp-mod__body,                                        6205
  .lp-mod:focus-within .lp-mod__body {                                6206
    grid-template-rows: 1fr;                                          6207
    margin-top: 7px;                                                  6208
  }                                                                   6209
  .lp-mod__desc {                                                     6210
    overflow: hidden;                                                 6211
    margin: 0;                                                        6212
    font-size: 12.5px;                                                6213
    line-height: 1.7;                                                 6214
    color: #6e6e73;                                                   6215
    opacity: 0;                                                       6216
    transition: opacity .3s ease;                                     6217
    text-wrap: pretty;                                                6218
  }                                                                   6219
  .lp-mod:hover .lp-mod__desc,                                        6220
  .lp-mod:focus-within .lp-mod__desc { opacity: 1; }                  6221
  /* 触摸设备没有悬停，直接把说明摊开 */                                  6222
  @media (hover: none) {                                              6223
    .lp-mod__body { grid-template-rows: 1fr; margin-top:              6224
    7px; }                                                            6225
    .lp-mod__desc { opacity: 1; }                                     6226
  }                                                                   6227
  .lp-foot {                                                          6228
    margin: 0;                                                        6229
    padding-top: 26px;                                                6230
    border-top: 1px solid #e6e6e8;                                    6231
    text-align: center;                                               6232
    font-size: 12.5px;                                                6233
    color: #86868b;                                                   6234
  }                                                                   6235
  /* ---------- 登录面板：点击后渐变浮现 ---------- */                    6236
  .lp-auth {                                                          6237
    position: fixed;                                                  6238
    inset: 0;                                                         6239
    z-index: 60;                                                      6240
    display: grid;                                                    6241
    place-items: center;                                              6242
    padding: 24px;                                                    6243
    opacity: 0;                                                       6244
    visibility: hidden;                                               6245
    transition: opacity .42s ease, visibility 0s linear               6246
    .42s;                                                             6247
  }                                                                   6248
  .lp-auth.is-open {                                                  6249
    opacity: 1;                                                       6250
```

```
    visibility: visible;                                                    6251
    transition: opacity .42s ease, visibility 0s;                           6252
}                                                                           6253
/* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */                                  6254
.lp-auth__scrim {                                                           6255
    position: absolute;                                                     6256
    inset: 0;                                                               6257
    background:                                                             6258
      radial-gradient(1000px 640px at 50% 0%,                               6259
                      rgba(31, 107, 69, .22), rgba(31, 107,                 6260
    69, 0) 70%),                                                            6261
      linear-gradient(180deg, rgba(255, 255, 255, .84),                     6262
                              rgba(246, 250, 247, .95));                    6263
    backdrop-filter: blur(16px) saturate(1.25);                             6264
    -webkit-backdrop-filter: blur(16px) saturate(1.25);                     6265
}                                                                           6266
.lp-auth__panel {                                                           6267
    position: relative;                                                     6268
    width: 100%;                                                            6269
    max-width: 396px;                                                       6270
    padding: 36px 36px 26px;                                                6271
    background: #fff;                                                       6272
    border-radius: 20px;                                                    6273
    box-shadow: 0 28px 80px rgba(14, 46, 32, .18);                          6274
    opacity: 0;                                                             6275
    transform: translateY(18px) scale(.975);                                6276
    transition: transform .55s cubic-bezier(.16, .84, .34,                  6277
    1) .05s,                                                                6278
                opacity .55s ease .05s;                                     6279
}                                                                           6280
.lp-auth.is-open .lp-auth__panel { opacity: 1; transform:                   6281
    none; }                                                                 6282
.lp-auth__close {                                                           6283
    position: absolute;                                                     6284
    top: 14px;                                                              6285
    right: 14px;                                                            6286
    display: grid;                                                          6287
    place-items: center;                                                    6288
    width: 30px;                                                            6289
    height: 30px;                                                           6290
    color: #86868b;                                                         6291
    background: transparent;                                                6292
    border: none;                                                           6293
    border-radius: 50%;                                                     6294
    cursor: pointer;                                                        6295
    transition: background-color .18s ease, color .18s ease;                6296
}                                                                           6297
.lp-auth__close:hover { background: #f5f5f7; color:                         6298
    #1d1d1f; }                                                              6299
.lp-auth__title {                                                           6300
```

```
  margin: 0 0 6px;
  text-align: center;
  font-size: 24px;
  font-weight: 700;
  letter-spacing: -.2px;
}
.lp-auth__sub {
  margin: 0 0 26px;
  text-align: center;
  font-size: 13px;
  color: #86868b;
}
/* 字段校验错误：贴着输入框显示，不做成会消失的浮层 —— 
   这类消息要对着输入框看，一闪而过会让人以为「点了没反应」 */
.lp-field__error {
  margin: 7px 0 0;
  font-size: 12.5px;
  color: #9c2b23;
}
.lp-field input[aria-invalid="true"] {
  background: #fdeeec;
  border-color: rgba(156, 43, 35, .35);
}
.lp-field { margin-bottom: 15px; }
.lp-field label {
  display: block;
  margin-bottom: 7px;
  font-size: 13px;
  color: #6e6e73;
}
.lp-field input {
  width: 100%;
  padding: 13px 16px;
  font-family: inherit;
  font-size: 16px;                /* 不小于 16px，避免手机端聚焦时自动放大 */
  color: #1d1d1f;
  background: #f5f5f7;
  border: 1px solid transparent;
  border-radius: 12px;
  outline: none;
  transition: background-color .18s ease, border-color
  .18s ease,
              box-shadow .18s ease;
}
.lp-field input::placeholder { color: #a1a1a6; }
.lp-field input:focus {
  background: #fff;
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);
}
```

```
.lp-remember {
  display: flex;
  align-items: center;
  gap: 8px;
  margin: 4px 0 20px;
  font-size: 13px;
  color: #6e6e73;
  cursor: pointer;
}
.lp-remember input {
  width: 16px;
  height: 16px;
  margin: 0;
  accent-color: var(--accent);
  cursor: pointer;
}
.lp-submit {
  width: 100%;
  padding: 13px 20px;
  font-family: inherit;
  font-size: 16px;
  font-weight: 500;
  color: #fff;
  background: var(--accent);
  border: none;
  border-radius: 980px;
  cursor: pointer;
  transition: background-color .2s ease, box-shadow .2s ease,
              transform .2s ease;
}
.lp-submit:hover {
  background: var(--accent-hover);
  box-shadow: 0 10px 24px rgba(31, 107, 69, .26);
}
.lp-submit:active { transform: translateY(1px);
  box-shadow: none; }
.lp-submit:focus-visible { outline: 3px solid rgba(31,
  107, 69, .40); outline-offset: 3px; }
.lp-auth__hint {
  margin: 18px 0 0;
  text-align: center;
  font-size: 12.5px;
  color: #a1a1a6;
}
.lp-auth__reg {
  color: var(--accent);
  font-weight: 500;
  text-decoration: none;
  border-bottom: 1px solid rgba(31, 107, 69, .35);
```

```
      transition: border-color .18s ease;                                  6401
    }                                                                      6402
    .lp-auth__reg:hover { border-bottom-color: var(--accent);              6403
      }                                                                    6404
    /* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */                6405
    @media (prefers-reduced-motion: reduce) {                              6406
      .lp-badge, .lp-title, .lp-cta, .lp-more { animation:                 6407
      none; }                                                              6408
      .lp-badge__svg { opacity: 1; filter: none; }                         6409
      .lp-ring, .lp-badge__mark { transform: none; }                       6410
      .lp-reveal { opacity: 1; transform: none !important; }               6411
      .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name               6412
      { color: inherit; }                                                  6413
      .lp-reveal.is-in h2 { color: #1d1d1f; }                              6414
      .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name {                6415
      color: var(--accent); }                                              6416
      .lp-auth, .lp-auth__panel { transition: opacity .2s                  6417
      linear; }                                                            6418
      .lp-auth__panel { transform: none; }                                 6419
    }                                                                      6420
    /* 窄屏退成两列、一列，避免卡片被挤变形 */                                6421
    @media (max-width: 900px) {                                            6422
      .lp-points, .lp-mods { grid-template-columns: repeat(2,              6423
      minmax(0, 1fr)); }                                                   6424
    }                                                                      6425
    @media (max-width: 620px) {                                            6426
      .lp-points, .lp-mods { grid-template-columns: minmax(0,              6427
      1fr); }                                                              6428
    }                                                                      6429
    @media (max-width: 560px) {                                            6430
      .lp-hero { padding: 36px 18px 48px; min-height: auto; }              6431
      .lp-badge { width: 160px; height: 160px; margin-bottom:              6432
      26px; }                                                              6433
      .lp-auth__panel { padding: 30px 24px 22px; }                         6434
    }                                                                      6435
</style>                                                                   6436
<div class="lp">                                                           6437
    {# ---------------- 首屏：徽标 + 标题 + 说明 + 登录入口                  6438
      ---------------- #}                                                  6439
    <section class="lp-hero">                                              6440
      <div class="lp-badge" id="lp-badge" aria-hidden="true">              6441
        {% include "_login_badge.svg" %}                                   6442
      </div>                                                               6443
      <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>          6444
      <button type="button" class="lp-cta" id="lp-open"                    6445
              aria-expanded="false" aria-controls="lp-auth">登录           6446
      </button>                                                            6447
      <a class="lp-more" href="#intro">                                    6448
        向下滑动，了解系统 <span aria-hidden="true">↓</span>                 6449
      </a>                                                                 6450
```

```
</section>
{# ---------------- 系统介绍：随滚动逐条弹入 ---------------- #}
<section class="lp-intro" id="intro">
  <h2 class="lp-intro__title lp-reveal"
  data-dir="up">一套把评价流程固定下来的工具</h2>
  <div class="lp-points">
    {% set pdirs = ['left', 'up', 'right'] %}
    {% set points = [
      ('sliders', '方法可溯',
       '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回看。'),
      ('shield', '流程可控',
       '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重建或人工辅助的结论。'),
      ('file', '结果可复算',
       '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')
    ] %}
    {% for ic, title, text in points %}
    <div class="lp-point lp-reveal" data-dir="{{
  pdirs[loop.index0 % 3] }}"
        style="--i: {{ loop.index0 }}">
      <span class="lp-point__icon">{{ icon(ic, 20)
  }}</span>
      <h3>{{ title }}</h3>
      <p>{{ text }}</p>
    </div>
    {% endfor %}
  </div>
  <h3 class="lp-sub-title lp-reveal"
  data-dir="up">九个功能模块</h3>
  <div class="lp-mods">
    {% set mdirs = ['left', 'up', 'right'] %}
    {% for no, name, desc in modules %}
    <div class="lp-mod lp-reveal" data-dir="{{
  mdirs[loop.index0 % 3] }}"
        style="--i: {{ loop.index0 % 3 }}" tabindex="0">
      <span class="lp-mod__no">{{ no }}</span>
      <span class="lp-mod__name">{{ name }}</span>
      <div class="lp-mod__body"><p class="lp-mod__desc">{{
  desc }}</p></div>
    </div>
    {% endfor %}
  </div>
  <p class="lp-foot lp-reveal" data-dir="up">
    历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０ ·
    运行于浏览器，数据保存在本机 ＳＱＬｉｔｅ 数据库
  </p>
</section>
{# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}
<div class="lp-auth" id="lp-auth" aria-hidden="true">
```

```
<div class="lp-auth__scrim" data-lp-close></div>
<div class="lp-auth__panel" role="dialog"
aria-modal="true"
     aria-labelledby="lp-auth-title">
  <button type="button" class="lp-auth__close"
data-lp-close
          aria-label="关闭登录框">
    <svg width="16" height="16" viewBox="0 0 24 24"
fill="none"
         stroke="currentColor" stroke-width="2"
stroke-linecap="round"
         aria-hidden="true"><path d="M6 6l12 12M18 6L6
18"/></svg>
  </button>
  <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>
  <p class="lp-auth__sub">使用系统账号登录</p>
  <form method="post" novalidate>
    {{ form.hidden_tag() }}
    <div class="lp-field">
      <label for="{{ form.username.id }}">账户名称</label>
      {{ form.username(id=form.username.id,
autocomplete="username",
                       placeholder="请输入账户名称",
                       **({'aria-invalid': 'true'} if
form.username.errors else {})) }}
      {% for e in form.username.errors %}
      <p class="lp-field__error">{{ e }}</p>
      {% endfor %}
    </div>
    <div class="lp-field">
      <label for="{{ form.password.id }}">密码</label>
      {{ form.password(id=form.password.id,
autocomplete="current-password",
                       placeholder="请输入密码",
                       **({'aria-invalid': 'true'} if
form.password.errors else {})) }}
      {% for e in form.password.errors %}
      <p class="lp-field__error">{{ e }}</p>
      {% endfor %}
    </div>
    <label class="lp-remember" for="{{ form.remember.id
}}">
      {{ form.remember(id=form.remember.id) }}
      <span>记住我（下次自动登录，一个月内有效）</span>
    </label>
    {{ form.submit(class_="lp-submit") }}
  </form>
  <p class="lp-auth__hint">
    还没有账号？<a class="lp-auth__reg"
                   href="{{ url_for('auth.register')
```

```
      }}">点我注册</a>                                                    6551
        </p>                                                             6552
        <p class="lp-auth__hint" style="margin-top:10px;">               6553
          演示账号：admin 密码：admin123                                  6554
        </p>                                                             6555
      </div>                                                             6556
    </div>                                                               6557
</div>                                                                   6558
{% endblock %}                                                           6559
{% block scripts %}                                                      6560
<script>                                                                 6561
(function () {                                                           6562
  var reduce = window.matchMedia &&                                      6563
    window.matchMedia('(prefers-reduced-motion:                          6564
    reduce)').matches;                                                   6565
  /* ---------- 登录面板 ---------- */                                    6566
  var btn = document.getElementById('lp-open');                          6567
  var auth = document.getElementById('lp-auth');                         6568
  var lastFocus = null;                                                  6569
  function openPanel() {                                                 6570
    lastFocus = document.activeElement;                                  6571
    auth.classList.add('is-open');                                       6572
    auth.setAttribute('aria-hidden', 'false');                           6573
    btn.setAttribute('aria-expanded', 'true');                           6574
    document.body.style.overflow = 'hidden';                             6575
    // 记到地址栏，直接访问 /login#login 也能展开登录框                  6576
    if (location.hash !== '#login') {                                    6577
      history.replaceState(null, '', location.pathname +                 6578
    '#login');                                                           6579
    }                                                                    6580
    var first = auth.querySelector('input');                             6581
    if (first) window.setTimeout(function () {                           6582
    first.focus(); }, 300);                                              6583
  }                                                                      6584
  function closePanel() {                                                6585
    auth.classList.remove('is-open');                                    6586
    auth.setAttribute('aria-hidden', 'true');                            6587
    btn.setAttribute('aria-expanded', 'false');                          6588
    document.body.style.overflow = '';                                   6589
    if (location.hash === '#login') {                                    6590
      history.replaceState(null, '', location.pathname);                 6591
    }                                                                    6592
    if (lastFocus && lastFocus.focus) lastFocus.focus();                 6593
  }                                                                      6594
  if (btn && auth) {                                                     6595
    btn.addEventListener('click', openPanel);                            6596
    Array.prototype.forEach.call(                                        6597
      auth.querySelectorAll('[data-lp-close]'),                          6598
      function (el) { el.addEventListener('click',                       6599
    closePanel); }                                                       6600
```

```
  });                                                                    6601
  document.addEventListener('keydown', function (e) {                   6602
    if (e.key === 'Escape' && auth.classList.contains('is-               6603
  open')) closePanel();                                                  6604
  });                                                                    6605
  // 登录失败回到本页、或直接带 #login 访问时，直接展开登录框。          6606
  // 失败可能落在两处：字段校验错误在表单里，凭据错误在浮层里。          6607
  if (location.hash === '#login'                                         6608
      || document.querySelector('.lp-field__error')                      6609
      || document.querySelector('.toast-center')) {                      6610
    openPanel();                                                         6611
  }                                                                      6612
}                                                                        6613
/* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- */          6614
   原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，        6615
   内容始终停在 opacity:0。改为直接按元素与视口的相对位置判断，        6616
   与徽标用同一次滚动回调，行为可预期、也便于排查。 */                  6617
var items = document.querySelectorAll('.lp-reveal');                    6618
var badge = document.getElementById('lp-badge');                        6619
var ticking = false;                                                     6620
function applyScroll() {                                                 6621
  ticking = false;                                                       6622
  var vh = window.innerHeight || 800;                                    6623
  /* 徽标：滚动前约三分之一屏内完成散开淡出 */                           6624
  if (badge) {                                                           6625
    var range = vh * 0.34;                                               6626
    var p = range > 0 ? window.scrollY / range : 0;                      6627
    badge.style.setProperty('--p', Math.min(1,                           6628
  Math.max(0, p)).toFixed(3));                                           6629
  }                                                                      6630
  /* 内容：先批量测量，再改类名，避免读写交替触发反复重排 */             6631
  if (reduce || !items.length) return;                                   6632
  var box = [];                                                          6633
  for (var i = 0; i < items.length; i++) {                               6634
    box.push(items[i].getBoundingClientRect());                          6635
  }                                                                      6636
  for (var j = 0; j < items.length; j++) {                               6637
    var r = box[j];                                                      6638
    // 上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内            6639
    var shown = r.top < vh * 0.88 && r.bottom > vh * 0.06;               6640
    if (shown) items[j].classList.add('is-in');                          6641
    else items[j].classList.remove('is-in');                             6642
  }                                                                      6643
}                                                                        6644
function onScroll() {                                                    6645
  if (!ticking) {                                                        6646
    ticking = true;                                                      6647
    window.requestAnimationFrame(applyScroll);                           6648
  }                                                                      6649
}                                                                        6650
```

```
  //  减少动态效果时不做滚动特效，内容直接显示
  if (reduce) {
    Array.prototype.forEach.call(items, function (el) {
      el.classList.add('is-in');
    });
    if (badge) badge.style.setProperty('--p', '0');
  } else {
    window.addEventListener('scroll', onScroll, { passive:
    true });
    window.addEventListener('resize', onScroll);
    applyScroll();
  }
})();
</script>
{% endblock %}
# ==== app/templates/auth/register.html ====
{% extends "base.html" %}
{% block title %}注册账号 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 注册页 ==================
     与登录页同一套观感：浅绿渐变底 + 白色卡片。
     必填只有账户名称与密码，其余收在「选填」分隔线以下，
     这样首次注册不会被一堆输入框挡住。 */
  .rg {
    min-height: 100dvh;
    display: grid;
    place-items: center;
    padding: 40px 20px 56px;
    background:
      radial-gradient(900px 520px at 50% 0%,
                      rgba(31, 107, 69, .14), rgba(31, 107,
    69, 0) 72%),
      #f6faf7;
  }
  .rg-card {
    width: 100%;
    max-width: 470px;
    padding: 38px 40px 30px;
    background: #fff;
    border-radius: 20px;
    box-shadow: 0 26px 70px rgba(14, 46, 32, .13);
    animation: rg-in .5s cubic-bezier(.16, .84, .34, 1)
    both;
  }
  @keyframes rg-in {
    from { opacity: 0; transform: translateY(16px)
    scale(.98); }
    to   { opacity: 1; transform: none; }
  }
```

```
.rg-title {
  margin: 0 0 6px;
  text-align: center;
  font-size: 25px;
  font-weight: 700;
  letter-spacing: -.2px;
}
.rg-sub {
  margin: 0 0 28px;
  text-align: center;
  font-size: 13px;
  color: #86868b;
}
.rg-field { margin-bottom: 15px; }
.rg-field > label {
  display: block;
  margin-bottom: 7px;
  font-size: 13px;
  color: #6e6e73;
}
.rg-field > label .rg-opt {
  margin-left: 5px;
  font-size: 12px;
  color: #a1a1a6;
}
.rg-field input[type="text"],
.rg-field input[type="password"] {
  width: 100%;
  padding: 12px 15px;
  font-family: inherit;
  font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */
  color: #1d1d1f;
  background: #f5f5f7;
  border: 1px solid transparent;
  border-radius: 12px;
  outline: none;
  transition: background-color .18s ease, border-color
  .18s ease,
              box-shadow .18s ease;
}
.rg-field input::placeholder { color: #a1a1a6; }
.rg-field input:focus {
  background: #fff;
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);
}
.rg-field input[aria-invalid="true"] {
  border-color: rgba(180, 60, 50, .55);
  background: #fff;
}
```

```
.rg-field select {
  width: 100%;
  padding: 12px 15px;
  font-family: inherit;
  font-size: 15px;
  color: #1d1d1f;
  background: #f5f5f7;
  border: 1px solid transparent;
  border-radius: 12px;
  outline: none;
}
.rg-field select:focus {
  background: #fff;
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);
}
.rg-err {
  margin: 6px 0 0;
  font-size: 12.5px;
  color: var(--danger);
}
/* 「选填」分隔线 */
.rg-split {
  display: flex;
  align-items: center;
  gap: 12px;
  margin: 24px 0 18px;
  font-size: 12.5px;
  color: #a1a1a6;
}
.rg-split::before,
.rg-split::after {
  content: "";
  flex: 1;
  height: 1px;
  background: #ececee;
}
.rg-pair {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 0 14px;
}
@media (max-width: 480px) { .rg-pair {
  grid-template-columns: 1fr; } }
/* 头像：圆形预览 + 选择文件 */
.rg-avatar {
  display: flex;
  align-items: center;
  gap: 14px;
  margin-bottom: 15px;
```

```
}
.rg-avatar__face {
  display: grid;
  place-items: center;
  flex: none;
  width: 54px;
  height: 54px;
  border-radius: 50%;
  background: #eef2f0;
  color: #9aa8a1;
  font-size: 12px;
  overflow: hidden;
  background-size: cover;
  background-position: center;
}
.rg-avatar__pick { min-width: 0; }
.rg-avatar__pick input[type="file"] {
  max-width: 100%;
  font-family: inherit;
  font-size: 12.5px;
  color: #6e6e73;
}
.rg-avatar__tip {
  margin: 4px 0 0;
  font-size: 12px;
  color: #a1a1a6;
}
.rg-submit {
  width: 100%;
  margin-top: 24px;
  padding: 13px 20px;
  font-family: inherit;
  font-size: 16px;
  font-weight: 500;
  color: #fff;
  background: var(--accent);
  border: none;
  border-radius: 980px;
  cursor: pointer;
  transition: background-color .2s ease, box-shadow .2s
  ease,
              transform .2s ease;
}
.rg-submit:hover {
  background: var(--accent-hover);
  box-shadow: 0 10px 24px rgba(31, 107, 69, .26);
}
.rg-submit:active { transform: translateY(1px);
  box-shadow: none; }
.rg-submit:focus-visible { outline: 3px solid rgba(31,
```

```
      107, 69, .40); outline-offset: 3px; }
    .rg-foot {
      margin: 20px 0 0;
      text-align: center;
      font-size: 13px;
      color: #86868b;
    }
    .rg-foot a { color: var(--accent); text-decoration: none;
      }
    .rg-foot a:hover { text-decoration: underline; }
    @media (prefers-reduced-motion: reduce) {
      .rg-card { animation: none; }
    }
</style>
<div class="rg">
  <div class="rg-card">
    <h1 class="rg-title">创建账号</h1>
    <p class="rg-sub">新账号为「操作员」，职务由管理员调整</p>
    <form method="post" enctype="multipart/form-data"
    novalidate>
      {{ form.hidden_tag() }}
      <div class="rg-field">
        <label for="{{ form.username.id }}">账户名称</label>
        {{ form.username(id=form.username.id,
    autocomplete="username",
                         placeholder="2–64 个字符",
                         **({'aria-invalid': 'true'} if
    form.username.errors else {})) }}
        {% for e in form.username.errors %}<p
    class="rg-err">{{ e }}</p>{% endfor %}
      </div>
      <div class="rg-field">
        <label for="{{ form.password.id }}">密码</label>
        {{ form.password(id=form.password.id,
    autocomplete="new-password",
                         placeholder="至少 6 位",
                         **({'aria-invalid': 'true'} if
    form.password.errors else {})) }}
        {% for e in form.password.errors %}<p
    class="rg-err">{{ e }}</p>{% endfor %}
      </div>
      <div class="rg-field">
        <label for="{{ form.confirm.id }}">确认密码</label>
        {{ form.confirm(id=form.confirm.id,
    autocomplete="new-password",
                        placeholder="再次输入密码",
                         **({'aria-invalid': 'true'} if
    form.confirm.errors else {})) }}
        {% for e in form.confirm.errors %}<p
    class="rg-err">{{ e }}</p>{% endfor %}
```

```
        </div>
        <p class="rg-split">以下均为选填</p>
        <div class="rg-avatar">
          <span class="rg-avatar__face" id="rg-face"
      aria-hidden="true">头像</span>
          <div class="rg-avatar__pick">
            <label class="visually-hidden"
      for="rg-avatar">选择头像图片</label>
            <input id="rg-avatar" type="file" name="avatar"
                   accept=".png,.jpg,.jpeg,.gif,.webp">
            <p class="rg-avatar__tip">PNG / JPG / GIF /
      WebP，不超过 4 MB</p>
          </div>
        </div>
        <div class="rg-pair">
          <div class="rg-field">
            <label for="{{ form.display_name.id }}">昵称</label>
            {{ form.display_name(id=form.display_name.id,
                                 placeholder="留空则显示账户名称") }}
          </div>
          <div class="rg-field">
            <label for="{{ form.gender.id }}">性别</label>
            {{ form.gender(id=form.gender.id) }}
          </div>
        </div>
        <div class="rg-field">
          <label for="{{ form.email.id }}">邮箱</label>
          {{ form.email(id=form.email.id,
      autocomplete="email",
                        placeholder="用于联系，可留空",
                        **({'aria-invalid': 'true'} if
      form.email.errors else {})) }}
          {% for e in form.email.errors %}<p
      class="rg-err">{{ e }}</p>{% endfor %}
        </div>
        <div class="rg-field" style="margin-bottom:0;">
          <label for="{{ form.phone.id }}">手机号</label>
          {{ form.phone(id=form.phone.id, autocomplete="tel",
      placeholder="可留空") }}
        </div>
        {{ form.submit_reg(class_="rg-submit") }}
      </form>
      <p class="rg-foot">已有账号？<a href="{{
      url_for('auth.login') }}">返回登录</a></p>
    </div>
  </div>
{% endblock %}
{% block scripts %}
<script>
//  选好头像后立刻在左侧圆框里预览，省得传上去才知道选错了
```

```
(function () {
  var input = document.getElementById('rg-avatar');
  var face = document.getElementById('rg-face');
  if (!input || !face || !window.FileReader) return;
  input.addEventListener('change', function () {
    var file = input.files && input.files[0];
    if (!file) return;
    var reader = new FileReader();
    reader.onload = function (e) {
      face.textContent = '';
      face.style.backgroundImage = 'url("' +
    e.target.result + '")';
    };
    reader.readAsDataURL(file);
  });
})();
</script>
{% endblock %}
# ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
{# ---------- 页首横幅 ---------- #}
<section class="hero">
  <div class="hero__body">
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>
    <p class="hero__desc">
      以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》
      建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，
      系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。
    </p>
    <div class="hero__actions">
      <a class="hero-btn hero-btn--ghost"
    href="/indicator/import">
        {{ icon("upload", 16) }}批量导入
      </a>
      <a class="hero-btn hero-btn--solid"
    href="/project/add">
        {{ icon("plus", 16) }}新建评价项目
      </a>
    </div>
  </div>
</section>
{# ---------- 数据概览 ---------- #}
<section class="stat-cards">
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">评价项目</span>
    <span class="stat-card__value">{{ dash.stats.projects
    }}<span class="stat-card__unit">个</span></span>
```

```
      <span class="stat-card__note">                                    7001
        {%- if projects_without_patch -%}                               7002
          其中 {{ projects_without_patch|length }} 个项目尚未建立图斑    7003
        {%- elif dash.progress.patches -%}                              7004
          已建立 {{ dash.progress.patches }} 个矿山图斑                 7005
        {%- else -%}                                                    7006
          尚未建立矿山图斑                                              7007
        {%- endif -%}                                                   7008
      </span>                                                           7009
    </a>                                                                7010
    <a class="stat-card" href="/project/">                              7011
      <span class="stat-card__label">矿山图斑</span>                    7012
      <span class="stat-card__value">{{ dash.stats.patches              7013
      }}<span class="stat-card__unit">个</span></span>                  7014
      <span class="stat-card__note">                                    7015
        {% if dash.progress.ready %}其中 {{ dash.progress.ready         7016
      }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}     7017
      </span>                                                           7018
    </a>                                                                7019
    <a class="stat-card" href="/unit/">                                 7020
      <span class="stat-card__label">微地质单元</span>                  7021
      <span class="stat-card__value">{{ dash.stats.unit_types           7022
      }}<span class="stat-card__unit">类</span></span>                  7023
      <span class="stat-card__note">                                    7024
        {% if dash.progress.units %}共划分 {{                           7025
      dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}    7026
      </span>                                                           7027
    </a>                                                                7028
    <a class="stat-card" href="/standard/">                             7029
      <span class="stat-card__label">标准与政策</span>                  7030
      <span class="stat-card__value">{{ dash.stats.standards            7031
      }}<span class="stat-card__unit">项</span></span>                  7032
      <span class="stat-card__note">地方标准与政策文件</span>           7033
    </a>                                                                7034
</section>                                                              7035
{# ---------- 评价流程 ---------- #}                                    7036
<section class="section">                                               7037
    <div class="section__head">                                         7038
      <h2 class="section__title">评价流程</h2>                          7039
      <p class="section__desc">                                         7040
        按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算        7041
      </p>                                                              7042
    </div>                                                              7043
    <div class="step-grid">                                             7044
      {% for s in dash.steps %}                                         7045
      {% if s.available %}                                              7046
      <a class="step-card" href="{{ s.href }}">                         7047
        <span class="step-card__no">{{ s.no }}</span>                   7048
        <span class="step-card__icon">{{ icon(s.icon, 18)               7049
      }}</span>                                                         7050
```

```
        <span class="step-card__title">{{ s.title }}</span>
        <span class="step-card__desc">{{ s.desc }}</span>
        {% if s.pending %}
        <span class="step-card__badge">待处理 {{ s.pending
    }}</span>
        {% endif %}
    </a>
    {% else %}
    <div class="step-card step-card--locked"
    aria-disabled="true">
        <span class="step-card__no">{{ s.no }}</span>
        <span class="step-card__icon">{{ icon(s.icon, 18)
    }}</span>
        <span class="step-card__title">{{ s.title }}</span>
        <span class="step-card__desc">{{ s.desc }}</span>
        <span class="step-card__lock">{{ s.note }}</span>
    </div>
    {% endif %}
    {% endfor %}
  </div>
</section>
{# ---------- 系统管理 ---------- #}
{# 个人设置、账号与操作日志的入口都在左侧导航栏，工作台不再重复列出 #}
{% endblock %}
# ==== app/templates/auth/add_user.html ====
{% extends "base.html" %}
{% block title %}新增账号{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>新增账号</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="username">账户名称</label
    >
      <input id="username" type="text" name="username"
    required
           maxlength="64" autocomplete="off"
    style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="password">密码</label>
      <input id="password" type="password" name="password"
    required
           autocomplete="new-password"
    style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="role">职务</label>
      <select id="role" name="role">
```

```
          {% for code, label in role_labels.items() %}                    7101
          <option value="{{ code }}" {% if code ==                        7102
      default_role %}selected{% endif %}>{{ label }}</option>             7103
          {% endfor %}                                                    7104
        </select>                                                         7105
        <p class="muted" style="margin:6px 0 0;">                         7106
          管理员可管理系统、维护基础数据并查看操作日志；操作员可使用全部业务功能。   7107
        </p>                                                              7108
      </div>                                                              7109
      <hr style="margin:22px 0;border:none;border-top:1px                 7110
      solid var(--line);">                                                7111
      <h3 style="margin:0 0 12px;font-size:15px;">个人资料（选填）</h3    7112
      >                                                                   7113
      <div class="field">                                                 7114
        <label class="field__label" for="display_name">昵称</lab          7115
      el>                                                                 7116
        <input id="display_name" type="text"                              7117
      name="display_name" maxlength="64"                                  7118
               placeholder="留空则显示账户名称">                          7119
      </div>                                                              7120
      <div class="field">                                                 7121
        <label class="field__label" for="gender">性别</label>             7122
        <select id="gender" name="gender">                                7123
          <option value="">不填</option>                                  7124
          {% for g in genders %}<option value="{{ g }}">{{ g              7125
      }}</option>{% endfor %}                                             7126
        </select>                                                         7127
      </div>                                                              7128
      <div class="field">                                                 7129
        <label class="field__label" for="email">邮箱</label>              7130
        <input id="email" type="text" name="email"                        7131
      maxlength="128">                                                    7132
      </div>                                                              7133
      <div class="field">                                                 7134
        <label class="field__label" for="phone">手机号</label>            7135
        <input id="phone" type="text" name="phone"                        7136
      maxlength="32">                                                     7137
      </div>                                                              7138
      <p class="actions" style="margin-top:18px;">                        7139
        <button type="submit" class="btn">保存</button>                   7140
        <a class="btn btn--ghost" href="{{                                7141
      url_for('auth.profile', tab='users') }}">返回</a>                   7142
      </p>                                                                7143
    </form>                                                               7144
</div>                                                                    7145
{% endblock %}                                                            7146
# ==== app/templates/auth/edit_user.html ====                             7147
{% extends "base.html" %}                                                 7148
{% block title %}编辑账号 · {{ user.username }}{% endblock %}             7149
{% block content %}                                                       7150
```

```
<div class="card" style="max-width:560px;">
  <h2>编辑账号 · {{ user.username }}</h2>
  <h3 style="margin:0 0 12px;font-size:15px;">个人资料</h3>
  <p class="muted" style="margin:0 0 14px;font-size:12.5px;">
    以下内容用户也可在「个人设置」中自行修改，此处为管理员代改。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="display_name">昵称</label>
      <input id="display_name" type="text" name="display_name" maxlength="64"
             value="{{ user.display_name or '' }}"
             placeholder="留空则显示账户名称">
    </div>
    <div class="field">
      <label class="field__label" for="gender">性别</label>
      <select id="gender" name="gender">
        <option value="">不填</option>
        {% for g in genders %}
        <option value="{{ g }}" {% if user.gender == g %}selected{% endif %}>{{ g }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="email">邮箱</label>
      <input id="email" type="text" name="email" maxlength="128"
             value="{{ user.email or '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="phone">手机号</label>
      <input id="phone" type="text" name="phone" maxlength="32"
             value="{{ user.phone or '' }}">
    </div>
    <hr style="margin:22px 0;border:none;border-top:1px solid var(--line);">
    <h3 style="margin:0 0 12px;font-size:15px;">职务与权限</h3>
    <p class="muted" style="margin:0 0 14px;font-size:12.5px;">
      职务只能由管理员设置：管理员可管理系统、维护基础数据并查看操作日志；
      操作员可使用全部业务功能。
    </p>
    <div class="field">
      <label class="field__label" for="role">职务</label>
      <select id="role" name="role">
```

```
          {% for code, label in role_labels.items() %}
          <option value="{{ code }}" {% if user.role == code
      %}selected{% endif %}>{{ label }}</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="checkbox" for="active">
          <input id="active" type="checkbox" name="active"
      value="1"
                 {% if user.active %}checked{% endif %}>
          <span>启用该账号</span>
        </label>
        <p class="muted" style="margin:6px 0 0;">
          停用后该账号无法登录，但其历史操作记录会保留。
          系统至少需保留一名启用状态的管理员。
        </p>
      </div>
      <div class="field">
        <label class="field__label" for="password">重置密码（留空则不修改
      ）</label>
        <input id="password" type="password" name="password"
               autocomplete="new-password"
      style="width:280px;"
               placeholder="留空表示保持原密码">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="{{
      url_for('auth.profile', tab='users') }}">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/auth/profile.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 个人设置（仿 Apple 账户页） ==================
     左栏是身份与分类，右栏是该分类下的功能卡。
     每张卡默认只显示当前值，点开才展开对应的修改表单 ——
     不点就不会误改，页面也不会一上来堆满输入框。 */
  .acct {
    display: grid;
    grid-template-columns: 244px 1fr;
    gap: 40px;
    align-items: start;
  }
```

```
@media (max-width: 900px) {
  .acct { grid-template-columns: 1fr; gap: 24px; }
}
/* ---------- 左栏 ---------- */
.acct-rail__avatar {
  display: grid;
  place-items: center;
  width: 88px;
  height: 88px;
  border-radius: 50%;
  background: #e9edeb;
  color: #97a09b;
  font-size: 32px;
  font-weight: 500;
  object-fit: cover;
}
.acct-rail__name {
  margin: 16px 0 2px;
  font-size: 16px;
  font-weight: 600;
}
.acct-rail__sub {
  margin: 0 0 24px;
  font-size: 13px;
  color: #86868b;
  word-break: break-all;
}
.acct-tabs {
  display: flex;
  flex-direction: column;
  gap: 2px;
  margin-bottom: 24px;
}
.acct-tabs a {
  padding: 7px 12px;
  border: 1px solid transparent;
  border-radius: 7px;
  font-size: 15px;
  color: #1d1d1f;
  text-decoration: none;
  transition: background-color .16s ease, border-color
  .16s ease;
}
.acct-tabs a:hover { background: #f5f5f7; }
.acct-tabs a.is-active {
  font-weight: 600;
  border-color: #bcc9c1;
  background: #fff;
}
/* 更换头像：折叠在左栏，点开才出现文件选择 */
```

```
.acct-avatar-edit > summary {
  display: inline-block;
  padding: 5px 12px;
  font-size: 13px;
  color: var(--accent);
  background: var(--accent-soft);
  border-radius: 7px;
  cursor: pointer;
  list-style: none;
}
.acct-avatar-edit > summary::-webkit-details-marker {
  display: none; }
.acct-avatar-edit > summary:hover { background: #dcebe2; }
.acct-avatar-edit__body { margin-top: 12px; }
.acct-avatar-edit__body .btn { width: 100%;
  margin-bottom: 8px; }
.acct-avatar-edit__file {
  margin: 0 0 10px;
  font-size: 12px;
  color: var(--ink-soft);
  word-break: break-all;
}
/* ---------- 右栏 ---------- */
.acct-title {
  margin: 0 0 8px;
  font-size: 26px;
  font-weight: 700;
  letter-spacing: -.3px;
}
.acct-lead {
  margin: 0 0 26px;
  font-size: 14px;
  color: #6e6e73;
}
/* 固定两列，与 Apple 账户页的 2×2 一致；
   卡片也宽一些，邮箱之类的长值不至于被拆成两行 */
.acct-grid {
  display: grid;
  grid-template-columns: repeat(2, minmax(0, 1fr));
  gap: 16px;
  align-items: start;
}
@media (max-width: 760px) {
  .acct-grid { grid-template-columns: 1fr; }
}
/* 功能卡：details/summary 实现，点开才展开表单 */
.acct-card {
  background: #fff;
  border: 1px solid #e6e6e8;
  border-radius: 14px;
```

```
    transition: border-color .2s ease, box-shadow .2s ease;
  }
  .acct-card[open] {
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
  }
  .acct-card > summary {
    position: relative;
    display: block;
    padding: 18px 56px 18px 20px;
    border-radius: 14px;
    cursor: pointer;
    list-style: none;
  }
  .acct-card > summary::-webkit-details-marker { display:
    none; }
  .acct-card > summary:hover { background: #fafbfa; }
  .acct-card > summary:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
  .acct-card__title {
    display: block;
    font-size: 15px;
    font-weight: 600;
  }
  .acct-card__value {
    display: block;
    margin-top: 3px;
    font-size: 14px;
    color: #6e6e73;
    overflow-wrap: anywhere;
  }
  .acct-card__icon {
    position: absolute;
    top: 17px;
    right: 17px;
    display: grid;
    place-items: center;
    width: 27px;
    height: 27px;
    border-radius: 8px;
    background: var(--accent-soft);
    color: var(--accent);
  }
  .acct-card__body {
    padding: 4px 20px 20px;
    border-top: 1px solid #f0f1f2;
    margin-top: 2px;
  }
```

```
  .acct-card__body .field:first-child { margin-top: 16px; }
  .acct-card__body .btn { width: 100%; }
  /* 只读信息列表 */
  .acct-kv {
    display: grid;
    grid-template-columns: 88px 1fr;
    gap: 7px 12px;
    margin: 16px 0 0;
    font-size: 13.5px;
  }
  .acct-kv dt { color: var(--ink-soft); }
  .acct-kv dd { margin: 0; }
  .field__error {
    margin: 5px 0 0;
    font-size: 12.5px;
    color: var(--danger);
  }
  /* 管理员分类（账号 / 操作日志）里的表格面板 */
  .acct-panel {
    background: #fff;
    border: 1px solid #e6e6e8;
    border-radius: 14px;
    overflow: hidden;
  }
  .acct-panel__head {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 12px;
    padding: 13px 20px;
    border-bottom: 1px solid #f0f1f2;
  }
  .acct-panel__count {
    margin: 0;
    font-size: 13.5px;
    color: #6e6e73;
  }
  .acct-panel table { margin: 0; }
  .acct-panel th,
  .acct-panel td { border: 0; border-bottom: 1px solid
    #f0f1f2; }
  .acct-panel thead th { background: #f7f9f8; }
  .acct-panel tbody tr:last-child td { border-bottom: 0; }
  .acct-panel tbody tr:hover { background: #fafbfa; }
</style>
<div class="acct">
  {# ---------------- 左栏：身份 + 分类 ---------------- #}
  <aside class="acct-rail">
    {% if current_user.avatar %}
    <img class="acct-rail__avatar"
```

```
      src="{{ url_for('static', filename=current_user.ava
tar) }}"
      alt="{{ current_user.shown_name }} 的头像">
{% else %}
<span class="acct-rail__avatar" aria-hidden="true">{{
current_user.initial }}</span>
{% endif %}
<p class="acct-rail__name">{{ current_user.shown_name
}}</p>
<p class="acct-rail__sub">
  {{ role_labels.get(current_user.role,
current_user.role) }}
  {% if current_user.email %} · {{ current_user.email
}}{% endif %}
</p>
<nav class="acct-tabs" aria-label="个人设置分类">
  <a href="{{ url_for('auth.profile', tab='info') }}"
     {% if tab == 'info' %}class="is-active"
aria-current="page"{% endif %}>个人信息</a>
  <a href="{{ url_for('auth.profile', tab='security')
}}"
     {% if tab == 'security' %}class="is-active"
aria-current="page"{% endif %}>登录与安全性</a>
  {% if current_user.is_admin %}
  <a href="{{ url_for('auth.profile', tab='users') }}"
     {% if tab == 'users' %}class="is-active"
aria-current="page"{% endif %}>账号</a>
  <a href="{{ url_for('auth.profile', tab='logs') }}"
     {% if tab == 'logs' %}class="is-active"
aria-current="page"{% endif %}>操作日志</a>
  {% endif %}
</nav>
<details class="acct-avatar-edit">
  <summary>更换头像</summary>
  <div class="acct-avatar-edit__body">
    <form method="post" action="/profile/avatar"
          enctype="multipart/form-data">
      {{ avatar_form.hidden_tag() }}
      <input type="hidden" name="tab" value="{{ tab }}">
      <label class="visually-hidden"
for="avatar-file">选择头像图片</label>
      <input id="avatar-file" type="file" name="avatar"
             accept=".png,.jpg,.jpeg,.gif,.webp">
      <p class="acct-avatar-edit__file">PNG / JPG / GIF
/ WebP，不超过 4 MB</p>
      <button type="submit" class="btn
btn--sm">上传头像</button>
    </form>
    {% if current_user.avatar %}
    <form method="post" action="/profile/avatar/delete"
```

```
                onsubmit="return confirm('确定移除当前头像吗？');">
            {{ avatar_form.hidden_tag() }}
            <input type="hidden" name="tab" value="{{ tab }}">
            <button type="submit" class="btn btn--sm
    btn--ghost">移除头像</button>
          </form>
          {% endif %}
        </div>
    </details>
</aside>
{# ---------------- 右栏：功能卡 ---------------- #}
<div class="acct-main">
    {% if tab == 'security' %}
    <h1 class="acct-title">登录与安全性</h1>
    <p class="acct-lead">管理与登录账号、账号安全有关的设置。</p>
    <div class="acct-grid">
      <details class="acct-card"
                {% if pwd_form.old_password.errors or
    pwd_form.new_password.errors or pwd_form.confirm.errors
    %}open{% endif %}>
          <summary>
            <span class="acct-card__title">密码</span>
            <span class="acct-card__value">修改登录密码</span>
            <span class="acct-card__icon">{{ icon("lock", 15)
    }}</span>
          </summary>
          <div class="acct-card__body">
            <form method="post">
              {{ pwd_form.hidden_tag() }}
              <input type="hidden" name="action"
    value="password">
              <input type="hidden" name="tab"
    value="security">
              <div class="field">
                <label class="field__label" for="{{
    pwd_form.old_password.id }}">当前密码</label>
                {{ pwd_form.old_password(id=pwd_form.old_passw
    ord.id,
    autocomplete="current-password") }}
                {% for e in pwd_form.old_password.errors %}
                <p class="field__error">{{ e }}</p>
                {% endfor %}
              </div>
              <div class="field">
                <label class="field__label" for="{{
    pwd_form.new_password.id }}">新密码</label>
                {{ pwd_form.new_password(id=pwd_form.new_passw
    ord.id,
```

```
autocomplete="new-password") }}                                    7551
            {% for e in pwd_form.new_password.errors %}            7552
            <p class="field__error">{{ e }}</p>                    7553
            {% endfor %}                                           7554
          </div>                                                   7555
          <div class="field">                                      7556
            <label class="field__label" for="{{                    7557
pwd_form.confirm.id }}">确认新密码</label>                          7558
            {{ pwd_form.confirm(id=pwd_form.confirm.id,            7559
                                autocomplete="new-password7560
") }}                                                              7561
            {% for e in pwd_form.confirm.errors %}                 7562
            <p class="field__error">{{ e }}</p>                    7563
            {% endfor %}                                           7564
          </div>                                                   7565
          <p class="actions">{{ pwd_form.submit_pwd(class_7566
="btn") }}</p>                                                     7567
        </form>                                                    7568
      </div>                                                       7569
    </details>                                                     7570
    <details class="acct-card">                                    7571
      <summary>                                                    7572
        <span class="acct-card__title">登录账号</span>             7573
        <span class="acct-card__value">{{                          7574
current_user.username }}</span>                                    7575
        <span class="acct-card__icon">{{ icon("user", 15)          7576
}}</span>                                                          7577
      </summary>                                                   7578
      <div class="acct-card__body">                                7579
        <dl class="acct-kv">                                       7580
          <dt>登录账号</dt><dd class="mono">{{                     7581
current_user.username }}</dd>                                      7582
          <dt>职务</dt>                                            7583
          <dd>{{ role_labels.get(current_user.role,                7584
current_user.role) }}</dd>                                         7585
          <dt>状态</dt>                                            7586
          <dd>                                                     7587
            {% if current_user.active %}                           7588
            <span class="tag tag--ok">启用</span>                  7589
            {% else %}                                             7590
            <span class="tag tag--warn">停用</span>                7591
            {% endif %}                                            7592
          </dd>                                                    7593
          <dt>创建时间</dt>                                        7594
          <dd class="mono">{{ current_user.created_at |            7595
localdt }}</dd>                                                    7596
        </dl>                                                      7597
      </div>                                                       7598
    </details>                                                     7599
</div>                                                             7600
```

```
{% elif tab == 'users' %}
<h1 class="acct-title">账号</h1>
<p class="acct-lead">管理本系统的账号、职务与启用状态。新注册的账号默认为「操作员」。</p>
{% include "auth/_users.html" %}
{% elif tab == 'logs' %}
<h1 class="acct-title">操作日志</h1>
<p class="acct-lead">记录各账号在本系统中的关键操作。</p>
{% include "auth/_logs.html" %}
{% else %}
<h1 class="acct-title">个人信息</h1>
<p class="acct-lead">管理你的昵称、性别以及可以联系到你的邮箱与手机号。</p>
<form method="post">
  {{ info_form.hidden_tag() }}
  <input type="hidden" name="action" value="info">
  <input type="hidden" name="tab" value="info">
  <div class="acct-grid">
    <details class="acct-card" {% if info_form.display_name.errors %}open{% endif %}>
      <summary>
        <span class="acct-card__title">昵称</span>
        <span class="acct-card__value">
          {{ current_user.display_name or "未设置" }}
        </span>
        <span class="acct-card__icon">{{ icon("user", 15) }}</span>
      </summary>
      <div class="acct-card__body">
        <div class="field">
          <label class="field__label" for="{{ info_form.display_name.id }}">
            昵称
          </label>
          {{ info_form.display_name(id=info_form.display_name.id,
                                    placeholder="留空则显示账户名称") }}
          {% for e in info_form.display_name.errors %}
          <p class="field__error">{{ e }}</p>
          {% endfor %}
        </div>
        <p class="actions">{{ info_form.submit_info(class_="btn") }}</p>
      </div>
    </details>
    <details class="acct-card" {% if info_form.gender.errors %}open{% endif %}>
      <summary>
        <span class="acct-card__title">性别</span>
        <span class="acct-card__value">{{
```

```
current_user.gender or "未设置" }}</span>                    7651
          <span class="acct-card__icon">{{ icon("user",      7652
15) }}</span>                                                7653
        </summary>                                           7654
        <div class="acct-card__body">                        7655
          <div class="field">                                7656
            <label class="field__label" for="{{              7657
info_form.gender.id }}">性别</label>                         7658
            {{ info_form.gender(id=info_form.gender.id) }}   7659
            {% for e in info_form.gender.errors %}           7660
            <p class="field__error">{{ e }}</p>              7661
            {% endfor %}                                     7662
          </div>                                             7663
          <p class="actions">{{ info_form.submit_info(clas   7664
s_="btn") }}</p>                                             7665
        </div>                                               7666
      </details>                                             7667
      <details class="acct-card" {% if                       7668
info_form.email.errors %}open{% endif %}>                    7669
        <summary>                                            7670
          <span class="acct-card__title">邮箱</span>         7671
          <span class="acct-card__value">{{                  7672
current_user.email or "未设置" }}</span>                     7673
          <span class="acct-card__icon">{{ icon("mail",      7674
15) }}</span>                                                7675
        </summary>                                           7676
        <div class="acct-card__body">                        7677
          <div class="field">                                7678
            <label class="field__label" for="{{              7679
info_form.email.id }}">邮箱</label>                          7680
            {{ info_form.email(id=info_form.email.id) }}     7681
            {% for e in info_form.email.errors %}            7682
            <p class="field__error">{{ e }}</p>              7683
            {% endfor %}                                     7684
          </div>                                             7685
          <p class="actions">{{ info_form.submit_info(clas   7686
s_="btn") }}</p>                                             7687
        </div>                                               7688
      </details>                                             7689
      <details class="acct-card" {% if                       7690
info_form.phone.errors %}open{% endif %}>                    7691
        <summary>                                            7692
          <span class="acct-card__title">手机号</span>       7693
          <span class="acct-card__value">{{                  7694
current_user.phone or "未设置" }}</span>                     7695
          <span class="acct-card__icon">{{ icon("phone",     7696
15) }}</span>                                                7697
        </summary>                                           7698
        <div class="acct-card__body">                        7699
          <div class="field">                                7700
```

```
                    <label class="field__label" for="{{                7701
      info_form.phone.id }}">手机号</label>                            7702
                    {{ info_form.phone(id=info_form.phone.id) }}       7703
                    {% for e in info_form.phone.errors %}              7704
                    <p class="field__error">{{ e }}</p>                7705
                    {% endfor %}                                       7706
                  </div>                                               7707
                  <p class="actions">{{ info_form.submit_info(clas     7708
      s_="btn") }}</p>                                                 7709
              </div>                                                   7710
            </details>                                                 7711
          </div>                                                       7712
      </form>                                                          7713
      {% endif %}                                                      7714
    </div>                                                             7715
</div>                                                                 7716
{% endblock %}                                                         7717
{% block scripts %}                                                    7718
<script>                                                               7719
//  同时只展开一张卡，避免点开多张后页面被撑得很长                    7720
(function () {                                                         7721
    var cards = document.querySelectorAll('.acct-card');               7722
    Array.prototype.forEach.call(cards, function (card) {              7723
      card.addEventListener('toggle', function () {                    7724
        if (!card.open) return;                                        7725
        Array.prototype.forEach.call(cards, function (other) {         7726
          if (other !== card && other.open) other.open =               7727
      false;                                                           7728
        });                                                            7729
      });                                                              7730
    });                                                                7731
})();                                                                  7732
</script>                                                              7733
{% endblock %}                                                         7734
#  ==== app/templates/auth/_users.html ====                            7735
{#  管理员的「账号」分类。由 auth/profile.html 内联引入，不单独成页。 #}  7736
<div class="acct-panel">                                               7737
    <div class="acct-panel__head">                                     7738
      <p class="acct-panel__count">共 {{ users|length }}               7739
      个账号</p>                                                       7740
      <a class="btn btn--sm" href="{{ url_for('auth.add_user')         7741
      }}">新增账号</a>                                                 7742
    </div>                                                             7743
    <table>                                                            7744
      <thead>                                                          7745
        <tr>                                                           7746
          <th scope="col">ID</th>                                      7747
          <th scope="col">账户名称</th>                                7748
          <th scope="col">昵称</th>                                    7749
          <th scope="col">职务</th>                                    7750
```

```
      <th scope="col">状态</th>                                          7751
      <th scope="col">操作</th>                                          7752
    </tr>                                                                7753
  </thead>                                                               7754
  <tbody>                                                                7755
    {% for u in users %}                                                 7756
    <tr>                                                                 7757
      <td class="num">{{ u.id }}</td>                                    7758
      <td class="mono">                                                  7759
        {{ u.username }}                                                 7760
        {% if u.id == current_user.id %}<span                            7761
class="muted">（当前登录）</span>{% endif %}                               7762
      </td>                                                              7763
      <td style="text-align:left;">                                      7764
        {% if u.display_name %}{{ u.display_name }}                      7765
        {% else %}<span class="muted">—</span>{% endif %}                7766
      </td>                                                              7767
      <td>                                                               7768
        <span class="tag {{ 'tag--ok' if u.role ==                       7769
'admin' else 'tag--muted' }}">                                           7770
          {{ role_labels.get(u.role, u.role) }}                          7771
        </span>                                                          7772
      </td>                                                              7773
      <td>                                                               7774
        {% if u.active %}<span class="tag                                7775
tag--ok">启用</span>                                                     7776
        {% else %}<span class="tag tag--muted">停用</span>{%             7777
endif %}                                                                 7778
      </td>                                                              7779
      <td class="nowrap">                                                7780
        <span class="row-actions">                                       7781
          <a class="link-btn" href="{{                                   7782
url_for('auth.edit_user', user_id=u.id) }}">编辑</a>                     7783
          <form method="post"                                            7784
                action="{{ url_for('auth.delete_user',                   7785
user_id=u.id) }}"                                                        7786
                onsubmit="return confirm('确定删除账号「{{               7787
u.username }}」吗？\n该操作无法恢复。');">                               7788
            {{ form.hidden_tag() }}                                      7789
            <button type="submit" class="link-btn                        7790
link-btn--danger"                                                        7791
                    {% if u.id == current_user.id                        7792
%}disabled                                                               7793
                    title="不能删除当前登录的账号"{% endif               7794
%}>删除</button>                                                         7795
          </form>                                                        7796
        </span>                                                          7797
      </td>                                                              7798
    </tr>                                                                7799
    {% endfor %}                                                         7800
```

```
      </tbody>
    </table>
</div>
#  ====  app/templates/auth/_logs.html  ====
{#  管理员的「操作日志」分类。由  auth/profile.html  内联引入，不单独成页。  #}
<div  class="acct-panel">
    <div  class="acct-panel__head">
      <p  class="acct-panel__count">
        共  {{  total  }}  条记录，当前第  {{  page  }}  /  {{  pages  }}  页
      </p>
      {%  if  pages  >  1  %}
      <nav  class="row-actions"  aria-label="日志分页">
        {%  if  page  >  1  %}
        <a  class="link-btn"
            href="{{  url_for('auth.profile',  tab='logs',
      page=page  -  1)  }}">上一页</a>
        {%  endif  %}
        {%  if  page  <  pages  %}
        <a  class="link-btn"
            href="{{  url_for('auth.profile',  tab='logs',
      page=page  +  1)  }}">下一页</a>
        {%  endif  %}
      </nav>
      {%  endif  %}
    </div>
    {%  if  not  records  %}
    <p  class="muted"  style="padding:18px  20px;">暂无操作记录。</p>
    {%  else  %}
    <table>
      <thead>
        <tr>
          <th  scope="col">时间</th>
          <th  scope="col">操作人</th>
          <th  scope="col">操作</th>
          <th  scope="col">对象</th>
          <th  scope="col">说明</th>
        </tr>
      </thead>
      <tbody>
        {%  for  r  in  records  %}
        <tr>
          <td  class="mono  nowrap">{{  r.created_at  |
      localdt("%Y-%m-%d  %H:%M:%S")  }}</td>
          <td>{{  users_map.get(r.user_id,  "—")  }}</td>
          <td  style="text-align:left;">{{  r.action  }}</td>
          <td>
            {%  if  r.target_type  %}{{  r.target_type  }}{%  if
      r.target_id  %}#{{  r.target_id  }}{%  endif  %}
            {%  else  %}—{%  endif  %}
          </td>
```

```
            <td style="text-align:left;">{{ r.detail or "—"
        }}</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
</div>
# ==== app/templates/project/index.html ====
{% extends "base.html" %}
{% block title %}项目与矿山档案{% endblock %}
{% block content %}
<div class="card">
  <h2>项目与矿山档案</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/add">新建项目</a>
  </p>
  {% if not projects %}
  <p class="muted">
    暂无项目。点击上方「新建项目」开始建立评价项目，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
  </p>
  {% else %}
  <table>
    <caption class="muted">共 {{ projects|length }}
    个项目</caption>
    <thead>
      <tr>
        <th scope="col">项目名称</th>
        <th scope="col">编号</th>
        <th scope="col">状态</th>
        <th scope="col">图斑数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in projects %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td class="mono">{{ p.code or "—" }}</td>
        <td><span class="tag tag--muted">{{ p.status
    }}</span></td>
        <td class="num">
          {% if p.patches.count() %}
            {{ p.patches.count() }}
          {% else %}
            <span class="tag tag--warn">0</span>
          {% endif %}
        </td>
```

```
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/project/{{ p.id }}/patches">{% if p.patches.count() %}矿山图斑{% else %}新建图斑{% endif %}</a>
              <a class="link-btn" href="/project/{{ p.id }}/edit">编辑</a>
              <form method="post" action="/project/{{ p.id }}/delete"
                    onsubmit="return confirm('确定删除项目「{{ p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn link-btn--danger">删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/project/edit.html ====
{% extends "base.html" %}
{% block title %}{{ "编辑项目" if project else "新建项目" }}{% endblock %}
{% block content %}
<div class="card" style="max-width:640px;">
    <h2>{{ "编辑项目" if project else "新建项目" }}</h2>
    <form method="post">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="name">项目名称</label>
        <input id="name" type="text" name="name"
               style="width:100%;" required
               value="{{ project.name if project else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="code">项目编号</label>
        <input id="code" type="text" name="code"
               style="width:240px;"
               value="{{ project.code or '' if project else '' }}">
      </div>
      {% if project %}
      <div class="field">
        <label class="field__label" for="status">状态</label>
        <select id="status" name="status">
```

```
            {% for s in ["进行中", "已完成", "已归档"] %}
            <option value="{{ s }}" {% if project.status == s
        %}selected{% endif %}>{{ s }}</option>
            {% endfor %}
          </select>
        </div>
        {% endif %}
        <div class="field">
          <label class="field__label" for="description">项目说明</la
        bel>
          <textarea id="description" name="description">{{
        project.description or '' if project else ''
        }}</textarea>
        </div>
        <p class="actions" style="margin-top:18px;">
          <button type="submit" class="btn">保存</button>
          <a class="btn btn--ghost" href="/project/">返回</a>
        </p>
      </form>
    </div>
    {% endblock %}
    # ==== app/templates/project/patches.html ====
    {% extends "base.html" %}
    {% block title %}矿山图斑 · {{ project.name }}{% endblock %}
    {% block content %}
    <div class="card">
      <h2>矿山图斑 · {{ project.name }}</h2>
      <p class="actions" style="margin-bottom:12px;">
        <a class="btn btn--sm" href="/project/{{ project.id
        }}/patch/add">新建图斑</a>
        <a class="btn btn--sm btn--ghost"
        href="/project/">返回项目列表</a>
      </p>
      {% if not project.patches.count() %}
      <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>
      {% else %}
      <table>
        <caption class="muted">共 {{ project.patches.count() }}
        个图斑</caption>
        <thead>
          <tr>
            <th scope="col">图斑名称</th>
            <th scope="col">行政区</th>
            <th scope="col">面积（公顷）</th>
            <th scope="col">开采矿种</th>
            <th scope="col">微地质单元</th>
            <th scope="col">操作</th>
          </tr>
        </thead>
        <tbody>
```

```
        {% for p in project.patches %}                                  8001
        <tr>                                                            8002
          <td style="text-align:left;">{{ p.name }}</td>                8003
          <td>{{ p.admin_region or "—" }}</td>                          8004
          <td class="num">{{ p.area_ha if p.area_ha is not              8005
      none else "—" }}</td>                                             8006
          <td>{{ p.mineral_type or "—" }}</td>                          8007
          <td class="num">{{ p.units.count() }}</td>                    8008
          <td class="nowrap">                                           8009
            <span class="row-actions">                                  8010
              <a class="link-btn" href="/unit/{{ p.id                   8011
      }}">微地质单元</a>                                                8012
              <a class="link-btn" href="/constraint/{{ p.id             8013
      }}">限制性判定</a>                                                8014
              <a class="link-btn" href="/project/patch/{{               8015
      p.id }}/edit">编辑</a>                                            8016
              <form method="post" action="/project/patch/{{             8017
      p.id }}/delete"                                                   8018
                    onsubmit="return confirm('确定删除图斑「{{          8019
      p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">  8020
                {{ form.hidden_tag() }}                                 8021
                <button type="submit" class="link-btn                   8022
      link-btn--danger">删除</button>                                   8023
              </form>                                                   8024
            </span>                                                     8025
          </td>                                                         8026
        </tr>                                                           8027
        {% endfor %}                                                    8028
      </tbody>                                                          8029
    </table>                                                            8030
    {% endif %}                                                         8031
</div>                                                                  8032
{% endblock %}                                                          8033
# ==== app/templates/project/patch_edit.html ====                       8034
{% extends "base.html" %}                                               8035
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%            8036
     endblock %}                                                        8037
{% block content %}                                                     8038
<div class="card" style="max-width:720px;">                             8039
    <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name     8040
      }}</h2>                                                           8041
    <form method="post">                                                8042
      {{ form.hidden_tag() }}                                           8043
      <div class="field">                                               8044
        <label class="field__label" for="patch-name">图斑名称</lab      8045
      el>                                                               8046
        <input id="patch-name" type="text" name="name"                  8047
      style="width:100%;" required                                      8048
               value="{{ patch.name if patch else '' }}">               8049
      </div>                                                            8050
```

```
<div class="field">
  <label class="field__label" for="admin-region">所在行政区</label>
  <input id="admin-region" type="text" name="admin_region" style="width:100%;"
         value="{{ patch.admin_region or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="area-ha">图斑面积（公顷）</label>
  <input id="area-ha" type="number" step="any" name="area_ha"
         value="{{ patch.area_ha if patch and patch.area_ha is not none else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mineral-type">开采矿种</label>
  <input id="mineral-type" type="text" name="mineral_type" style="width:240px;"
         value="{{ patch.mineral_type or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mining-method">开采方式</label>
  <input id="mining-method" type="text" name="mining_method" style="width:240px;"
         value="{{ patch.mining_method or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="stop-years">停产年限</label>
  <input id="stop-years" type="number" step="any" name="stop_years"
         value="{{ patch.stop_years if patch and patch.stop_years is not none else '' }}">
</div>
<div class="field">
  <label class="checkbox" for="responsible-missing">
    <input id="responsible-missing" type="checkbox" name="responsible_entity_missing" value="1"
           {% if not patch or patch.responsible_entity_missing %}checked{% endif %}>
    <span>责任主体缺失</span>
  </label>
</div>
<h3>参照生态系统（指标标准化基准）</h3>
```

```
      <p class="muted">                                                    8101
        论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。  8102
        该基准与图斑绑定管理，可按不同参照生态系统分别设定。                8103
      </p>                                                                 8104
      <div class="field">                                                  8105
        <label class="field__label" for="coverage-benchmark">植            8106
      被覆盖度基准（%）</label>                                            8107
        <input id="coverage-benchmark" type="number"                       8108
      step="any" name="coverage_benchmark"                                 8109
               value="{{ reference.coverage_benchmark if                   8110
      reference else 75 }}">                                               8111
      </div>                                                               8112
      <div class="field">                                                  8113
        <label class="field__label" for="shannon-benchmark">Sh             8114
      annon-Wiener 多样性指数基准</label>                                  8115
        <input id="shannon-benchmark" type="number"                        8116
      step="any" name="shannon_benchmark"                                  8117
               value="{{ reference.shannon_benchmark if                    8118
      reference else 0.7 }}">                                              8119
      </div>                                                               8120
      <div class="field">                                                  8121
        <label class="field__label" for="dominant-species">建群种          8122
      </label>                                                             8123
        <input id="dominant-species" type="text"                           8124
      name="dominant_species" style="width:100%;"                          8125
               value="{{ reference.dominant_species or '' if               8126
      reference else '' }}">                                               8127
      </div>                                                               8128
      <p class="actions" style="margin-top:18px;">                         8129
        <button type="submit" class="btn">保存</button>                    8130
        <a class="btn btn--ghost" href="/project/{{                        8131
      project.id }}/patches">返回</a>                                      8132
      </p>                                                                 8133
    </form>                                                                8134
</div>                                                                     8135
{% if patch %}                                                             8136
<div class="card" style="max-width:720px;">                                8137
    <h2>遥感影像挂接</h2>                                                  8138
    <p class="muted">                                                      8139
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。  8140
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。          8141
    </p>                                                                   8142
    {% if patch.image_path %}                                              8143
    <figure style="margin:0 0 16px;">                                      8144
      <img src="{{ url_for('static', filename=patch.image_path             8145
      ) }}"                                                                8146
           alt="图斑「{{ patch.name }}」已挂接的遥感影像"                  8147
           style="max-width:100%;border:1px solid                          8148
      var(--line);border-radius:6px;">                                     8149
      <figcaption class="muted" style="margin-top:6px;">                   8150
```

```
        {{ patch.image_note or "遥感影像" }}                                8151
        {% if patch.image_uploaded_at %}                                    8152
        · 上传于 {{ patch.image_uploaded_at | localdt }}                    8153
        {% endif %}                                                         8154
      </figcaption>                                                         8155
    </figure>                                                               8156
    <form method="post" action="/project/patch/{{ patch.id                  8157
      }}/image/delete"                                                      8158
          onsubmit="return confirm('确定移除该遥感影像吗？');">             8159
      {{ form.hidden_tag() }}                                               8160
      <button type="submit" class="btn btn--sm                              8161
      btn--danger">移除影像</button>                                        8162
    </form>                                                                 8163
    <hr style="margin:16px 0;border:none;border-top:1px solid               8164
      var(--line);">                                                        8165
    {% endif %}                                                             8166
    <form method="post" action="/project/patch/{{ patch.id                  8167
      }}/image"                                                             8168
          enctype="multipart/form-data">                                    8169
      {{ form.hidden_tag() }}                                               8170
      <div class="field">                                                   8171
        <label class="field__label" for="image-file">                       8172
          {{ "替换影像文件" if patch.image_path else "选择影像文件" }}      8173
        </label>                                                            8174
        <input id="image-file" type="file" name="image"                     8175
               accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t              8176
      iff" required>                                                        8177
      </div>                                                                8178
      <div class="field">                                                   8179
        <label class="field__label" for="image-note">影像说明（如来源、     8180
      时相）</label>                                                        8181
        <input id="image-note" type="text" name="image_note"                8182
      style="width:100%;"                                                   8183
               placeholder="如：2025 年 6 月无人机正射影像"                 8184
               value="{{ patch.image_note or '' }}">                        8185
      </div>                                                                8186
      <p class="actions">                                                   8187
        <button type="submit" class="btn">{{ "替换影像" if                  8188
      patch.image_path else "上传并挂接" }}</button>                        8189
      </p>                                                                  8190
    </form>                                                                 8191
</div>                                                                      8192
{% endif %}                                                                 8193
{% endblock %}                                                              8194
# ==== app/templates/unit/list.html ====                                    8195
{% extends "base.html" %}                                                   8196
{% block title %}微地质单元{% endblock %}                                   8197
{% block content %}                                                         8198
<div class="card">                                                          8199
  <h2>微地质单元</h2>                                                       8200
```

```
  {% if not rows %}                                                    8201
  <p class="muted">                                                    8202
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。                  8203
  </p>                                                                 8204
  {% else %}                                                           8205
  <table>                                                              8206
    <thead>                                                            8207
      <tr>                                                             8208
        <th scope="col">所属项目</th>                                  8209
        <th scope="col">矿山图斑</th>                                  8210
        <th scope="col">已划分单元</th>                                8211
        <th scope="col">占比合计</th>                                  8212
        <th scope="col">操作</th>                                      8213
      </tr>                                                            8214
    </thead>                                                           8215
    <tbody>                                                            8216
      {% for r in rows %}                                              8217
      <tr>                                                             8218
        <td>{{ r.patch.project.name if r.patch.project else            8219
    "—" }}</td>                                                        8220
        <td style="text-align:left;">{{ r.patch.name }}</td>           8221
        <td class="num">{{ r.count }} 个</td>                          8222
        <td class="num">                                               8223
          {% if r.count %}                                             8224
            {{ "%.2f%%"|format(r.ratio_sum * 100) }}                   8225
            {% if r.ratio_sum < 0.999 or r.ratio_sum >                 8226
    1.001 %}                                                           8227
              <span class="warn">（不等于 100%）</span>                8228
            {% endif %}                                                8229
          {% else %}—{% endif %}                                       8230
        </td>                                                          8231
        <td class="nowrap">                                            8232
          <span class="row-actions">                                   8233
            <a class="link-btn" href="/unit/{{ r.patch.id              8234
    }}">划分与维护</a>                                                 8235
            <a class="link-btn" href="/indicator/">录入指标数据</a     8236
    >                                                                  8237
          </span>                                                      8238
        </td>                                                          8239
      </tr>                                                            8240
      {% endfor %}                                                     8241
    </tbody>                                                           8242
  </table>                                                             8243
  {% endif %}                                                          8244
</div>                                                                 8245
{% endblock %}                                                         8246
# ==== app/templates/unit/index.html ====                              8247
{% extends "base.html" %}                                              8248
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}           8249
{% block content %}                                                    8250
```

```
<div class="card">
  <h2>微地质单元 · {{ patch.name }}</h2>
  {% if units %}
  <table>
    <caption class="muted">
      共 {{ units|length }} 个单元；
      面积合计 <span class="num">{{ "%.4f"|format(total_area)
    }}</span>；
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *
    100) }}</span>
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >
    1.001) %}
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>
      {% endif %}
    </caption>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积</th>
        <th scope="col">面积占比</th>
        <th scope="col">说明</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ u.area if u.area is not none
    else "—" }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ u.description or
    "—" }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/edit/{{ u.id
    }}">编辑</a>
            <form method="post" action="/unit/delete/{{
    u.id }}"
                  onsubmit="return confirm('确定删除微地质单元「{{
    u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
```

```
        </tr>                                                                    8301
        {% endfor %}                                                             8302
      </tbody>                                                                   8303
    </table>                                                                     8304
    {% else %}                                                                   8305
    <p class="muted">尚未划分微地质单元。请在下方添加。</p>                                  8306
    {% endif %}                                                                  8307
    <h3>添加微地质单元</h3>                                                          8308
    <form method="post" action="/unit/{{ patch.id }}/add">                       8309
      {{ form.hidden_tag() }}                                                    8310
      <div class="actions">                                                      8311
        <label class="field__label" for="add-unit-type"                          8312
      style="margin:0;">类型</label>                                               8313
        <select id="add-unit-type" name="unit_type">                             8314
          {% for t in unit_types %}<option value="{{ t }}">{{                    8315
      t }}</option>{% endfor %}                                                  8316
        </select>                                                                8317
        <label class="field__label" for="add-unit-area"                          8318
      style="margin:0;">面积</label>                                               8319
        <input id="add-unit-area" type="number" step="any"                       8320
      name="area">                                                               8321
        <label class="field__label" for="add-unit-desc"                          8322
      style="margin:0;">说明</label>                                               8323
        <input id="add-unit-desc" type="text"                                    8324
      name="description" style="width:220px;">                                   8325
        <button type="submit" class="btn btn--sm">添加</button>                  8326
      </div>                                                                     8327
    </form>                                                                      8328
    <p class="actions" style="margin-top:20px;">                                 8329
      <a class="btn btn--sm btn--ghost" href="/project/{{                        8330
      patch.project_id }}/patches">返回图斑列表</a>                                  8331
      <a class="btn btn--sm btn--ghost"                                          8332
      href="/indicator/">去录入指标数据</a>                                          8333
    </p>                                                                         8334
</div>                                                                           8335
{% endblock %}                                                                   8336
# ==== app/templates/unit/edit.html ====                                         8337
{% extends "base.html" %}                                                        8338
{% block title %}编辑微地质单元{% endblock %}                                        8339
{% block content %}                                                              8340
<div class="card">                                                               8341
    <h2>编辑微地质单元 · {{ patch.name }}</h2>                                       8342
    <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>                    8343
    <form method="post">                                                         8344
      {{ form.hidden_tag() }}                                                    8345
      <div class="field">                                                        8346
        <label class="field__label" for="unit-type">微地质单元类型</l 8347
      abel>                                                                      8348
        <select id="unit-type" name="unit_type">                                 8349
          {% for t in unit_types %}                                              8350
```

```
          <option value="{{ t }}" {% if unit.unit_type == t     8351
      %}selected{% endif %}>{{ t }}</option>                    8352
          {% endfor %}                                          8353
        </select>                                               8354
      </div>                                                    8355
      <div class="field">                                       8356
        <label class="field__label" for="unit-area">面积</label>8357
        <input id="unit-area" type="number" step="any"          8358
      name="area"                                               8359
               value="{{ unit.area if unit.area is not none     8360
      else '' }}">                                              8361
      </div>                                                    8362
      <div class="field">                                       8363
        <label class="field__label" for="unit-desc">说明</label>8364
        <input id="unit-desc" type="text" name="description"    8365
      style="width:420px;"                                      8366
               value="{{ unit.description or '' }}">            8367
      </div>                                                    8368
      <p class="actions" style="margin-top:16px;">              8369
        <button type="submit" class="btn">保存</button>         8370
        <a class="btn btn--ghost" href="/unit/{{                8371
      unit.patch_id }}">返回</a>                                8372
      </p>                                                      8373
    </form>                                                     8374
</div>                                                          8375
{% endblock %}                                                  8376
# ==== app/templates/standard/index.html ====                   8377
{% extends "base.html" %}                                       8378
{% block title %}政策与标准知识库{% endblock %}                  8379
{% block content %}                                             8380
<div class="card">                                              8381
    <h2>政策与标准知识库</h2>  <p class="actions"               8382
      style="margin-bottom:14px;">                              8383
      <a class="btn btn--sm" href="/standard/add">添加标准</a>   8384
      <a class="btn btn--sm btn--ghost"                         8385
      href="/standard/compare">省际口径比对</a>                 8386
    </p>                                                        8387
    <h3>地方标准</h3>                                          8388
    {% if local %}                                              8389
    <table>                                                     8390
      <thead>                                                   8391
        <tr>                                                    8392
          <th scope="col">省份</th>                             8393
          <th scope="col">名称</th>                             8394
          <th scope="col">编号</th>                             8395
          <th scope="col">规则数</th>                           8396
          <th scope="col">操作</th>                             8397
        </tr>                                                   8398
      </thead>                                                  8399
      <tbody>                                                   8400
```

```
        {% for s in local %}
        <tr>
          <td>{{ s.province }}</td>
          <td style="text-align:left;">{{ s.name }}</td>
          <td class="mono">{{ s.doc_number or "—" }}</td>
          <td class="num">{{ s.rules|length }}</td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
              <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
              <form method="post" action="/standard/{{ s.id
    }}/delete"
                    onsubmit="return confirm('确定删除「{{
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
    link-btn--danger">删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% else %}
    <p class="muted">暂无地方标准。运行 <span class="mono">python
      seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>
    {% endif %}
    <h3>政策文件</h3>
    {% if policy %}
    <table>
      <thead>
        <tr>
          <th scope="col">省份</th>
          <th scope="col">名称</th>
          <th scope="col">文号</th>
          <th scope="col">规则数</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for s in policy %}
        <tr>
          <td>{{ s.province }}</td>
          <td style="text-align:left;">{{ s.name }}</td>
          <td class="mono">{{ s.doc_number or "—" }}</td>
          <td class="num">{{ s.rules|length }}</td>
```

```
            <td class="nowrap">                                                    8451
              <span class="row-actions">                                           8452
                <a class="link-btn" href="/standard/{{ s.id                        8453
      }}/rules">规则</a>                                                           8454
                <a class="link-btn" href="/standard/{{ s.id                        8455
      }}/edit">编辑</a>                                                            8456
                <form method="post" action="/standard/{{ s.id                      8457
      }}/delete"                                                                   8458
                      onsubmit="return confirm('确定删除「{{                       8459
      s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length               8460
      }} 条规则将一并删除，且无法恢复。');">                                       8461
                  {{ form.hidden_tag() }}                                          8462
                  <button type="submit" class="link-btn                            8463
      link-btn--danger">删除</button>                                              8464
                </form>                                                            8465
              </span>                                                              8466
            </td>                                                                  8467
          </tr>                                                                    8468
          {% endfor %}                                                             8469
        </tbody>                                                                   8470
      </table>                                                                     8471
      {% else %}                                                                   8472
      <p class="muted">暂无政策文件。</p>                                          8473
      {% endif %}                                                                  8474
</div>                                                                             8475
{% endblock %}                                                                     8476
# ==== app/templates/standard/edit.html ====                                       8477
{% extends "base.html" %}                                                          8478
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%                    8479
    endblock %}                                                                    8480
{% block content %}                                                                8481
<div class="card" style="max-width:800px;">                                        8482
  <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                            8483
  <form method="post">                                                             8484
    {{ form.hidden_tag() }}                                                        8485
    <div class="field">                                                            8486
      <label class="field__label" for="province">省份</label>                      8487
      <input id="province" type="text" name="province"                             8488
    style="width:160px;" required                                                  8489
             value="{{ standard.province if standard else                          8490
    '' }}">                                                                        8491
    </div>                                                                         8492
    <div class="field">                                                            8493
      <label class="field__label" for="std-name">名称</label>                      8494
      <input id="std-name" type="text" name="name"                                 8495
    style="width:100%;" required                                                   8496
             value="{{ standard.name if standard else ''                           8497
    }}">                                                                           8498
    </div>                                                                         8499
    <div class="field">                                                            8500
```

```
  <label class="field__label" for="doc-number">标准编号 /
文号</label>
  <input id="doc-number" type="text" name="doc_number"
style="width:320px;"
         value="{{ standard.doc_number or '' if
standard else '' }}">
</div>
<div class="field">
  <label class="field__label" for="category">类别</label>
  <select id="category" name="category">
    {% for c in ["地方标准", "政策文件"] %}
    <option value="{{ c }}" {% if standard and
standard.category == c %}selected{% endif %}>{{ c
}}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="publish-date">发布时间</l
abel>
  <input id="publish-date" type="text"
name="publish_date" style="width:200px;"
         placeholder="如 2023-03-30"
         value="{{ standard.publish_date or '' if
standard else '' }}">
</div>
<div class="field">
  <label class="field__label" for="implement-date">实施时间<
/label>
  <input id="implement-date" type="text"
name="implement_date" style="width:200px;"
         placeholder="如 2023-04-30（未明确可留空）"
         value="{{ standard.implement_date or '' if
standard else '' }}">
</div>
<div class="field">
  <label class="field__label" for="summary">特点 /
判定标准</label>
  <textarea id="summary" name="summary"
style="min-height:96px;">{{ standard.summary or '' if
standard else '' }}</textarea>
</div>
<p class="actions" style="margin-top:18px;">
  <button type="submit" class="btn">保存</button>
  {% if standard %}
  <form method="post" action="/standard/{{ standard.id
}}/delete" style="display:inline;"
        onsubmit="return confirm('确定删除「{{
standard.province }} {{ standard.name
}}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
```

```
          {{ form.hidden_tag() }}                                          8551
          <button type="submit" class="btn                                 8552
      btn--danger">删除该标准</button>                                      8553
        </form>                                                            8554
        {% endif %}                                                        8555
        <a class="btn btn--ghost" href="/standard/">返回</a>               8556
      </p>                                                                 8557
    </form>                                                                8558
</div>                                                                     8559
{% endblock %}                                                             8560
# ==== app/templates/standard/rules.html ====                              8561
{% extends "base.html" %}                                                  8562
{% block title %}结构化规则 · {{ standard.province }}{% endblock           8563
    %}                                                                     8564
{% block content %}                                                        8565
<div class="card">                                                         8566
    <h2>结构化规则 · {{ standard.province }} {{ standard.name              8567
      }}</h2>                                                              8568
    {% if rules %}                                                         8569
    <table>                                                                8570
      <caption class="muted">共 {{ rules|length }}                         8571
      条规则</caption>                                                     8572
      <thead>                                                              8573
        <tr>                                                               8574
          <th scope="col">判定项</th>                                      8575
          <th scope="col">条件</th>                                        8576
          <th scope="col">条件类型</th>                                    8577
          <th scope="col">适用分区</th>                                    8578
          <th scope="col">备注</th>                                        8579
          <th scope="col">操作</th>                                        8580
        </tr>                                                              8581
      </thead>                                                             8582
      <tbody>                                                              8583
        {% for r in rules %}                                               8584
        <tr>                                                               8585
          <td style="text-align:left;">{{ r.item }}</td>                   8586
          <td class="mono nowrap">{{ r.operator }} {{                      8587
      r.threshold or "" }}</td>                                            8588
          <td>{{ r.condition_type }}</td>                                  8589
          <td>{{ r.region or "—" }}</td>                                   8590
          <td style="text-align:left;">{{ r.note or "—"                    8591
      }}</td>                                                              8592
          <td>                                                             8593
            <form method="post" action="/standard/rule/{{                  8594
      r.id }}/delete"                                                      8595
                  onsubmit="return confirm('确定删除规则「{{ r.item        8596
      }} {{ r.operator }} {{ r.threshold or '' }}」吗？');">               8597
              {{ form.hidden_tag() }}                                      8598
              <button type="submit" class="btn btn--sm                     8599
      btn--danger">删除</button>                                           8600
```

```
            </form>                                                    8601
          </td>                                                        8602
        </tr>                                                          8603
        {% endfor %}                                                   8604
      </tbody>                                                         8605
    </table>                                                           8606
    {% else %}                                                         8607
    <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>                  8608
    {% endif %}                                                        8609
    <h3>添加规则</h3>                                                  8610
    <form method="post">                                               8611
      {{ form.hidden_tag() }}                                          8612
      <div class="actions">                                            8613
        <div>                                                          8614
          <label class="field__label" for="r-item">判定项</label>      8615
          <input id="r-item" type="text" name="item"                   8616
      placeholder="如：植被覆盖度" required>                             8617
        </div>                                                         8618
        <div>                                                          8619
          <label class="field__label" for="r-op">比较符</label>        8620
          <select id="r-op" name="operator">                           8621
            {% for o in [">=", ">", "<=", "<", "=="]                   8622
      %}<option value="{{ o }}">{{ o }}</option>{% endfor %}           8623
          </select>                                                    8624
        </div>                                                         8625
        <div>                                                          8626
          <label class="field__label" for="r-thr">阈值</label>         8627
          <input id="r-thr" type="text" name="threshold"               8628
      placeholder="如：50" style="width:110px;">                       8629
        </div>                                                         8630
        <div>                                                          8631
          <label class="field__label" for="r-cond">条件类型</label     8632
      >                                                                8633
          <select id="r-cond" name="condition_type">                   8634
            <option value="必要条件">必要条件</option>                   8635
            <option value="充分条件">充分条件</option>                   8636
          </select>                                                    8637
        </div>                                                         8638
        <div>                                                          8639
          <label class="field__label" for="r-region">适用分区</lab     8640
      el>                                                              8641
          <input id="r-region" type="text" name="region"               8642
      placeholder="如宁夏分区" style="width:130px;">                   8643
        </div>                                                         8644
        <div>                                                          8645
          <label class="field__label" for="r-note">备注</label>        8646
          <input id="r-note" type="text" name="note"                   8647
      style="width:220px;">                                            8648
        </div>                                                         8649
        <div>                                                          8650
```

```
            <button type="submit" class="btn">添加规则</button>
          </div>
        </div>
      </form>
      <p class="actions" style="margin-top:20px;">
        <a class="btn btn--sm btn--ghost"
        href="/standard/">返回标准列表</a>
        <a class="btn btn--sm btn--ghost"
        href="/standard/compare">省际口径比对</a>
      </p>
</div>
{% endblock %}
# ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
{% block content %}
<div class="card">
  <h2>省际口径比对</h2>
  {% if not groups %}
  <p class="muted">
    暂无规则数据。请先在「标准知识库」中为标准录入规则，
    或运行 <span class="mono">python seed_demo.py</span> 载入论文表
    1 数据。
  </p>
  {% else %}
  {% for item, rs in groups.items() %}
  <h3>{{ item }}</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">标准 / 文件</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">备注</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rs %}
      <tr>
        <td>{{ r.standard.province }}</td>
        <td style="text-align:left;">{{ r.standard.name
    }}</td>
        <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td style="text-align:left;">{{ r.note or "—"
    }}</td>
      </tr>
      {% endfor %}
```

```
      </tbody>                                                                  8701
    </table>                                                                    8702
    {% endfor %}                                                                8703
    {% endif %}                                                                 8704
    <p class="actions" style="margin-top:16px;">                                8705
      <a class="btn btn--ghost" href="/standard/">返回标准列表</a>              8706
    </p>                                                                        8707
</div>                                                                          8708
{% endblock %}                                                                  8709
# ==== app/templates/constraint/index.html ====                                 8710
{% extends "base.html" %}                                                       8711
{% from "_macros.html" import patch_guide %}                                    8712
{% block title %}限制性指标判定{% endblock %}                                   8713
{% block content %}                                                             8714
<div class="card">                                                              8715
    <h2>限制性指标判定</h2>                                                     8716
    {{ patch_guide(projects_without_patch) }}                                   8717
    {% if not patches %}                                                        8718
    <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>    8719
    {% else %}                                                                  8720
    <table>                                                                     8721
      <thead>                                                                   8722
        <tr>                                                                    8723
          <th scope="col">所属项目</th>                                         8724
          <th scope="col">矿山图斑</th>                                         8725
          <th scope="col">判定结果</th>                                         8726
          <th scope="col">操作</th>                                             8727
        </tr>                                                                   8728
      </thead>                                                                  8729
      <tbody>                                                                   8730
        {% for p in patches %}                                                  8731
        {% set r = latest.get(p.id) %}                                          8732
        <tr>                                                                    8733
          <td>{{ p.project.name if p.project else "—" }}</td>                   8734
          <td style="text-align:left;">{{ p.name }}</td>                        8735
          <td style="text-align:left;">                                         8736
            {% if r %}                                                          8737
              {% if r.passed %}                                                 8738
              <span class="tag tag--ok">通过</span>                             8739
              {% else %}                                                        8740
              <span class="tag tag--warn">不通过</span>                         8741
              {% endif %}                                                       8742
              <span class="muted">{{ r.reason }}</span>                         8743
            {% else %}                                                          8744
            <span class="tag tag--muted">未判定</span>                          8745
            {% endif %}                                                         8746
          </td>                                                                 8747
          <td class="nowrap">                                                   8748
            <span class="row-actions">                                          8749
              <a class="link-btn" href="/constraint/{{ p.id                     8750
```

```
      }}">执行判定</a>                                                    8751
            </span>                                                      8752
          </td>                                                          8753
        </tr>                                                            8754
        {% endfor %}                                                     8755
      </tbody>                                                           8756
    </table>                                                             8757
    {% endif %}                                                          8758
</div>                                                                   8759
{% endblock %}                                                           8760
# ==== app/templates/constraint/check.html ====                          8761
{% extends "base.html" %}                                                8762
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}          8763
{% block content %}                                                      8764
<div class="card" style="max-width:760px;">                              8765
    <h2>限制性条件判定 · {{ patch.name }}</h2>                           8766
    <form method="post">                                                 8767
      {{ form.hidden_tag() }}                                            8768
      {% for field, name, val in field_states %}                         8769
      <fieldset style="border:1px solid                                  8770
      var(--line);border-radius:6px;padding:10px                         8771
      14px;margin:0 0 10px;">                                            8772
        <legend style="padding:0 6px;font-weight:600;">{{                8773
      name }}</legend>                                                   8774
        <div class="actions">                                            8775
          <label class="checkbox">                                       8776
            <input type="radio" name="{{ field }}" value="1"             8777
                   {% if val is none or val %}checked{% endif            8778
      %}>                                                                8779
            <span>通过</span>                                            8780
          </label>                                                       8781
          <label class="checkbox">                                       8782
            <input type="radio" name="{{ field }}" value="0"             8783
                   {% if val is not none and not val                     8784
      %}checked{% endif %}>                                              8785
            <span>不通过</span>                                          8786
          </label>                                                       8787
        </div>                                                           8788
      </fieldset>                                                        8789
      {% endfor %}                                                       8790
      <p class="actions" style="margin-top:16px;">                       8791
        <button type="submit" class="btn">保存判定</button>              8792
        <a class="btn btn--ghost" href="/constraint/">返回</a>           8793
      </p>                                                               8794
    </form>                                                              8795
    {% if record %}                                                      8796
    <div style="margin-top:18px;padding:14px;border-radius:6px           8797
      ;border-left:4px solid;                                            8798
                 {% if record.passed %}background:var(--ok-soft          8799
      );border-left-color:var(--ok);                                     8800
```

```
                {% else %}background:var(--danger-soft);border
    -left-color:var(--danger);{% endif %}">
    <p style="margin:0 0 4px;font-weight:600;">
      判定结论：
      {% if record.passed %}
      <span class="ok">通过，可进入定量评价</span>
      {% else %}
      <span class="warn">不通过 —— 建议生态重建或人工辅助</span>
      {% endif %}
    </p>
    <p class="muted" style="margin:0;">{{ record.reason
    }}</p>
  </div>
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/ahp/index.html ====
{% extends "base.html" %}
{% block title %}AHP 权重计算{% endblock %}
{% block content %}
<div class="card" style="max-width:720px;">
  <h2>AHP 权重计算 · 选择指标层级</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="{{ form.unit_type.id
    }}">微地质单元类型</label>
      {{ form.unit_type(id=form.unit_type.id) }}
    </div>
    <div class="field">
      <label class="field__label" for="{{ form.level.id
    }}">指标层级</label>
      {{ form.level(id=form.level.id) }}
    </div>
    <div class="field">
      <label class="field__label" for="{{ form.l1.id }}">
        一级指标（层级选「二级指标」时需选择）
      </label>
      {{ form.l1(id=form.l1.id) }}
    </div>
    <p class="actions" style="margin-top:16px;">
      {{ form.submit(class_="btn") }}
      <a class="btn btn--ghost" href="/ahp/indicators">指标定义维
    护</a>
      <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a
    >
    </p>
  </form>
</div>
<div class="card">
```

```
  <h2>各微地质单元当前生效权重</h2>                                                8851
  <table>                                                                          8852
    <thead>                                                                        8853
      <tr>                                                                         8854
        <th scope="col">微地质单元</th>                                            8855
        <th scope="col">当前权重来源</th>                                          8856
        <th scope="col">操作</th>                                                  8857
      </tr>                                                                        8858
    </thead>                                                                       8859
    <tbody>                                                                        8860
      {% for r in weight_rows %}                                                   8861
      <tr>                                                                         8862
        <td style="text-align:left;">{{ r.unit_type }}</td>                        8863
        <td style="text-align:left;">                                              8864
          {% if r.custom %}                                                        8865
          <span class="tag tag--warn">{{ r.label }}</span>                         8866
          {% else %}                                                               8867
          <span class="tag tag--muted">{{ r.label }}</span>                        8868
          {% endif %}                                                              8869
        </td>                                                                      8870
        <td class="nowrap">                                                        8871
          <span class="row-actions">                                               8872
            <a class="link-btn" href="/ahp/matrix?unit_type=                       8873
    {{ r.unit_type }}&level=1">一级指标权重</a>                                    8874
            <a class="link-btn" href="/ahp/matrix?unit_type=                       8875
    {{ r.unit_type }}&level=1&calc=1">查看一致性</a>                               8876
            {% if r.custom %}                                                      8877
            <form method="post" action="/ahp/weights/reset"                        8878
                  onsubmit="return confirm('确定把「{{                             8879
    r.unit_type }}」恢复为论文表 3 的默认权重吗？');">                             8880
              <input type="hidden" name="csrf_token"                               8881
    value="{{ csrf_token() }}">                                                    8882
              <input type="hidden" name="unit_type"                                8883
    value="{{ r.unit_type }}">                                                     8884
              <button type="submit" class="link-btn                                8885
    link-btn--danger">恢复默认</button>                                            8886
            </form>                                                                8887
            {% endif %}                                                            8888
          </span>                                                                  8889
        </td>                                                                      8890
      </tr>                                                                        8891
      {% endfor %}                                                                 8892
    </tbody>                                                                       8893
  </table>                                                                         8894
</div>                                                                             8895
{% endblock %}                                                                     8896
# ==== app/templates/ahp/matrix.html ====                                          8897
{% extends "base.html" %}                                                          8898
{% block title %}判断矩阵录入{% endblock %}                                        8899
{% block content %}                                                                8900
```

```
<div class="card">
  <h2>
    判断矩阵录入
    · {{ unit_type }}
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%
    endif %}
  </h2>
  <p class="muted">
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
    矩阵已按论文表 3 的权重预填。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <table style="width:auto;">
      <thead>
        <tr>
          <th></th>
          {% for name in names %}<th scope="col">{{ name
    }}</th>{% endfor %}
        </tr>
      </thead>
      <tbody>
        {% for i in range(names|length) %}
        <tr>
          <th scope="row">{{ names[i] }}</th>
          {% for j in range(names|length) %}
            {% if i == j %}
            <td class="muted num">1</td>
            {% elif i < j %}
            <td>
              <label class="muted" style="position:absolute;
    left:-9999px;"
                     for="m-{{ i }}-{{ j }}">{{ names[i] }}
    相对 {{ names[j] }} 的重要性</label>
              <input id="m-{{ i }}-{{ j }}" type="number"
    step="0.01" min="0.01"
                     name="m_{{ i }}_{{ j }}" value="{{
    default[i][j] }}">
            </td>
            {% else %}
            <td class="muted num">{{ default[i][j] }}</td>
            {% endif %}
          {% endfor %}
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="actions">
      <button type="submit" class="btn" name="action"
```

```
    value="calc">计算权重</button>                                    8951
      {% if result and result.consistent %}                          8952
      <button type="submit" class="btn" name="action"                8953
    value="save">保存并生效</button>                                  8954
      {% endif %}                                                    8955
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>             8956
    </p>                                                             8957
</form>                                                              8958
<p class="muted" style="margin-top:10px;">                           8959
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label         8960
    }}</strong>。                                                    8961
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；  8962
    如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。    8963
</p>                                                                 8964
{% if result %}                                                      8965
<hr style="margin:22px 0;border:none;border-top:1px solid            8966
    var(--line);">                                                   8967
<h3>计算结果</h3>                                                    8968
<table style="width:auto;">                                          8969
    <thead>                                                          8970
      <tr>                                                           8971
        <th scope="col">指标</th>                                    8972
        {% for name in names %}<th scope="col">{{ name               8973
    }}</th>{% endfor %}                                              8974
      </tr>                                                          8975
    </thead>                                                         8976
    <tbody>                                                          8977
      <tr>                                                           8978
        <th scope="row">权重</th>                                    8979
        {% for w in result.weights %}<td class="mono">{{ w           8980
    }}</td>{% endfor %}                                              8981
      </tr>                                                          8982
    </tbody>                                                         8983
</table>                                                             8984
<div class="stat-row">                                               8985
    <div>                                                            8986
      <div class="stat__label">最大特征值 λmax</div>                 8987
      <div class="stat__value mono">{{ result.lambda_max             8988
    }}</div>                                                         8989
    </div>                                                           8990
    <div>                                                            8991
      <div class="stat__label">一致性指标 CI</div>                   8992
      <div class="stat__value mono">{{                               8993
    "%.4f"|format(result.ci) }}</div>                                8994
    </div>                                                           8995
    <div>                                                            8996
      <div class="stat__label">一致性比例 CR</div>                   8997
      <div class="stat__value mono">{{                               8998
    "%.4f"|format(result.cr) }}</div>                                8999
    </div>                                                           9000
```

```
      <div>                                                                    9001
        <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>          9002
        <div class="stat__value">                                              9003
          {% if result.consistent %}<span class="ok">通过</span>               9004
          {% else %}<span class="warn">不通过</span>{% endif %}                9005
        </div>                                                                 9006
      </div>                                                                   9007
    </div>                                                                     9008
    {% if not result.consistent and result.most_inconsistent                   9009
      %}                                                                       9010
    <p style="margin-top:14px;padding:12px;border-radius:6px;b                 9011
      ackground:var(--danger-soft);                                            9012
              border-left:4px solid var(--danger);">                           9013
      <strong class="warn">建议优先调整：</strong>                             9014
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                        9015
      result.most_inconsistent.j + 1 }} 列的比较                               9016
      （当前 <span class="mono num">{{ result.most_inconsistent.v              9017
      alue|round(3) }}</span>，                                                9018
      理想约 <span class="mono num">{{ result.most_inconsistent.e              9019
      xpected|round(3) }}</span>）                                             9020
    </p>                                                                       9021
    {% endif %}                                                                9022
  {% endif %}                                                                  9023
</div>                                                                         9024
{% endblock %}                                                                 9025
# ==== app/templates/ahp/indicators.html ====                                  9026
{% extends "base.html" %}                                                      9027
{% block title %}指标定义维护{% endblock %}                                    9028
{% block content %}                                                            9029
<div class="card">                                                             9030
  <h2>指标体系维护</h2>                                                       9031
  <p class="actions" style="margin-bottom:14px;">                              9032
    <a class="btn btn--sm btn--ghost"                                          9033
    href="/ahp/">返回权重计算</a>                                              9034
    <a class="btn btn--sm btn--ghost"                                          9035
    href="/ahp/experts">多专家矩阵</a>                                         9036
    <form method="post" action="/ahp/indicators/reset"                         9037
    style="display:inline;"                                                    9038
          onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。 9039
    确定继续？');">                                                            9040
      {{ form.hidden_tag() }}                                                  9041
      <button type="submit" class="btn btn--sm                                 9042
    btn--danger">按论文取值重置</button>                                       9043
    </form>                                                                    9044
  </p>                                                                         9045
  {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,                9046
    "三级指标")] %}                                                            9047
  {% set level_items = all_ind|selectattr("level",                             9048
    "equalto", level)|list %}                                                  9049
  {% if level_items %}                                                         9050
```

```
  <h3>{{ title }}（{{ level_items|length }} 项）</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">指标</th>
        <th scope="col">上级指标</th>
        <th scope="col">单位</th>
        <th scope="col">标准化类型</th>
        <th scope="col">理想取值</th>
        <th scope="col">启用单元</th>
        <th scope="col">限制性</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for i in level_items %}
      <tr>
        <td style="text-align:left;">
          <span class="mono muted">{{ i.code
}}</span><br>{{ i.name }}
        </td>
        <td>{{ (all_ind|selectattr("code", "equalto",
i.parent_code)|list|first).name if i.parent_code else
"—" }}</td>
        <td>{{ i.unit or "—" }}</td>
        <td>{{ type_label(i.ideal_type) }}</td>
        <td class="mono">{{ i.ideal_value or "—" }}</td>
        <td style="text-align:left;">{{ i.enabled_units or
"—" }}</td>
        <td>
          {% if i.is_constraint %}<span class="tag
tag--warn">是</span>
          {% else %}<span class="tag tag--muted">否</span>{%
endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/ahp/indicators/{{
i.code }}/edit">编辑</a>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
  {% endfor %}
</div>
{% endblock %}
# ==== app/templates/ahp/indicator_edit.html ====
```

```
{% extends "base.html" %}
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:680px;">
  <h2>编辑指标 · {{ ind.name }}</h2>
  <p class="muted">
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{ ind.level }} 级，
    由论文指标体系确定，不可修改。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="name">指标名称</label>
      <input id="name" type="text" name="name" style="width:100%;" required
             value="{{ ind.name }}">
    </div>
    <div class="field">
      <label class="field__label" for="unit">计量单位</label>
      <input id="unit" type="text" name="unit" style="width:180px;"
             value="{{ ind.unit or '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="ideal-type">标准化类型</label>
      <select id="ideal-type" name="ideal_type">
        {% for code, label in [("benefit","越大越优"),("cost","越小越优"),("mid","适中型"),("discrete","离散赋值")] %}
        <option value="{{ code }}" {% if ind.ideal_type == code %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="ideal-value">理想取值</label>
      <input id="ideal-value" type="text" name="ideal_value" style="width:100%;"
             value="{{ ind.ideal_value or '' }}">
      <p class="muted" style="margin:4px 0 0;">
        如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。
      </p>
    </div>
    <fieldset style="border:1px solid var(--line);border-radius:6px;padding:10px 14px;margin:0 0 12px;">
      <legend style="padding:0 6px;font-weight:600;">启用该指标的微地质单元</legend>
```

```
        {% for t in unit_types %}                                          9151
        <label class="checkbox" style="margin-right:18px;">               9152
          <input type="checkbox" name="enabled_units"                      9153
      value="{{ t }}"                                                      9154
                 {% if t in ind.units_list() %}checked{%                   9155
      endif %}>                                                            9156
          <span>{{ t }}</span>                                             9157
        </label>                                                           9158
        {% endfor %}                                                       9159
        <p class="muted" style="margin:8px 0 0;">                          9160
          论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。                  9161
        </p>                                                               9162
      </fieldset>                                                          9163
      <div class="field">                                                  9164
        <label class="checkbox" for="is-constraint">                       9165
          <input id="is-constraint" type="checkbox"                        9166
      name="is_constraint" value="1"                                       9167
                 {% if ind.is_constraint %}checked{% endif %}>             9168
          <span>同时作为限制性指标（参与一票否决判定）</span>                      9169
        </label>                                                           9170
      </div>                                                               9171
      <div class="field">                                                  9172
        <label class="field__label" for="description">取值依据说明</       9173
      label>                                                               9174
        <textarea id="description" name="description"                      9175
      style="min-height:88px;">{{ ind.description or ''                    9176
      }}</textarea>                                                        9177
      </div>                                                               9178
      <p class="actions" style="margin-top:16px;">                         9179
        <button type="submit" class="btn">保存</button>                    9180
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列         9181
      表</a>                                                               9182
      </p>                                                                 9183
    </form>                                                                9184
</div>                                                                     9185
{% endblock %}                                                             9186
# ==== app/templates/ahp/experts.html ====                                 9187
{% extends "base.html" %}                                                  9188
{% block title %}多专家判断矩阵{% endblock %}                                9189
{% block content %}                                                        9190
<div class="card">                                                         9191
  <h2>多专家判断矩阵 · 录入与汇总</h2>                                        9192
  <form method="get" class="actions" style="margin-bottom:14               9193
    px;">                                                                  9194
    <div>                                                                  9195
      <label class="field__label" for="unit-type">微地质单元类型</l        9196
    abel>                                                                  9197
      <select id="unit-type" name="unit_type"                              9198
    onchange="this.form.submit()">                                         9199
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}                        9200
```

```
          <option value="{{ t }}" {% if unit_type == t                    9201
    %}selected{% endif %}>{{ t }}</option>                                9202
          {% endfor %}                                                    9203
        </select>                                                         9204
    </div>                                                                9205
    <div>                                                                 9206
        <label class="field__label" for="level">指标层级</label>          9207
        <select id="level" name="level"                                   9208
    onchange="this.form.submit()">                                        9209
          <option value="1" {% if level == 1 %}selected{%                 9210
    endif %}>一级指标</option>                                            9211
          <option value="2" {% if level == 2 %}selected{%                 9212
    endif %}>二级指标</option>                                            9213
        </select>                                                         9214
    </div>                                                                9215
</form>                                                                   9216
<p class="muted">                                                         9217
    当前范围：<strong>{{ unit_type }}</strong> ·                          9218
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif     9219
    %}。                                                                  9220
    若选二级指标，请先到                                                  9221
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。                 9222
</p>                                                                      9223
{% if rows %}                                                             9224
<table>                                                                   9225
    <caption class="muted">已录入 {{ rows|length }}                       9226
    位专家的矩阵</caption>                                                9227
    <thead>                                                               9228
        <tr>                                                              9229
          <th scope="col">专家</th>                                       9230
          <th scope="col">矩阵规模</th>                                   9231
          <th scope="col">该矩阵 CR</th>                                  9232
          <th scope="col">一致性</th>                                     9233
          <th scope="col">录入时间</th>                                   9234
          <th scope="col">操作</th>                                       9235
        </tr>                                                             9236
    </thead>                                                              9237
    <tbody>                                                               9238
        {% for r in rows %}                                               9239
        <tr>                                                              9240
          <td>{{ r.expert_name }}</td>                                    9241
          <td class="num">{{ n }} × {{ n }}</td>                          9242
          <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is              9243
    not none else "—" }}</td>                                             9244
          <td>                                                            9245
            {% if r.cr is not none and r.cr < 0.1 %}<span                 9246
    class="tag tag--ok">通过</span>                                       9247
            {% else %}<span class="tag tag--warn">不通过</span>{%         9248
    endif %}                                                              9249
          </td>                                                           9250
```

```
        <td class="mono nowrap">{{ r.created_at | localdt }}</td>
        <td>
          <form method="post" action="/ahp/experts/{{ r.id }}/delete"
                onsubmit="return confirm('确定删除专家「{{ r.expert_name }}」的矩阵吗？');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  <p class="actions" style="margin-bottom:10px;">
    <a class="btn" href="/ahp/experts/aggregate?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">
      汇总计算（几何平均）
    </a>
  </p>
{% else %}
  <p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>
{% endif %}

<h3>录入一位专家的判断矩阵</h3>
<form method="post">
  {{ form.hidden_tag() }}
  <input type="hidden" name="action" value="add">
  <input type="hidden" name="unit_type" value="{{ unit_type }}">
  <input type="hidden" name="level" value="{{ level }}">
  {% if l1 %}<input type="hidden" name="l1" value="{{ l1 }}">{% endif %}
  <div class="field">
    <label class="field__label" for="expert-name">专家姓名</label>
    <input id="expert-name" type="text" name="expert_name"
           style="width:220px;" placeholder="如：张工">
  </div>
  <table style="width:auto;">
    <thead>
      <tr>
        <th></th>
        {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
```

```
          {% for i in range(n) %}                                          9301
          <tr>                                                            9302
            <th scope="row">{{ names[i] }}</th>                           9303
            {% for j in range(n) %}                                       9304
              {% if i == j %}                                             9305
              <td class="muted num">1</td>                                9306
              {% elif i < j %}                                            9307
              <td>                                                        9308
                <label class="muted" style="position:absolute;            9309
    left:-9999px;"                                                        9310
                       for="e-{{ i }}-{{ j }}">{{ names[i] }}             9311
    相对 {{ names[j] }} 的重要性</label>                                  9312
                <input id="e-{{ i }}-{{ j }}" type="number"               9313
    step="0.01" min="0.01"                                                9314
                       name="m_{{ i }}_{{ j }}" value="{{                 9315
    default[i][j] }}">                                                    9316
              </td>                                                       9317
              {% else %}                                                  9318
              <td class="muted num">{{ default[i][j] }}</td>              9319
              {% endif %}                                                 9320
            {% endfor %}                                                  9321
          </tr>                                                           9322
          {% endfor %}                                                    9323
        </tbody>                                                          9324
    </table>                                                              9325
    <p class="actions" style="margin-top:12px;">                          9326
      <button type="submit" class="btn">录入该专家矩阵</button>           9327
      <a class="btn btn--ghost" href="/ahp/indicators">指标维护<          9328
    /a>                                                                   9329
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>             9330
    </p>                                                                  9331
  </form>                                                                 9332
</div>                                                                    9333
{% endblock %}                                                            9334
# ==== app/templates/ahp/experts_aggregate.html ====                      9335
{% extends "base.html" %}                                                 9336
{% block title %}多专家矩阵汇总{% endblock %}                             9337
{% block content %}                                                       9338
<div class="card">                                                        9339
  <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{%   9340
    else %}二级指标（{{ l1 }}）{% endif %}</h2>                           9341
  {% if not result %}                                                     9342
  <p class="muted">                                                       9343
    该范围内尚未录入专家矩阵。                                           9344
    <a href="/ahp/experts?unit_type={{ unit_type                          9345
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif                   9346
    %}">去录入</a>                                                       9347
  </p>                                                                    9348
  {% else %}                                                              9349
  <div class="stat-row">                                                  9350
```

```
    <div>                                                                    9351
      <div class="stat__label">参与汇总的专家</div>                          9352
      <div class="stat__value">{{ result.count }} 位</div>                   9353
    </div>                                                                   9354
    <div>                                                                    9355
      <div class="stat__label">λmax</div>                                    9356
      <div class="stat__value mono">{{ result.lambda_max                     9357
    }}</div>                                                                 9358
    </div>                                                                   9359
    <div>                                                                    9360
      <div class="stat__label">CI</div>                                      9361
      <div class="stat__value mono">{{                                       9362
    "%.4f"|format(result.ci) }}</div>                                        9363
    </div>                                                                   9364
    <div>                                                                    9365
      <div class="stat__label">CR（要求 &lt; 0.1）</div>                     9366
      <div class="stat__value mono">{{                                       9367
    "%.4f"|format(result.cr) }}</div>                                        9368
    </div>                                                                   9369
    <div>                                                                    9370
      <div class="stat__label">一致性检验</div>                              9371
      <div class="stat__value">                                              9372
        {% if result.consistent %}<span class="ok">通过</span>               9373
        {% else %}<span class="warn">不通过</span>{% endif %}                9374
      </div>                                                                 9375
    </div>                                                                   9376
  </div>                                                                     9377
  <h3>汇总后权重</h3>                                                        9378
  <table style="width:auto;">                                                9379
    <thead>                                                                  9380
      <tr>                                                                   9381
        <th scope="col">指标</th>                                            9382
        {% for name in names %}<th scope="col">{{ name                       9383
    }}</th>{% endfor %}                                                      9384
      </tr>                                                                  9385
    </thead>                                                                 9386
    <tbody>                                                                  9387
      <tr>                                                                   9388
        <th scope="row">权重</th>                                            9389
        {% for w in result.weights %}<td class="mono">{{ w                   9390
    }}</td>{% endfor %}                                                      9391
      </tr>                                                                  9392
    </tbody>                                                                 9393
  </table>                                                                   9394
  <h3>各专家矩阵的一致性</h3>                                                9395
  <table style="width:auto;">                                                9396
    <thead>                                                                  9397
      <tr><th scope="col">专家</th><th scope="col">该矩阵                    9398
    CR</th><th scope="col">判定</th></tr>                                    9399
    </thead>                                                                 9400
```

```
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.expert_name }}</td>
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is
    not none else "—" }}</td>
        <td>
          {% if r.cr is not none and r.cr < 0.1 %}<span
    class="ok">通过</span>
          {% else %}<span class="warn">不通过</span>{% endif %}
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% if not result.consistent and result.most_inconsistent
    %}
  <p style="padding:12px;border-radius:6px;background:var(--
    danger-soft);
              border-left:4px solid var(--danger);">
    <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{
    result.most_inconsistent.j + 1 }} 列的比较
    （当前 <span class="mono">{{ result.most_inconsistent.value
    |round(3) }}</span>，
    理想约 <span class="mono">{{ result.most_inconsistent.expec
    ted|round(3) }}</span>）。
    该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。
  </p>
  {% endif %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <form method="post"
          action="/ahp/experts?unit_type={{ unit_type
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"
          style="display:inline;"
          onsubmit="return confirm('保存后，「{{ unit_type
    }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">
      {{ form.hidden_tag() }}
      <input type="hidden" name="action" value="save_agg">
      <button type="submit" class="btn" {% if not
    result.consistent %}disabled{% endif %}>
        保存为生效权重
      </button>
    </form>
    <a class="btn btn--ghost" href="/ahp/experts?unit_type={
    { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1
    }}{% endif %}">返回录入</a>
    <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
  </p>
```

```
</div>
{% endblock %}
# ==== app/templates/indicator/index.html ====
{% extends "base.html" %}
{% block title %}指标数据采集{% endblock %}
{% block content %}
<div class="card">
  <h2>指标数据采集与管理</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/indicator/import">批量导入（CSV / Excel）</a>
  </p>
  {% if not patches %}
  <p class="muted">
    暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
  </p>
  {% else %}
  {% for p in patches %}
  <div class="patch-panel">
    <div class="patch-panel__head">
      <h3 class="patch-panel__name">{{ p.name }}</h3>
      <span class="patch-panel__meta">
        所属项目：{{ p.project.name if p.project else "—" }}
      </span>
      <span class="patch-panel__meta">
        微地质单元：{{ p.units.count() }} 个
      </span>
      {% if p.units.count() %}
      <span class="patch-panel__meta patch-panel__tail">
        <a class="link-btn" href="/indicator/import">批量导入本图斑数据</a>
      </span>
      {% endif %}
    </div>
    {% if not p.units.count() %}
    <div class="notice">
      <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
      <p style="margin:0;">
        指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
        请先完成划分，再回到本页录入。
      </p>
      <p style="margin:10px 0 0;">
        <a class="btn btn--sm" href="/unit/{{ p.id }}">去划分微地质单元</a>
      </p>
    </div>
    {% else %}
    <table>
```

```
        <thead>                                                                  9501
          <tr>                                                                   9502
            <th scope="col">微地质单元</th>                                      9503
            <th scope="col">面积占比</th>                                        9504
            <th scope="col">数据状态</th>                                        9505
            <th scope="col">操作</th>                                            9506
          </tr>                                                                  9507
        </thead>                                                                 9508
        <tbody>                                                                  9509
          {% for u in p.units %}                                                 9510
          <tr>                                                                   9511
            <td>{{ u.unit_type }}</td>                                           9512
            <td class="num">                                                     9513
              {% if u.ratio is not none %}{{                                     9514
      "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}                    9515
            </td>                                                                9516
            <td>                                                                 9517
              {% if u.indicator_data %}                                          9518
              <span class="tag tag--ok">已录入</span>                            9519
              {% else %}                                                         9520
              <span class="tag tag--muted">未录入</span>                         9521
              {% endif %}                                                        9522
            </td>                                                                9523
            <td class="nowrap">                                                  9524
              <span class="row-actions">                                         9525
                <a class="link-btn" href="/indicator/{{ u.id                     9526
      }}">录入 / 编辑</a>                                                        9527
              </span>                                                            9528
            </td>                                                                9529
          </tr>                                                                  9530
          {% endfor %}                                                           9531
        </tbody>                                                                 9532
      </table>                                                                   9533
      {% endif %}                                                                9534
    </div>                                                                       9535
    {% endfor %}                                                                 9536
    {% endif %}                                                                  9537
</div>                                                                           9538
{% endblock %}                                                                   9539
# ==== app/templates/indicator/edit.html ====                                    9540
{% extends "base.html" %}                                                        9541
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}               9542
{% block content %}                                                              9543
<div class="card" style="max-width:820px;">                                      9544
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>                                   9545
  <p class="muted">                                                              9546
    输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。  9547
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。    9548
  </p>                                                                           9549
  <form method="post">                                                           9550
```

```
{{ form.hidden_tag() }}
{% for group_name, fields, has_hm in groups %}
<h3>{{ group_name }}</h3>
{% for fname, cname, unit_ in fields %}
<div class="field">
  <label class="field__label" for="f-{{ fname }}">
    {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}
  </label>
  {% if fname == "aspect" %}
  <select id="f-{{ fname }}" name="aspect">
    <option value="">— 不适用 —</option>
    {% for a in aspect_types %}
    <option value="{{ a }}" {% if current.get("aspect")
== a %}selected{% endif %}>{{ a }}</option>
    {% endfor %}
  </select>
  {% else %}
  <input id="f-{{ fname }}" type="number" step="any"
name="{{ fname }}"
         value="{{ current.get(fname) if
current.get(fname) is not none else '' }}">
  {% endif %}
</div>
{% endfor %}
{% endfor %}
{% for group_name, fields, has_hm in groups %}
  {% if has_hm %}
  <h3>重金属含量（mg/kg）</h3>
  <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>
  <table style="width:auto;">
    <thead>
      <tr>
        {% for m in heavy_metals %}<th scope="col">{{ m
}}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      <tr>
        {% for m in heavy_metals %}
        <td>
          <label class="muted" style="position:absolute;
left:-9999px;"
                 for="hm-{{ m }}">{{ m }} 含量</label>
          <input id="hm-{{ m }}" type="number"
step="any" name="hm_{{ m }}"
                 value="{{ current.get('heavy',
{}).get(m) if current.get('heavy', {}).get(m) is not
none else '' }}">
        </td>
        {% endfor %}
```

```
            </tr>                                                    9601
          </tbody>                                                   9602
        </table>                                                     9603
        {% endif %}                                                  9604
      {% endfor %}                                                   9605
      <p class="actions" style="margin-top:18px;">                   9606
        <button type="submit" class="btn">保存</button>              9607
        <a class="btn btn--ghost" href="/indicator/">返回</a>        9608
      </p>                                                           9609
    </form>                                                          9610
</div>                                                               9611
{% endblock %}                                                       9612
# ==== app/templates/indicator/import.html ====                      9613
{% extends "base.html" %}                                            9614
{% block title %}批量导入指标数据{% endblock %}                      9615
{% block content %}                                                  9616
<div class="card" style="max-width:860px;">                          9617
  <h2>批量导入指标数据</h2>                                          9618
  <p class="muted">                                                  9619
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，    9620
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。            9621
    先下载模板按格式填写，可减少出错。                              9622
  </p>                                                               9623
  <p class="actions" style="margin-bottom:14px;">                    9624
    <a class="btn btn--sm btn--ghost"                                9625
    href="/indicator/import/template">下载导入模板（CSV）</a>        9626
    <a class="btn btn--sm btn--ghost"                                9627
    href="/indicator/">返回指标数据</a>                              9628
  </p>                                                               9629
  <form method="post" enctype="multipart/form-data">                 9630
    {{ form.hidden_tag() }}                                          9631
    <div class="field">                                              9632
      <label class="field__label" for="patch-id">目标矿山图斑</lab9633
    el>                                                              9634
      <select id="patch-id" name="patch_id" required>                9635
        <option value="">— 请选择 —</option>                         9636
        {% for p in patches %}                                       9637
        <option value="{{ p.id }}">{{ p.name }} ({{                  9638
    p.units.count() }} 个单元）</option>                             9639
        {% endfor %}                                                 9640
      </select>                                                      9641
    </div>                                                           9642
    <div class="field">                                              9643
      <label class="field__label" for="datafile">数据文件</label9644
    >                                                                9645
      <input id="datafile" type="file" name="datafile"               9646
             accept=".csv,.xlsx,.xls" required>                      9647
    </div>                                                           9648
    <p class="actions" style="margin-top:16px;">                     9649
      <button type="submit" class="btn">开始导入</button>            9650
```

```
    </p>
  </form>
  <h3>模板列说明</h3>
  <p class="muted">
    共 {{ headers|length }} 列。留空表示该指标「不适用」。
    坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
    （按论文注释视为未受污染）。
  </p>
  <p class="mono muted" style="line-height:1.9;">{{
    headers|join(" , ") }}</p>
  <h3>导入规则</h3>
  <ul class="muted" style="padding-left:20px;line-height:1.9
    ;">
    <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
    <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
    <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
    <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
    <span class="mono">10cm</span>）。</li>
  </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
  <h2>导入结果</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">目标图斑</div>
      <div style="font-size:15px;">{{ report.patch }}</div>
    </div>
    <div>
      <div class="stat__label">成功导入</div>
      <div class="stat__value ok">{{ report.saved }}
    个单元</div>
    </div>
    <div>
      <div class="stat__label">问题行</div>
      <div class="stat__value {% if report.errors %}warn{%
    endif %}">{{ report.errors|length }}</div>
    </div>
  </div>
  {% if report.errors %}
  <h3>未导入的行</h3>
  <ul class="muted" style="padding-left:20px;line-height:1.9
    ;">
    {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
  </ul>
  {% else %}
  <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
```

```
    </p>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/evaluate/index.html ====
{% extends "base.html" %}
{% block title %}适宜性定量评价{% endblock %}
{% block content %}
<div class="card">
    <h2>适宜性定量评价</h2>
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">所属项目</th>
          <th scope="col">矿山图斑</th>
          <th scope="col">前置条件</th>
          <th scope="col">最近评价结果</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in patches %}
        {% set pr = progress.get(p.id) %}
        <tr>
          <td>{{ p.project.name if p.project else "—" }}</td>
          <td style="text-align:left;">{{ p.name }}</td>
          <td style="text-align:left;">
            {% if pr and pr.ready %}
            <span class="tag tag--ok">已具备</span>
            {% elif pr %}
            <span class="tag tag--warn">缺：{{
    pr.blocking[0].label }}</span>
            <span class="muted">{{ pr.blocking[0].reason
    }}</span>
            {% endif %}
          </td>
          <td>
            {% if latest.get(p.id) %}
            F = <span class="mono num">{{
    latest[p.id].final_score }}</span>,
            <span class="tag tag--ok">{{ latest[p.id].grade
    }}</span>
            {% else %}
            <span class="tag tag--muted">未评价</span>
            {% endif %}
          </td>
          <td class="nowrap">
```

```
            <span class="row-actions">
              <a class="link-btn" href="/evaluate/{{ p.id
    }}">执行评价</a>
              {% if latest.get(p.id) %}
              <a class="link-btn" href="/evaluate/result/{{
    latest[p.id].id }}">查看结果</a>
              {% endif %}
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/evaluate/run.html ====
{% extends "base.html" %}
{% import "_macros.html" as ui %}
{% block title %}执行评价 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:880px;">
    <h2>执行适宜性评价 · {{ patch.name }}</h2>
    {{ ui.workflow_steps(progress) }}
    {{ ui.workflow_blocking(progress) }}
    {% if progress.ready %}
    <form method="post">
      {{ form.hidden_tag() }}
      <p class="actions">
        <button type="submit" class="btn">开始评价</button>
        <a class="btn btn--ghost" href="/evaluate/">返回</a>
      </p>
    </form>
    {% else %}
    <p class="actions">
      <button type="button" class="btn" disabled>开始评价</button>
      <a class="btn btn--ghost" href="/evaluate/">返回</a>
    </p>
    {% endif %}
</div>
<div class="card" style="max-width:880px;">
    <h2>当前流程状态</h2>
    <table>
      <thead>
        <tr><th scope="col">步骤</th><th scope="col">状态</th><th
    scope="col">说明</th></tr>
      </thead>
      <tbody>
        <tr>
          <td>限制性条件判定</td>
```

```
          <td>                                                                  9801
            {% if progress.constraint.passed %}<span                            9802
      class="tag tag--ok">已通过</span>                                         9803
            {% elif progress.constraint.done %}<span                            9804
      class="tag tag--warn">未通过</span>                                       9805
            {% else %}<span class="tag tag--muted">待完成</span>{               9806
      % endif %}                                                                9807
          </td>                                                                 9808
          <td style="text-align:left;">{{                                       9809
      progress.constraint.detail }}</td>                                        9810
        </tr>                                                                   9811
        <tr>                                                                    9812
          <td>划分微地质单元</td>                                               9813
          <td>                                                                  9814
            {% if progress.units.done %}<span class="tag                        9815
      tag--ok">已完成</span>                                                    9816
            {% else %}<span class="tag tag--muted">待完成</span>{               9817
      % endif %}                                                                9818
          </td>                                                                 9819
          <td style="text-align:left;">{{                                       9820
      progress.units.detail }}</td>                                             9821
        </tr>                                                                   9822
        <tr>                                                                    9823
          <td>录入指标数据</td>                                                 9824
          <td>                                                                  9825
            {% if progress.data.ready %}<span class="tag                        9826
      tag--ok">已完成</span>                                                    9827
            {% else %}<span class="tag tag--muted">待完成</span>{               9828
      % endif %}                                                                9829
          </td>                                                                 9830
          <td style="text-align:left;">{{                                       9831
      progress.data.detail }}</td>                                              9832
        </tr>                                                                   9833
      </tbody>                                                                  9834
    </table>                                                                    9835
</div>                                                                          9836
{% if weight_sources %}                                                         9837
<div class="card" style="max-width:880px;">                                     9838
    <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">            9839
      <thead>                                                                   9840
        <tr><th scope="col">微地质单元</th><th                                  9841
      scope="col">权重来源</th></tr>                                            9842
      </thead>                                                                  9843
      <tbody>                                                                   9844
        {% for ut, label in weight_sources.items() %}                           9845
        <tr>                                                                    9846
          <td style="text-align:left;">{{ ut }}</td>                           9847
          <td style="text-align:left;">                                         9848
            {% if "自定义" in label %}                                          9849
            <span class="tag tag--warn">{{ label }}</span>                      9850
```

```
            {% else %}                                                    9851
            <span class="tag tag--muted">{{ label }}</span>               9852
            {% endif %}                                                   9853
          </td>                                                           9854
        </tr>                                                             9855
        {% endfor %}                                                      9856
      </tbody>                                                            9857
    </table>                                                              9858
</div>                                                                    9859
{% endif %}                                                               9860
{% endblock %}                                                            9861
# ==== app/templates/evaluate/result.html ====                            9862
{% extends "base.html" %}                                                 9863
{% block title %}评价结果{% endblock %}                                    9864
{% block content %}                                                       9865
<div class="card">                                                        9866
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>              9867
  <div class="stat-row">                                                  9868
    <div>                                                                 9869
      <div class="stat__label">最终评价因子 F</div>                        9870
      <div class="stat__value mono">{{ r.final_score                      9871
    }}</div>                                                              9872
    </div>                                                                9873
    <div>                                                                 9874
      <div class="stat__label">适宜性等级</div>                           9875
      <div class="stat__value ok">{{ r.grade }}</div>                     9876
    </div>                                                                9877
    <div style="flex:1;min-width:240px;">                                 9878
      <div class="stat__label">推荐措施</div>                             9879
      <div style="font-size:15px;">{{ r.measures }}</div>                 9880
    </div>                                                                9881
  </div>                                                                  9882
  {% if unit_scores %}                                                    9883
  <h3>各微地质单元得分</h3>                                               9884
  <table style="max-width:860px;">                                        9885
    <thead>                                                               9886
      <tr>                                                                9887
        <th scope="col">微地质单元</th>                                   9888
        <th scope="col">面积占比</th>                                     9889
        <th scope="col">单元评价因子 F'</th>                              9890
        <th scope="col">加权贡献</th>                                     9891
        <th scope="col">权重来源</th>                                     9892
      </tr>                                                               9893
    </thead>                                                              9894
    <tbody>                                                               9895
      {% for u in unit_scores %}                                          9896
      <tr>                                                                9897
        <td>{{ u.unit_type }}</td>                                        9898
        <td class="num">{{ "%.2f%%"|format(u.ratio * 100)                 9899
    }}</td>                                                               9900
```

```
            <td class="mono">{{ u.unit_score }}</td>                    9901
            <td class="mono">{{ "%.4f"|format(u.ratio *                 9902
        u.unit_score) }}</td>                                           9903
            <td class="muted">{{ u.weight_source or "论文表 3           9904
        默认权重" }}</td>                                               9905
          </tr>                                                         9906
          {% endfor %}                                                  9907
        </tbody>                                                        9908
      </table>                                                          9909
      {% endif %}                                                       9910
      <p class="actions" style="margin-top:18px;">                      9911
        <a class="btn btn--ghost" href="/report/{{ r.patch_id           9912
        }}">查看得分明细</a>                                           9913
        <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>    9914
      </p>                                                              9915
</div>                                                                  9916
{% endblock %}                                                          9917
# ==== app/templates/report/index.html ====                             9918
{% extends "base.html" %}                                               9919
{% block title %}报告输出{% endblock %}                                 9920
{% block content %}                                                     9921
<div class="card">                                                      9922
  <h2>报告与结果输出</h2>                                               9923
  {% if not patches %}                                                  9924
  <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>  9925
  {% else %}                                                            9926
  <table>                                                               9927
    <thead>                                                             9928
      <tr>                                                              9929
        <th scope="col">所属项目</th>                                   9930
        <th scope="col">矿山图斑</th>                                   9931
        <th scope="col">可生成报告</th>                                 9932
        <th scope="col">操作</th>                                       9933
      </tr>                                                             9934
    </thead>                                                            9935
    <tbody>                                                             9936
      {% for p in patches %}                                            9937
      {% set pr = progress.get(p.id) %}                                 9938
      <tr>                                                              9939
        <td>{{ p.project.name if p.project else "—" }}</td>             9940
        <td style="text-align:left;">{{ p.name }}</td>                  9941
        <td style="text-align:left;">                                   9942
          {% if pr and pr.ready %}                                      9943
          <span class="tag tag--ok">可以</span>                         9944
          {% elif pr %}                                                 9945
          <span class="tag tag--warn">暂不可</span>                     9946
          <span class="muted">{{ pr.blocking[0].reason                  9947
    }}</span>                                                           9948
          {% endif %}                                                   9949
        </td>                                                           9950
```

```
            <td class="nowrap">
              {% if pr and pr.ready %}
              <span class="row-actions">
                <a class="link-btn" href="/report/{{ p.id
    }}">查看得分明细</a>
                <a class="link-btn" href="/report/{{ p.id
    }}/pdf">导出 PDF</a>
                <a class="link-btn" href="/report/{{ p.id
    }}/word">导出 Word</a>
              </span>
              {% else %}
              <span class="muted">—</span>
              {% endif %}
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
    </div>
    {% endblock %}
    # ==== app/templates/report/detail.html ====
    {% extends "base.html" %}
    {% block title %}得分明细 · {{ patch.name }}{% endblock %}
    {% block content %}
    <div class="card">
      <h2>得分明细 · {{ patch.name }}</h2>
      <div class="stat-row">
        <div>
          <div class="stat__label">最终评价因子 F</div>
          <div class="stat__value mono">{{
        "%.4f"|format(result.final_score) }}</div>
        </div>
        <div>
          <div class="stat__label">适宜性等级</div>
          <div class="stat__value ok">{{ result.grade }}</div>
        </div>
        <div style="flex:1;min-width:240px;">
          <div class="stat__label">推荐措施</div>
          <div style="font-size:15px;">{{
        result.grade_info.measure }}</div>
        </div>
      </div>
      <p class="muted" style="margin-top:12px;">
        本页与导出报告均取自
        <strong>{% if record %}{{ record.created_at |
        localdt("%Y-%m-%d %H:%M:%S") }}{% endif %}
        记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else
        "—" }}），
        不会因事后修改指标数据而变化。
```

```
  </p>                                                                  10001
  <p class="actions" style="margin-top:16px;">                          10002
    <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF           10003
    报告</a>                                                            10004
    <a class="btn btn--ghost" href="/report/{{ patch.id                 10005
    }}/word">导出 Word 报告</a>                                         10006
    <a class="btn btn--ghost" href="/report/">返回</a>                  10007
  </p>                                                                  10008
</div>                                                                  10009
<div class="card">                                                      10010
  <h2>结果图表</h2>                                                     10011
  <h3>各微地质单元得分对比</h3>                                         10012
  <div>{{ chart_units|safe }}</div>                                     10013
  <h3>各一级指标平均得分</h3>                                           10014
  <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>  10015
  <div>{{ chart_weight|safe }}</div>                                    10016
</div>                                                                  10017
{% for u in result.unit_results %}                                      10018
<div class="card">                                                      10019
  <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *             10020
    100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>           10021
  {% if u.weight_source %}                                              10022
  <p class="muted" style="margin-top:-6px;">权重来源：{{                10023
    u.weight_source }}</p>                                              10024
  {% endif %}                                                           10025
  <table>                                                               10026
    <thead>                                                             10027
      <tr>                                                              10028
        <th scope="col">一级指标</th>                                   10029
        <th scope="col">二级指标</th>                                   10030
        <th scope="col">实测值</th>                                     10031
        <th scope="col">标准化值</th>                                   10032
        <th scope="col">权重</th>                                       10033
        <th scope="col">贡献</th>                                       10034
      </tr>                                                             10035
    </thead>                                                            10036
    <tbody>                                                             10037
      {% for d in u.details %}                                          10038
      <tr>                                                              10039
        <td>{{ d.l1 }}</td>                                            10040
        <td style="text-align:left;">{{ d.l2 }}</td>                    10041
        <td class="mono" style="text-align:left;">{{ "—" if 10042
    d.raw is none else d.raw }}</td>                                    10043
        <td class="mono">{{ "—" if d.std_value is none else 10044
    "%.4f"|format(d.std_value) }}</td>                                  10045
        <td class="mono">{{ "%.4f"|format(d.weight) }}</td> 10046
        <td class="mono">{{ "—" if d.contribution is none   10047
    else "%.4f"|format(d.contribution) }}</td>                          10048
      </tr>                                                             10049
      {% endfor %}                                                      10050
```

```
      </tbody>
    </table>
    <p class="muted" style="margin:0;">
      一级指标得分：
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span
      class="num">{{ "%.4f"|format(v) }}</span>{% if not
      loop.last %};{% endif %}{% endfor %}
    </p>
</div>
{% endfor %}
{% endblock %}
# ==== app/templates/report/print.html ====
<!DOCTYPE html>
<html lang="zh-CN">
<head>
<meta charset="UTF-8">
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>
<style>
  @page { size: A4; margin: 18mm 16mm; }
  * { box-sizing: border-box; }
  body {
    margin: 0; color: #23262b; background: #fff;
    font-family: "Microsoft YaHei", "SimSun", sans-serif;
    font-size: 10.5pt; line-height: 1.65;
  }
  h1 { font-size: 17pt; text-align: center; margin: 0 0
    6px; }
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;
       border-bottom: 1px solid #dfe3e8; padding-bottom:
    4px; }
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0
    6px; }
  .sub { text-align: center; color: #6b7280; font-size:
    9.5pt; margin-bottom: 18px; }
  table { width: 100%; border-collapse: collapse; margin:
    8px 0 14px;
          font-variant-numeric: tabular-nums; }
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;
    text-align: center; }
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace; }
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
```

```
  svg { max-width: 100%; }                                                    10101
</style>                                                                      10102
</head>                                                                       10103
<body>                                                                        10104
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>                                    10105
  <p class="sub">                                                             10106
    {{ patch.name }}                                                          10107
    {% if patch.admin_region %} · {{ patch.admin_region                       10108
    }}{% endif %}                                                             10109
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0 ·                                10110
    生成时间 {{ generated_at | localdt }}                                      10111
  </p>                                                                        10112
  <h2>一、图斑基本信息</h2>                                                       10113
  <table class="kv">                                                          10114
    <tr><th>图斑名称</th><td>{{ patch.name }}</td></tr>                          10115
    <tr><th>所在行政区</th><td>{{ patch.admin_region or "—"                    10116
    }}</td></tr>                                                              10117
    <tr><th>图斑面积（公顷）</th><td>{{ patch.area_ha if                        10118
    patch.area_ha is not none else "—" }}</td></tr>                           10119
    <tr><th>开采矿种</th><td>{{ patch.mineral_type or "—"                     10120
    }}</td></tr>                                                              10121
    <tr><th>开采方式</th><td>{{ patch.mining_method or "—"                    10122
    }}</td></tr>                                                              10123
    <tr><th>责任主体缺失</th><td>{{ "是" if patch.responsible_entit10124
    y_missing else "否" }}</td></tr>                                          10125
    {% if reference %}                                                        10126
    <tr><th>参照生态系统</th>                                                  10127
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%,                10128
            多样性指数基准 {{ reference.shannon_benchmark                     10129
    }}</td></tr>                                                              10130
    {% endif %}                                                               10131
  </table>                                                                    10132
  <h2>二、评价结论</h2>                                                         10133
  <table class="kv">                                                          10134
    <tr><th>最终评价因子 F</th><td class="mono"><strong>{{                    10135
    "%.4f"|format(result.final_score) }}</strong></td></tr>10136
    <tr><th>适宜性等级</th><td><strong>{{ result.grade                        10137
    }}</strong></td></tr>                                                     10138
    <tr><th>判定依据</th><td>{{ result.grade_info.meaning                     10139
    }}</td></tr>                                                              10140
    <tr><th>推荐措施</th><td>{{ result.grade_info.measure                     10141
    }}</td></tr>                                                              10142
  </table>                                                                    10143
  <h2>三、各微地质单元得分</h2>                                                   10144
  <table>                                                                     10145
    <thead>                                                                   10146
      <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子                    10147
    F'</th><th>加权贡献</th></tr>                                              10148
    </thead>                                                                  10149
    <tbody>                                                                   10150
```

```
    {% for u in result.unit_results %}
    <tr>
      <td>{{ u.unit_type }}</td>
      <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
}}</td>
      <td class="mono">{{ "%.4f"|format(u.unit_score)
}}</td>
      <td class="mono">{{ "%.4f"|format(u.weighted)
}}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
<div class="chart">{{ chart_units|safe }}</div>
<h2>四、一级指标权重分布</h2>
<div class="chart">{{ chart_weight|safe }}</div>
<h2>五、得分明细</h2>
{% for u in result.unit_results %}
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *
100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
<table>
  <thead>
    <tr>
      <th>一级指标</th><th>二级指标</th><th>实测值</th>
      <th>标准化值</th><th>权重</th><th>贡献</th>
    </tr>
  </thead>
  <tbody>
    {% for d in u.details %}
    <tr>
      <td>{{ d.l1 }}</td>
      <td class="left">{{ d.l2 }}</td>
      <td class="mono left">{{ "—" if d.raw is none else
d.raw }}</td>
      <td class="mono">{{ "—" if d.std_value is none else
"%.4f"|format(d.std_value) }}</td>
      <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
      <td class="mono">{{ "—" if d.contribution is none
else "%.4f"|format(d.contribution) }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
<p class="note">
  一级指标得分：
  {% for k, v in u.l1_scores.items() %}{{ k }} = {{
  "%.4f"|format(v) }}{% if not loop.last %}；{% endif %}{%
  endfor %}
</p>
{% endfor %}
```

```
    <h2>六、评价依据说明</h2>
    <ul class="note">
      <li>指标标准化按论文式（8） 执行，按指标性质分为越大越优、越小越优、
          适中型与离散赋值四类。</li>
      <li>单元评价因子按论文式（9） 由二级指标权重加权求得；
          整矿评价因子由各微地质单元按面积占比加权合成。</li>
      <li>适宜性等级按论文表 4 划分：F&lt;0.3 不适宜、0.3≤F≤0.7 基本适宜、F≥0.7
      适宜。</li>
      <li>指标权重取自论文表 3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>
    </ul>
    <p class="sig">
      评价人（签字）：_____　　日期：_____
    </p>
  </body>
</html>
# ==== app/templates/search.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="card">
  <h2>搜索结果</h2>
  {% if not q %}
  <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区 / 矿种、
    标准名称与文号。</p>
  {% else %}
  <p class="muted">
    关键词「<strong>{{ q }}</strong>」共找到
    <span class="num">{{ total }}</span> 条结果。
  </p>
  {% if not total %}
  <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>
  {% endif %}
  {% endif %}
</div>
{% if projects %}
<div class="card">
  <h2>评价项目（{{ projects|length }}）</h2>
  <table>
    <thead>
      <tr>
        <th scope="col">项目名称</th>
        <th scope="col">编号</th>
        <th scope="col">状态</th>
        <th scope="col">图斑数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in projects %}
```

```
        <tr>                                                                    10251
          <td style="text-align:left;">{{ p.name }}</td>                        10252
          <td class="mono">{{ p.code or "—" }}</td>                             10253
          <td><span class="tag tag--muted">{{ p.status                          10254
    }}</span></td>                                                              10255
          <td class="num">{{ p.patches.count() }}</td>                          10256
          <td class="nowrap">                                                   10257
            <span class="row-actions">                                          10258
              <a class="link-btn" href="/project/{{ p.id                        10259
    }}/patches">矿山图斑</a>                                                    10260
              <a class="link-btn" href="/project/{{ p.id                        10261
    }}/edit">编辑</a>                                                           10262
            </span>                                                             10263
          </td>                                                                 10264
        </tr>                                                                   10265
        {% endfor %}                                                            10266
      </tbody>                                                                  10267
    </table>                                                                    10268
</div>                                                                          10269
{% endif %}                                                                     10270
{% if patches %}                                                                10271
<div class="card">                                                              10272
  <h2>矿山图斑（{{ patches|length }}）</h2>                                     10273
  <table>                                                                       10274
    <thead>                                                                     10275
      <tr>                                                                      10276
        <th scope="col">所属项目</th>                                           10277
        <th scope="col">图斑名称</th>                                           10278
        <th scope="col">行政区</th>                                             10279
        <th scope="col">矿种</th>                                               10280
        <th scope="col">操作</th>                                               10281
      </tr>                                                                     10282
    </thead>                                                                    10283
    <tbody>                                                                     10284
      {% for m in patches %}                                                    10285
      <tr>                                                                      10286
        <td>{{ m.project.name if m.project else "—" }}</td>                     10287
        <td style="text-align:left;">{{ m.name }}</td>                          10288
        <td>{{ m.admin_region or "—" }}</td>                                    10289
        <td>{{ m.mineral_type or "—" }}</td>                                    10290
        <td class="nowrap">                                                     10291
          <span class="row-actions">                                            10292
            <a class="link-btn" href="/constraint/{{ m.id                       10293
    }}">限制性判定</a>                                                          10294
            <a class="link-btn" href="/unit/{{ m.id                             10295
    }}">微地质单元</a>                                                          10296
            <a class="link-btn" href="/evaluate/{{ m.id                         10297
    }}">执行评价</a>                                                            10298
          </span>                                                               10299
        </td>                                                                   10300
```

```
        </tr>                                                                  10301
        {% endfor %}                                                           10302
      </tbody>                                                                 10303
    </table>                                                                   10304
</div>                                                                         10305
{% endif %}                                                                    10306
{% if standards %}                                                             10307
<div class="card">                                                             10308
  <h2>标准与政策文件（{{ standards|length }}）</h2>                              10309
  <table>                                                                      10310
    <thead>                                                                    10311
      <tr>                                                                     10312
        <th scope="col">类别</th>                                              10313
        <th scope="col">省份</th>                                              10314
        <th scope="col">名称</th>                                              10315
        <th scope="col">文号</th>                                              10316
        <th scope="col">操作</th>                                              10317
      </tr>                                                                    10318
    </thead>                                                                   10319
    <tbody>                                                                    10320
      {% for s in standards %}                                                 10321
      <tr>                                                                     10322
        <td>{{ s.category }}</td>                                              10323
        <td>{{ s.province }}</td>                                              10324
        <td style="text-align:left;">{{ s.name }}</td>                         10325
        <td class="mono">{{ s.doc_number or "—" }}</td>                        10326
        <td class="nowrap">                                                    10327
          <span class="row-actions">                                           10328
            <a class="link-btn" href="/standard/{{ s.id                        10329
    }}/rules">规则</a>                                                         10330
          </span>                                                              10331
        </td>                                                                  10332
      </tr>                                                                    10333
      {% endfor %}                                                             10334
    </tbody>                                                                   10335
  </table>                                                                     10336
</div>                                                                         10337
{% endif %}                                                                    10338
{% endblock %}                                                                 10339
#  ==== seed_demo.py ====                                                      10340
#  -*- coding: utf-8 -*-                                                       10341
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。                              10342
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。                         10343
运行：python seed_demo.py                  幂等，只补写缺失的演示数据，          10344
                                               不会删除用户自建的项目与图斑      10345
      python seed_demo.py --reset          连同标准库与指标定义一并重建          10346
"""                                                                            10347
import json                                                                    10348
import sys                                                                     10349
sys.stdout.reconfigure(encoding="utf-8")                                       10350
```

```
from app import create_app
from app.models import (db, User, Project, MinePatch,
    ReferenceEcosystem,
                        MicroUnit, IndicatorData, Standard,
    StandardRule,
                        ConstraintCheck, EvaluationResult,
    ScoreDetail)
app = create_app()
#  内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」
DEMO_CODE = "DEMO-001"
#  论文表 1：11 项地方标准与政策文件
STANDARDS = [
    dict(category="地方标准", province="辽宁",
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="内蒙古",
    name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
    publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="江西",
    name="废弃矿山自然恢复技术规范",
        doc_number="DB36/T 2221—2026",
    publish_date="2026-02-02",
        implement_date="2026-08-01",
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),
    dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",
        doc_number=None, publish_date="2026-06-18",
    implement_date=None,
```

```
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",                    10401
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),                    10402
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),          10403
               ("与参照系差距", "<", "20", "充分条件", "%")]),                      10404
dict(category="政策文件", province="安徽",                                         10405
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",    10406
        doc_number="皖自然资修函[2020]34号",                                       10407
publish_date="2020-03-03", implement_date=None,                                   10408
        summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",10409
        rules=[("责任主体", "==", "政府", "必要条件", None),                       10410
               ("无地质灾害", "==", "是", "必要条件", None),                       10411
               ("无水土污染", "==", "是", "必要条件", None)]),                     10412
dict(category="政策文件", province="山东",                                         10413
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",10414
        doc_number="鲁自然资字[2022]104号",                                        10415
publish_date="2022-08-02", implement_date=None,                                   10416
        summary="条件与安徽省相同，另增加自然恢复认定标准",                        10417
        rules=[("无地质灾害", "==", "是", "必要条件", None),                       10418
               ("无水土污染", "==", "是", "必要条件", None)]),                     10419
dict(category="政策文件", province="湖南",                                         10420
        name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",      10421
        doc_number="湘自资办发[2023]58号",                                         10422
publish_date="2023-07-20", implement_date=None,                                   10423
        summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度10424
>75%、连续3年正向增长",                                                          10425
        rules=[("植被覆盖度", ">", "75", "必要条件", "%"),                         10426
               ("植被覆盖率趋势", "==", "连续3年正向增长", "必要条件", None),     10427
               ("无地质灾害", "==", "是", "必要条件", None),                       10428
               ("符合国土空间规划", "==", "是", "必要条件", None)]),               10429
dict(category="政策文件", province="新疆",                                         10430
        name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",        10431
        doc_number="新自然资源办函〔2024〕219号",                                  10432
publish_date="2024-12-20", implement_date=None,                                   10433
        summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",10434
        rules=[("植被覆盖度", ">=", "周边30%", "必要条件", None),                  10435
               ("无地质灾害", "==", "是", "必要条件", None),                       10436
               ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),               10437
dict(category="政策文件", province="广西",                                         10438
        name="广西壮族自治区历史遗留矿山自然恢复技术指南",                        10439
        doc_number="桂自然资办〔2024〕250号",                                      10440
publish_date="2024-12-30", implement_date=None,                                   10441
        summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",  10442
        rules=[("无地质灾害", "==", "是", "必要条件", None),                       10443
               ("植被自然恢复趋势", "==", "明显", "必要条件", None),               10444
               ("生产设备拆除完毕", "==", "是", "必要条件", None)]),               10445
dict(category="政策文件", province="西藏",                                         10446
        name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",        10447
        doc_number="藏自然资发〔2025〕19号",                                       10448
publish_date="2025-09-24", implement_date=None,                                   10449
        summary="含修复成本高、生态修复效益不显著等适用条件",                      10450
```

```
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
                     "slope": 28.75, "aspect": "阳坡"},
        },
        {
            "unit_type": "平台", "ratio": 0.2319,
            "data": {"soil_ratio": 0.56, "thickness": 30,
    "ph": 8.72,
                     "organic_matter": 10.92,
                     "heavy_metals": {"Hg": 0.02, "As":
    3.13, "Ni": 12.33,
                                      "Cu": 11.14, "Cd":
    0.09, "Pb": 9.98},
                     "coverage": 58.42, "shannon": 0.72},
        },
    ],
}
def clean_case(reset=False):
    """清理数据。
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，
    以便把演示环境恢复到初始状态。
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目
    会在每次启动时被删除。
    """
    from app.models import EvaluationResult, ScoreDetail,
    ConstraintCheck
```

```
    demo_projects = Project.query.filter_by(code=DEMO_CODE).
    all()
    demo_ids = [p.id for p in demo_projects]
    patch_ids = []
    if demo_ids:
        patch_ids = [p.id for p in MinePatch.query.filter(
            MinePatch.project_id.in_(demo_ids)).all()]
    if patch_ids:
        unit_ids = [u.id for u in MicroUnit.query.filter(
            MicroUnit.patch_id.in_(patch_ids)).all()]
        if unit_ids:
            IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).delete(
                synchronize_session=False)
        result_ids = [r.id for r in EvaluationResult.query.f
    ilter(
            EvaluationResult.patch_id.in_(patch_ids)).all()]
        if result_ids:
            ScoreDetail.query.filter(
                ScoreDetail.result_id.in_(result_ids)).delet
    e(
                synchronize_session=False)
        EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).delete
    (
            synchronize_session=False)
        ConstraintCheck.query.filter(
            ConstraintCheck.patch_id.in_(patch_ids)).delete(
            synchronize_session=False)
        ReferenceEcosystem.query.filter(
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele
    te(
            synchronize_session=False)
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_
    ids)).delete(
            synchronize_session=False)
        MinePatch.query.filter(MinePatch.id.in_(patch_ids)).
    delete(
            synchronize_session=False)
    if demo_ids:
        Project.query.filter(Project.id.in_(demo_ids)).delet
    e(
            synchronize_session=False)
    if reset:
        StandardRule.query.delete()
        Standard.query.delete()
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，
    #  避免长期运行后 score_details 表堆积无效数据。
    _purge_orphan_details()
    db.session.commit()
```

```
    return len(demo_ids)
def demo_patch():
    """取内置演示案例的图斑。
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。
    没有演示案例时才退回第一个图斑。
    """
    proj = Project.query.filter_by(code=DEMO_CODE).first()
    if proj is not None:
        m = MinePatch.query.filter_by(project_id=proj.id).first()
        if m is not None:
            return m
    return MinePatch.query.first()
def _purge_orphan_details():
    """清理指向已不存在评价记录的孤儿得分明细。
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。
    """
    live_ids = [r.id for r in EvaluationResult.query.all()]
    if live_ids:
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(
    live_ids)).delete(
            synchronize_session=False)
    else:
        ScoreDetail.query.delete()
    db.session.commit()
def seed(reset=False):
    with app.app_context():
        db.create_all()          # 全新环境下先建表，保证本脚本可独立运行
        _purge_orphan_details()
        admin = User.query.filter_by(username="admin").first
    ()
        if not admin:
            from werkzeug.security import
    generate_password_hash
            admin = User(username="admin",
                         password_hash=generate_password_has
    h("admin123"),
                         role="admin")
            db.session.add(admin)
            db.session.commit()
        # 幂等：演示案例已存在就不重复写入，更不清空用户自己的项目
        existing = Project.query.filter_by(code=DEMO_CODE).f
    irst()
        if existing and not reset:
            print("演示案例已存在，跳过写入（用户自建项目不受影响）")
            _report()
            _seed_reference_data(reset=False)
            return
        clean_case(reset=reset)
```

```
    project = Project(name=CASE["name"], code=DEMO_CODE,
                      description="论文第 5 节典型案例",
created_by=admin.id)
    db.session.add(project)
    db.session.flush()
    patch = MinePatch(project_id=project.id,
name=CASE["patch_name"],
                      admin_region="河北省唐山市玉田县",
                      area_ha=CASE["area_ha"],
                      mineral_type=CASE["mineral_type"],
                      mining_method="露天开采",
stop_years=None,
                      responsible_entity_missing=True)
    db.session.add(patch)
    db.session.flush()
    ref = ReferenceEcosystem(patch_id=patch.id,
                             coverage_benchmark=75.0,
shannon_benchmark=0.7)
    db.session.add(ref)
    for u in CASE["units"]:
        unit = MicroUnit(patch_id=patch.id,
unit_type=u["unit_type"],
                         ratio=u["ratio"], area=None)
        db.session.add(unit)
        db.session.flush()
        d = u["data"]
        ind = IndicatorData(
            unit_id=unit.id,
            soil_ratio=d.get("soil_ratio"),
            thickness=d.get("thickness"),
            ph=d.get("ph"),
            organic_matter=d.get("organic_matter"),
            heavy_metals=json.dumps(d["heavy_metals"],
ensure_ascii=False)
                         if d.get("heavy_metals") else
None,
            coverage=d.get("coverage"),
            shannon=d.get("shannon"),
            slope=d.get("slope"),
            aspect=d.get("aspect"),
        )
        db.session.add(ind)
    #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、
    #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。
    db.session.add(ConstraintCheck(
        patch_id=patch.id,
        policy_pass=True, location_pass=True,
pollution_pass=True,
        geo_pass=True, farmland_pass=True,
        passed=True, reason="全部通过"))
```

```
        db.session.commit()                                                    10651
        print("已写入案例：", CASE["name"])                                    10652
        print("   图斑：", CASE["patch_name"])                                 10653
        print("   微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台          10654
    23.19%")                                                                   10655
        print("   管理员账号：admin / admin123")                               10656
        _seed_reference_data(reset=reset)                                      10657
        _report()                                                              10658
def _seed_reference_data(reset=False):                                         10659
    """写入标准库与指标定义。                                                  10660
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护           10661
    的标准条目与指标定义。                                                     10662
    """                                                                        10663
    if reset or Standard.query.count() == 0:                                   10664
        seed_standards()                                                       10665
    from app.services.indicator_service import                                 10666
    seed_indicators                                                            10667
    n = seed_indicators()                                                      10668
    print("指标定义：%d 项" % n)                                              10669
def _report():                                                                 10670
    total_p = Project.query.count()                                            10671
    total_patch = MinePatch.query.count()                                      10672
    print()                                                                    10673
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))         10674
    if total_p > total_patch:                                                  10675
        print("   提示：项目是归档单位，评价针对的是图斑。")                   10676
        print("         新项目需先建立图斑，才会出现在后续各模块中。")         10677
def seed_standards():                                                          10678
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""                  10679
    StandardRule.query.delete()                                                10680
    Standard.query.delete()                                                    10681
    db.session.commit()                                                        10682
    for s in STANDARDS:                                                        10683
        std = Standard(category=s["category"],                                 10684
    province=s["province"],                                                    10685
                       name=s["name"],                                         10686
    doc_number=s["doc_number"],                                                10687
                       publish_date=s["publish_date"],                         10688
                       implement_date=s["implement_date"],                     10689
                       summary=s["summary"])                                   10690
        db.session.add(std)                                                    10691
        db.session.flush()                                                     10692
        for item, op, thr, ctype, note in s["rules"]:                          10693
            db.session.add(StandardRule(standard_id=std.id,                    10694
    item=item,                                                                 10695
                                        operator=op,                           10696
    threshold=thr,                                                             10697
                                                                               10698
    condition_type=ctype, note=note))                                          10699
    db.session.commit()                                                        10700
```

```
    print("已写入标准/政策文件：%d 项，规则 %d 条"                                10701
          % (len(STANDARDS), sum(len(s["rules"]) for s in                    10702
    STANDARDS)))                                                            10703
if __name__ == "__main__":                                                  10704
    # --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。                  10705
    # 无论是否 --reset，都不会删除用户自建的项目与图斑。                             10706
    seed(reset="--reset" in sys.argv[1:])                                   10707
# ==== 测试/test_ahp.py ====                                                10708
# -*- coding: utf-8 -*-                                                     10709
"""地基自检：验证 AHP 引擎与应用可启动。"""                                        10710
import sys                                                                  10711
sys.stdout.reconfigure(encoding="utf-8")                                    10712
import numpy as np                                                          10713
import os                                                                   10714
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                          10715
# 使 `from app import ...` 与 app/app.db 的定位不受影响。                     10716
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil                10717
    e__)))                                                                  10718
if ROOT not in sys.path:                                                    10719
    sys.path.insert(0, ROOT)                                                10720
from app.services import ahp_service as ahp                                 10721
LINE = "=" * 66                                                             10722
# ------------------------------------------------------------------1
    ------------------                                                      10724
# 1） 完全一致矩阵 → 权重应精确还原，CR=0                                        10725
# ------------------------------------------------------------------1
    ------------------                                                      10727
print(LINE)                                                                 10728
print("测试 1  完全一致判断矩阵的权重还原与一致性")                                10729
print(LINE)                                                                 10730
# 由权重构造完全一致矩阵：a_ij = w_i / w_j                                      10731
w_true = np.array([0.44, 0.31, 0.25])                                       10732
A = np.outer(w_true, 1.0 / w_true)                                          10733
r = ahp.weights_and_consistency(A)                                          10734
print("  判断矩阵 A：")                                                        10735
for row in A:                                                               10736
    print("    " + "  ".join(f"{v:8.4f}" for v in row))                     10737
print()                                                                     10738
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in                          10739
    r["weights"]))                                                          10740
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))                 10741
print("  λmax = %.6f   CI = %.3e   CR = %.3e   一致: %s"                     10742
      % (r["lambda_max"], r["ci"], r["cr"],                                 10743
    r["consistent"]))                                                       10744
assert np.allclose(r["weights"], w_true, atol=1e-6),                        10745
    "权重未精确还原"                                                          10746
assert r["consistent"], "完全一致矩阵应通过检验"                                  10747
# ------------------------------------------------------------------1
    ------------------                                                      10749
# 2） 上三角输入 → 完整矩阵 → 一致性定位                                         10750
```

```
# ----------------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 2  上三角录入与最不一致项定位")
print(LINE)
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）
M = ahp.matrix_from_upper([2, 5, 3])
r2 = ahp.weights_and_consistency(M)
bad = ahp.find_most_inconsistent(M)
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")
for row in M:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print("  权重：", "  ".join(f"{v:.4f}" for v in r2["weights"]),
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))
print("  最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],
    bad["expected"]))
# ----------------------------------------------------------------------
    ----------------
# 3) 多专家聚合（几何平均）
# ----------------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 3  多专家判断矩阵聚合（几何平均）")
print(LINE)
m1 = ahp.matrix_from_upper([2, 4, 3])
m2 = ahp.matrix_from_upper([3, 5, 2])
agg = ahp.aggregate_matrices([m1, m2])
print("  专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")
print("  聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"
      % (agg[0, 1], agg[0, 2], agg[1, 2]))
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"
# ----------------------------------------------------------------------
    ----------------
# 4) 论文表 3 权重可还原性
# ----------------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f   CR = %.3e   一致: %s" %
    (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ----------------------------------------------------------------------
    ----------------
# 5) 应用可启动 + 建表
```

```
# ----------------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表：" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检：用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ----------------------------------------------------------------------
    ----------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ----------------------------------------------------------------------
    ----------------
# 三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
```

```
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
# ------------------------------------------------------------------
    ------------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
```

```
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据   ：%s" % res["grade_info"]["meaning"])
print("  推荐措施   ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式（8）（9） 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
#  ==== 测试/test_web_ahp.py ====
#  -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
#  测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
#  建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
```

```
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_
    hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码:", r.status_code, "(应 200)")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码:", r.status_code, "(应 200)")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」:", "判断矩阵录入" in body)
print("  含三个一级指标(土壤/植被/地形):", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角:
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
```

```
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
```

```
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
# 工作台：页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」：", "评价流程" in body)
print("  含账户 admin：", "admin" in body)
print("  含页首横幅：", 'class="hero"' in body)
print("  含数据概览：", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码：", r.status_code)
print("  含案例项目：", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
# 新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」：", "测试项目A" in r.get_data(as_text=True
    ))
assert "测试项目A" in r.get_data(as_text=True)
# 清理测试数据，避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
    irst()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元：", all(t in body for t in ("岩质边坡",
    "碎石边坡", "平台")))
print("  含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
```

```
print()                                                                    11101
print(LINE)                                                                11102
print("测试 5  标准知识库与省际口径比对")                                    11103
print(LINE)                                                                11104
r = client.get("/standard/")                                               11105
body = r.get_data(as_text=True)                                            11106
print("  含辽宁/湖南/新疆标准:", all(p in body for p in ("辽宁", "湖南",    11107
    "新疆")))                                                              11108
assert "辽宁" in body                                                      11109
r = client.get("/standard/compare")                                        11110
body = r.get_data(as_text=True)                                            11111
print("  口径比对页含「植被覆盖度」分组:", "植被覆盖度" in body)             11112
print("  含多省份并列:", "湖南" in body and "新疆" in body)                  11113
assert "植被覆盖度" in body                                                11114
print()                                                                    11115
print(LINE)                                                                11116
print("测试 6  限制性指标判定（一票否决）")                                  11117
print(LINE)                                                                11118
r = client.get("/constraint/%d" % pid)                                     11119
body = r.get_data(as_text=True)                                            11120
print("  含五项条件:", all(x in body for x in ("政策约束", "区位约束",      11121
    "土壤污染", "地质灾害", "耕地属性")))                                  11122
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",         11123
    "地质灾害", "耕地属性"))                                               11124
# 全部通过                                                                 11125
client.post("/constraint/%d" % pid, data={                                 11126
    "policy_pass": "1", "location_pass": "1",                              11127
    "pollution_pass": "1",                                                 11128
    "geo_pass": "1", "farmland_pass": "1"})                                11129
r = client.get("/constraint/%d" % pid)                                     11130
body = r.get_data(as_text=True)                                            11131
print("  全通过后显示「通过，可进入定量评价」:", "可进入定量评价" in body)   11132
assert "可进入定量评价" in body                                            11133
# 一项不通过                                                               11134
client.post("/constraint/%d" % pid, data={                                 11135
    "policy_pass": "1", "location_pass": "0",                              11136
    "pollution_pass": "1",                                                 11137
    "geo_pass": "1", "farmland_pass": "1"})                                11138
r = client.get("/constraint/%d" % pid)                                     11139
body = r.get_data(as_text=True)                                            11140
print("  一项不通过后显示「不通过」:", "不通过" in body)                   11141
print("  含失败项「区位约束」:", "区位约束" in body)                       11142
assert "不通过" in body                                                    11143
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤             11144
client.post("/constraint/%d" % pid, data={                                 11145
    "policy_pass": "1", "location_pass": "1",                              11146
    "pollution_pass": "1",                                                 11147
    "geo_pass": "1", "farmland_pass": "1"})                                11148
r = client.get("/constraint/%d" % pid)                                     11149
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True)) 11150
```

```
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
#  ==== 测试/test_web_evaluate.py ====
#  -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
#  与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
#  登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
```

```
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.i
    d.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
```

```
import io                                                              11251
sys.stdout.reconfigure(encoding="utf-8")                               11252
#  本脚本位于  测试/  子目录，先把仓库根目录加入模块搜索路径，                  11253
#  使 `from app import ...` 与 app/app.db 的定位不受影响。                 11254
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil           11255
    e__)))                                                             11256
if ROOT not in sys.path:                                               11257
    sys.path.insert(0, ROOT)                                           11258
from app import create_app                                             11259
LINE = "=" * 66                                                        11260
class TestConfig:                                                      11261
    SECRET_KEY = "test"                                                11262
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(             11263
        ROOT, "app", "app.db")                                         11264
    SQLALCHEMY_TRACK_MODIFICATIONS = False                             11265
    WTF_CSRF_ENABLED = False                                           11266
    TESTING = True                                                     11267
app = create_app(TestConfig)                                           11268
client = app.test_client()                                             11269
#  登录（案例种子已创建  admin  用户）                                       11270
client.post("/login", data={"username": "admin",                       11271
    "password": "admin123"})                                           11272
from app.models import MinePatch, EvaluationResult                     11273
with app.app_context():                                                11274
    import seed_demo                                                   11275
    patch = seed_demo.demo_patch()                                     11276
    pid = patch.id                                                     11277
#  报告取自已保存的评价记录，因此先按网页路径真实执行一次评价                      11278
from app.views.evaluate import load_patch_input,                       11279
    save_evaluation                                                    11280
with app.app_context():                                                11281
    units = load_patch_input(pid)                                      11282
    rec = save_evaluation(pid, units)                                  11283
    rid = rec.id                                                       11284
    print("前置：已执行评价并保存记录  id=%d，F=%.4f（%s）"                    11285
          % (rid, rec.final_score, rec.grade))                         11286
    from app.models import ScoreDetail                                 11287
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co           11288
    unt()                                                              11289
    print("         逐层得分明细已落库：%d  条" % n_detail)                    11290
print()                                                                11291
print(LINE)                                                            11292
print("测试  1   报告列表")                                                11293
print(LINE)                                                            11294
r = client.get("/report/")                                             11295
print("   状态码：", r.status_code)                                        11296
body = r.get_data(as_text=True)                                        11297
print("   含「所属项目」列：", "所属项目" in body)                             11298
assert r.status_code == 200                                            11299
print()                                                                11300
```

```
print(LINE)                                                          11301
print("测试 2  得分明细页")                                          11302
print(LINE)                                                          11303
r = client.get("/report/%d" % pid)                                   11304
body = r.get_data(as_text=True)                                      11305
print("  状态码:", r.status_code)                                    11306
print("  含「得分明细」:", "得分明细" in body)                       11307
print("  含「最终评价因子 F」:", "最终评价因子 F" in body)           11308
print("  含二级指标明细（植被覆盖度/坡度/坡向）:", all(n in body for n in   11309
    ("植被覆盖度", "坡度", "坡向")))                                 11310
print("  含权重/标准化值列:", "标准化值" in body and "贡献" in body)  11311
assert r.status_code == 200                                          11312
assert "植被覆盖度" in body and "标准化值" in body                   11313
print()                                                              11314
print(LINE)                                                          11315
print("测试 3  Word 报告导出")                                       11316
print(LINE)                                                          11317
r = client.get("/report/%d/word" % pid)                              11318
print("  状态码:", r.status_code)                                    11319
print("  Content-Type:", r.headers.get("Content-Type",               11320
    "")[:60])                                                        11321
data = r.data                                                        11322
print("  文件大小:", len(data), "字节")                              11323
print("  是否为有效 docx（PK 头）:", data[:2] == b"PK")              11324
assert r.status_code == 200 and data[:2] == b"PK"                    11325
# 用 python-docx 打开验证内容                                       11326
import docx                                                          11327
d = docx.Document(io.BytesIO(data))                                  11328
text = "\n".join(p.text for p in d.paragraphs)                       11329
print("  含报告标题:", "自然恢复适宜性评价报告" in text)             11330
print("  含评价结论章节:", "评价结论" in text)                       11331
print("  含得分明细章节:", "得分明细" in text)                       11332
print("  段落数:", len(d.paragraphs), " 表格数:", len(d.tables))     11333
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3       11334
print()                                                              11335
print(LINE)                                                          11336
print("报告输出全部测试通过 √")                                     11337
print(LINE)                                                          11338
# ==== 测试/test_ui_fixes.py ====                                    11339
# -*- coding: utf-8 -*-                                              11340
"""界面修复自检:「记住我」与各模块删除功能。"""                      11341
import os                                                            11342
import re                                                            11343
import sys                                                           11344
# 本脚本位于 测试/ 子目录,先把仓库根目录加入模块搜索路径,            11345
# 使 `from app import ...` 与 app/app.db 的定位不受影响。            11346
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil         11347
    e__)))                                                           11348
if ROOT not in sys.path:                                             11349
    sys.path.insert(0, ROOT)                                         11350
```

```
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ----------------------------------------------------------------
    -----------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
    "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
```

```
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
     "password": "admin123"},
                follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
     h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
     else "[不符合]"))
assert not rt2
# 4) 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
                rt[0].split(";")[0].split("=", 1)[1],
     domain="localhost")
r3 = c3.get("/", follow_redirects=False)
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]",
     r3.status_code))
assert r3.status_code == 200
# ------------------------------------------------------------------
     ----------------
print()
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
     "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
     "[不符合]"))
assert "confirm(" in html
# 图斑列表：地址用的是项目主键，不是图斑主键
with app.app_context():
     from app.models import MinePatch, Project
     import seed_demo
     patch = seed_demo.demo_patch()
     patch_id = patch.id
     project_id = patch.project_id
html = client.get("/project/%d/patches" %
     project_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
     html and "btn--danger" in html else "[不符合]"))
assert "/project/patch/" in html and "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
     ue)
```

```
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__
    e__)))
if ROOT not in sys.path:
```

```
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ----------------------------------------------------------------
    ----------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
```

```
        last = OperationLog.query.order_by(OperationLog.id.desc(
        )).first()
    ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
        after))
    ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
    ok("project" in (last.target_type or ""), "日志记录了对象类型",
        last.target_type)
    r = c.get("/profile?tab=logs")
    body = r.get_data(as_text=True)
    ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
    ok("新建项目" in body, "日志页显示操作记录")
    # 账号编辑 / 删除（职务只有管理员与操作员两种）
    with app.app_context():
        u = User(username="tmp_user", role="operator",
                 password_hash=User.query.first().password_hash)
        db.session.add(u); db.session.commit()
        uid = u.id
    r = c.get("/users/%d/edit" % uid)
    ok(r.status_code == 200, "账号编辑页可访问")
    # 保护校验要趁系统里只有一名管理员时做：
    # 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
    with app.app_context():
        admin_id = User.query.filter_by(role="admin").first().id
    c.post("/users/%d/edit" % admin_id,
           data={"role": "operator", "active": "1"})
    with app.app_context():
        ok(db.session.get(User, admin_id).role == "admin",
        "拒绝把最后一个管理员降级")
    c.post("/users/%d/edit" % uid, data={"role": "admin",
        "active": "1"})
    with app.app_context():
        ok(db.session.get(User, uid).role == "admin",
        "职务已修改并保存")
    c.post("/users/%d/delete" % uid, follow_redirects=True)
    with app.app_context():
        ok(db.session.get(User, uid) is None, "账号删除生效")
    # 清理日志核对用的项目
    with app.app_context():
        for p in Project.query.filter_by(name="日志核对项目").all():
            db.session.delete(p)
        db.session.commit()
    # ------------------------------------------------------------
        ------------------
    print()
    print(LINE)
    print("遥感影像挂接")
    print(LINE)
    doc = fitz.open()
    pg = doc.new_page(width=320, height=200)
    pg.insert_text((20, 60), "RS image", fontname="china-s",
```

```
        fontsize=16)                                                    11601
png = doc[0].get_pixmap(dpi=100).tobytes("png")                         11602
doc.close()                                                             11603
r = c.post("/project/patch/%d/image" % patch_id,                        11604
                data={"image": (io.BytesIO(png), "影像.png"),           11605
                      "image_note": "2025年6月无人机正射影像"},          11606
                content_type="multipart/form-data",                     11607
        follow_redirects=True)                                          11608
body = r.get_data(as_text=True)                                         11609
ok("已挂接" in body and "<img" in body, "影像上传并显示")                 11610
with app.app_context():                                                 11611
        rel = db.session.get(MinePatch, patch_id).image_path            11612
ok(bool(rel), "数据库记录影像路径", rel)                                  11613
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,                       11614
        rel.replace("uploads/", "", 1))                                 11615
ok(os.path.isfile(abs_path), "影像文件落盘")                              11616
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL            11617
    访问")                                                              11618
r = c.post("/project/patch/%d/image" % patch_id,                        11619
                data={"image": (io.BytesIO(b"x"), "bad.txt")},          11620
                content_type="multipart/form-data",                     11621
        follow_redirects=True)                                          11622
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")                11623
c.post("/project/patch/%d/image/delete" % patch_id,                     11624
        follow_redirects=True)                                          11625
with app.app_context():                                                 11626
        ok(db.session.get(MinePatch, patch_id).image_path is            11627
        None, "影像移除后字段清空")                                       11628
ok(not os.path.isfile(abs_path), "影像文件已删除")                        11629
# ------------------------------------------------------------------    11630
        ----------------                                                11631
print()                                                                 11632
print(LINE)                                                             11633
print("指标定义维护与多专家矩阵")                                         11634
print(LINE)                                                             11635
r = c.get("/ahp/indicators")                                            11636
body = r.get_data(as_text=True)                                         11637
with app.app_context():                                                 11638
        n_ind = Indicator.query.count()                                 11639
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d      11640
    项" % n_ind)                                                        11641
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),          11642
    "指标树按层级展示")                                                 11643
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")     11644
c.post("/ahp/indicators/coverage/edit",                                 11645
          data={"name": "植被覆盖度", "unit": "%", "ideal_type":        11646
        "benefit",                                                      11647
                "ideal_value": "75", "enabled_units": ["岩质边坡"],    11648
                "description": "自检修改"})                             11649
with app.app_context():                                                 11650
```

```
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
        "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="cover
    ge").first().description or ""),
        "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
        "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------
    ------------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,,\n"
```

```
)                                                                    11701
r = c.post("/indicator/import",                                      11702
           data={"patch_id": str(patch_id),                          11703
                 "datafile": (io.BytesIO(csv_text.encode("ut         11704
    f-8")), "d.csv")},                                               11705
           content_type="multipart/form-data",                       11706
    follow_redirects=True)                                           11707
body = r.get_data(as_text=True)                                      11708
ok("导入结果" in body, "导入结果页展示")                                11709
ok("不存在单元" in body, "错误行被报告")                                11710
with app.app_context():                                              11711
    u = MicroUnit.query.filter_by(patch_id=patch_id,                 11712
    unit_type="碎石边坡").first()                                      11713
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(         11714
    )                                                                11715
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and                         11716
    abs(ind.coverage - 87.92) < 1e-6,                                11717
        "导入数据已写入数据库")                                          11718
r = c.post("/indicator/import",                                      11719
           data={"patch_id": str(patch_id),                          11720
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},           11721
           content_type="multipart/form-data",                       11722
    follow_redirects=True)                                           11723
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")          11724
# ------------------------------------------------------------------11725
    ----------------                                                 11726
print()                                                              11727
print(LINE)                                                          11728
print("图表与 PDF 导出")                                              11729
print(LINE)                                                          11730
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,      11731
    "0.8376")])                                                      11732
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")          11733
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,    11734
    "0.4000")])                                                      11735
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")              11736
# 报告是评价结论的呈现，需先真实执行一次评价才能打开                          11737
with app.app_context():                                              11738
    from app.views.evaluate import load_patch_input,                 11739
    save_evaluation                                                  11740
    save_evaluation(patch_id, load_patch_input(patch_id))            11741
r = c.get("/report/%d" % patch_id)                                   11742
body = r.get_data(as_text=True)                                      11743
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %          11744
    body.count("<svg"))                                              11745
exe = browser.browser_path()                                         11746
if exe is None:                                                      11747
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")                      11748
else:                                                                11749
    r = c.get("/report/%d/pdf" % patch_id)                           11750
```

```
    data = r.data                                                    11751
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %          11752
    len(data))                                                       11753
    d = fitz.open(stream=data, filetype="pdf")                       11754
    txt = "".join(d[i].get_text() for i in                           11755
    range(d.page_count))                                             11756
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")            11757
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")  11758
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)    11759
    d.close()                                                        11760
print()                                                              11761
print(LINE)                                                          11762
print("补齐模块自检全部通过 √")                                     11763
print(LINE)                                                          11764
# ==== 测试/test_workflow.py ====                                    11765
# -*- coding: utf-8 -*-                                              11766
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。              11767
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。  11768
本测试逐个构造「缺前置」的场景，验证系统不会放行。                  11769
"""                                                                  11770
import os                                                            11771
import sys                                                           11772
sys.stdout.reconfigure(encoding="utf-8")                             11773
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，          11774
# 使 `from app import ...` 与 app/app.db 的定位不受影响。            11775
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil         11776
    e__)))                                                           11777
if ROOT not in sys.path:                                             11778
    sys.path.insert(0, ROOT)                                         11779
from app import create_app                                           11780
from app.models import (db, Project, MinePatch, MicroUnit,           11781
    IndicatorData,                                                   11782
                        ConstraintCheck, EvaluationResult)           11783
from app.services import workflow as wf                              11784
LINE = "=" * 72                                                      11785
BASE = ROOT                                                          11786
class TestConfig:                                                    11787
    SECRET_KEY = "test"                                              11788
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +                         11789
    os.path.join(BASE, "app", "app.db")                              11790
    SQLALCHEMY_TRACK_MODIFICATIONS = False                           11791
    WTF_CSRF_ENABLED = False                                         11792
    TESTING = True                                                   11793
app = create_app(TestConfig)                                         11794
c = app.test_client()                                                11795
c.post("/login", data={"username": "admin", "password":              11796
    "admin123"})                                                     11797
_fails = []                                                          11798
def ok(flag, label, extra=""):                                       11799
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",       11800
```

```
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
        from app.models import ScoreDetail
        # 先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
```

```
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        pr = Project.query.filter_by(name="门禁-" +
    tag).first()
        if pr:
            db.session.delete(pr)
        db.session.commit()
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ----------------------------------------------------------------
    ------------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
```

```
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
   "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
   "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 100%")
pid = new_patch("5"); add_constraint(pid, passed=True)
with app.app_context():
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
```

```
        ratio=0.5)                                                    11951
        db.session.add(u); db.session.commit()                        11952
        db.session.add(IndicatorData(unit_id=u.id,                    11953
        soil_ratio=0.3, thickness=10,                                 11954
                                              ph=8.0,                 11955
        organic_matter=5.0, coverage=80.0,                            11956
                                              shannon=0.6, slope=20.0,11957
        aspect="阳坡"))                                               11958
        db.session.commit()                                           11959
r = post_eval(pid)                                                    11960
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")              11961
cleanup(pid, "5")                                                     11962
print()                                                               11963
print(LINE)                                                           11964
print("二、前置条件齐备时才放行")                                      11965
print(LINE)                                                           11966
pid = new_patch("OK"); add_units(pid); add_constraint(pid,            11967
        passed=True); add_data(pid)                                   11968
with app.app_context():                                               11969
        pr = wf.patch_progress(pid)                                   11970
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")           11971
r = post_eval(pid)                                                    11972
n = results_of(pid)                                                   11973
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)                11974
print("   重定向到:", r.headers.get("Location"))                      11975
print()                                                               11976
print(LINE)                                                           11977
print("三、报告不得先于评价产生")                                      11978
print(LINE)                                                           11979
pid2 = new_patch("R")     # 什么都不做                                11980
r = get_report(pid2)                                                  11981
ok(r.status_code == 302 and "/report/" ==                             11982
        (r.headers.get("Location") or ""),                            11983
    "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)   11984
for suffix in ("/pdf", "/word"):                                      11985
        rr = c.get("/report/%d%s" % (pid2, suffix),                   11986
        follow_redirects=False)                                       11987
        ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP11988
        %d" % rr.status_code)                                         11989
cleanup(pid2, "R")                                                    11990
cleanup(pid, "OK")                                                    11991
print()                                                               11992
print(LINE)                                                           11993
print("四、前置齐备但从未执行评价时，报告仍不得产生")                  11994
print(LINE)                                                           11995
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，  11996
#  也不应凭空推算出结论。                                               11997
pid3 = new_patch("N")                                                 11998
add_units(pid3); add_constraint(pid3, passed=True);                   11999
        add_data(pid3)                                                12000
```

```
with app.app_context():
    pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
   "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
   "直接访问 /report/<id> 被退回，且不产生任何评价结果",
   "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
   "报告数值取自评价记录，而非重新计算",
   "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
#  导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
#  演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
```

```
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
# 论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
# 允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
           follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
           follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
       follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first(
    ).id
r = c.post("/unit/edit/%d" % uid,
           data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
```

```
                    MicroUnit.query.filter_by(patch_id=pid4).
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
```

```
c.post("/login", data={"username": "admin", "password": "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]", label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s" % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s" % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id, unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).first()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10, ph=9.24,
            organic_matter=4.89, coverage=87.92, shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True, location_pass=True,
            pollution_pass=True, geo_pass=True, farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        return patch.id
def cleanup(pid):
    with app.app_context():
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.in_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delete()
```

```
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        p = Project.query.filter_by(code="WEIGHT-%s" %
    TAG).first()
        if p:
            db.session.delete(p)
        db.session.commit()
def run_eval(pid):
    c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ) \
            .order_by(EvaluationResult.id.desc()).first().fi
    nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
    label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",
   "土壤指标 %.2f" % w["l1"]["土壤指标"])
pid = build_case()
base_score = run_eval(pid)
print("  默认权重下的评价值 F = %.4f" % base_score)
print()
print(LINE)
print("二、保存 AHP 算出的权重后，评价应改用新权重")
print(LINE)
with app.app_context():
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",
    0.20),
                                ("terrain", 0.10)])
    n_items = WeightItem.query.count()
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)
with app.app_context():
```

```
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",
    label2)
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,
   "未覆盖的二级权重仍保留论文取值",
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])
new_score = run_eval(pid)
print("  新权重下的评价值 F = %.4f" % new_score)
ok(abs(new_score - base_score) > 1e-4,
   "权重变化确实改变了评价结果",
   "%.4f → %.4f" % (base_score, new_score))
print()
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
        .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
   "评价记录中标注了权重来源",
   stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到论文取值")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",
    label3)
back_score = run_eval(pid)
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
    base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
```

```
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/timefmt.py ====
# -*- coding: utf-8 -*-
"""时间显示：库里一律存 UTC，展示时换成服务器本地时间。
为什么不在写库时直接用本地时间：一个系统里时间基准只能有一个。
`app/models.py` 里所有 `created_at` / `checked_at` 都用
    `datetime.utcnow`，
存的是 UTC；如果写库改成本地时间，已经存进库的几千条记录会新旧混着差 8 小时。
所以统一在「展示」这一层换算，老数据不用动。
用法（Jinja）：
    {{ row.created_at | localdt }}                    →
    2026-09-26 23:33
    {{ row.created_at | localdt("%Y-%m-%d %H:%M:%S") }} →
    2026-09-26 23:33:28
"""
from datetime import datetime, timezone
DEFAULT_FORMAT = "%Y-%m-%d %H:%M"
def to_local(value, fmt=DEFAULT_FORMAT, empty="—"):
    """把 UTC 时间换算成服务器本地时间并格式化。
```

```
    传进来的值按「无时区即 UTC」处理；带时区的原样换算，便于以后改用
    带时区的时间戳。空值返回占位符，免得页面上出现 None。
    """
    if value is None:
        return empty
    if value.tzinfo is None:
        value = value.replace(tzinfo=timezone.utc)
    return value.astimezone().strftime(fmt)
def now_utc():
    """取当前 UTC 时间（带时区），用于只作展示、不落库的时间戳。"""
    return datetime.now(timezone.utc)
# ==== app/views/__init__.py ====
# -*- coding: utf-8 -*-
```